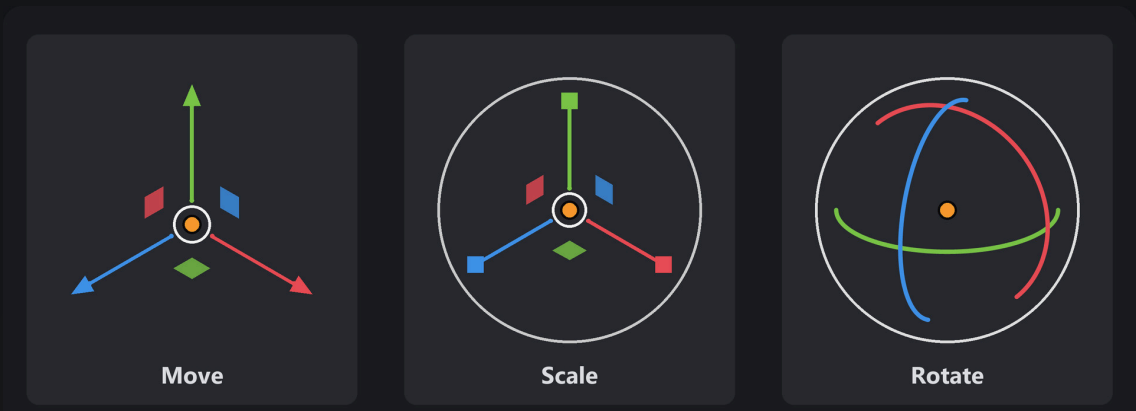
Blendon

Blender-style Scene view, gizmos and pie menus for the Unity Editor



User Manual

Version 1.0.0

Unity 6000.0 (Unity 6) or newer

DRYGIEL

blendon.drygiel.com

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1 Introduction

Blendon rebinds and extends the Unity Scene view so that it behaves like Blender. You orbit around your selection, pan and zoom the way Blender does, replace the Move, Rotate, Scale and Transform gizmos with Blender-style ones, type exact values in the middle of a drag, drag a box that selects everything it touches, reach for utilities such as isolate view and snap-to-floor, open radial pie menus on a single key, and right-click for a searchable menu that still holds every entry Unity's own menu has.

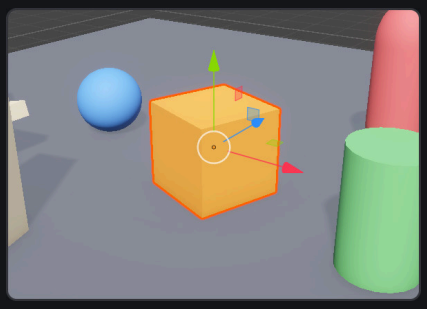


Figure 1.1 The Move gizmo, drawn the Blender way.

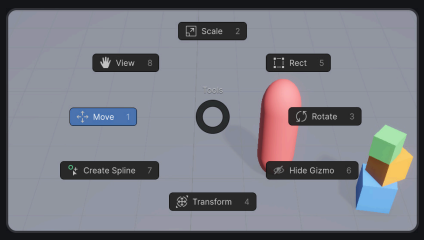


Figure 1.2 The Tools pie menu, opened on Q.

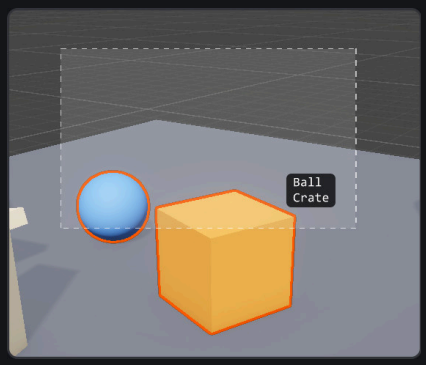


Figure 1.3 Box Select picks everything the box touches.

Every feature is optional and fully configurable from a single settings window. Nothing is forced on you, and nothing is written to your project until you ask for it.

1.1 Editor-only by design

Blendon is an Editor-only extension. It adds nothing to player builds, contains no runtime components, makes no network requests, collects no analytics and installs no additional packages. Its assemblies are compiled for the Editor platform only, so no part of Blendon can end up in a shipped game.

1.2 No scripting required

Blendon is used entirely through the Unity Editor interface: the Scene view, the settings window and Unity's own Shortcut Manager. You never have to write, modify or reference a single line of code to use any feature in this manual, and Blendon exposes no public scripting API for you to call. For that reason this manual contains no script reference and no code samples.

1.3 How to use this manual

If you have just installed Blendon, read Installation and setup and then work through the in-Editor Scene tutorial. Everything after the setup chapter is reference material: each chapter covers one group of features and can be read on its own, in any order.

Throughout this manual, menu paths are written as **Tools -> Blendon -> Settings**, and keys are written as **Shift + Z**. Unity's Shortcuts window opens from **Edit -> Shortcuts** on Windows and Linux and from **Unity -> Shortcuts** on macOS; this manual writes it as **Edit -> Shortcuts**. Where a default binding is given, it can always be changed; see Settings window. Screenshots show the Editor's dark skin with Blendon's default settings unless the caption says otherwise.

The settings window, its tooltips and Unity's own Shortcuts window spell keys the Unity way: `Shift+Z` for **Shift + Z**, `Num 1` for **Numpad 1**, and mouse buttons by number. `Mouse 1` is the right button, `Mouse 2` the middle one, and `Mouse 3` and `Mouse 4` are the side buttons. The settings window draws the mouse buttons as small mouse icons.

2 Requirements

- **Unity 6000.0 (Unity 6) or newer.** Everything introduced after Unity 6.0 (`EntityId`, the separate angle and scale snap toggles, the Grid handle orientation, the movable Scene grid, overlay groups) is routed through a compatibility layer, so Unity 6.0 and the newest release behave identically wherever the Editor itself allows it.
- **Any render pipeline.** Blendon draws through the Editor's own Handles and gizmo systems and does not depend on Built-in, URP or HDRP.
- **Windows, macOS or Linux.** Modifier labels are platform-correct: macOS users see Cmd and Option rather than Ctrl and Alt.
- **A numeric keypad with NumLock enabled** for the default keys of Numpad Views, Frame Selected (**Numpad .**), Isolate View (**Numpad /**) and keyboard zoom (**Numpad +** and **-**). All of them can be rebound to other keys, and everything else works without a keypad.

3 Installation and setup

This chapter is the step-by-step setup guide. Follow steps 1 to 5 in order; the whole process takes about two minutes.

3.1 Step 1 - Import the package

1. In Unity, open **Window -> Package Manager**.
2. Select **My Assets** from the packages dropdown at the top left.
3. Find **Blendon** in the list, then click **Download** and, once it finishes, **Import**.
4. Leave every file selected in the import dialog and click **Import**.

Blendon works from wherever it lands: directly under `Assets/`, inside a subfolder of your own, or as a package under `Packages/`. Nothing in the plugin references a fixed path, so you are free to move the folder afterwards.

Wait for Unity to finish compiling before continuing. The progress bar in the bottom right corner of the Editor disappears when compilation is done.

3.2 Step 2 - Choose a keyboard layout

Blendon's shortcuts ship with the Blender keys (G, R, S, the middle mouse button, the right-click menu and so on), and they work at once on whatever shortcut profile the Editor is on, Unity's read-only **Default** included. Nothing is written to your profile for that. While the Scene view has focus, these keys answer Blendon; a handful of Editor commands on the same keys (Cycle Tool Modes on **G**, the Scale tool on **R**, Show Tool Settings on **S**, ...) wait there, and keep working everywhere else.

On its first load Blendon asks one question, in a window titled **Blendon - keyboard presets** that stays in front of the Editor until you answer. It names how many Editor shortcuts the Blender keys cover and shows three cards, **Blender keys** on the left, **Unity keys** in the middle and **No changes** on the right, each with its keys and what the answer changes. Either preset leaves Blendon ready to use, **Skip** leaves your shortcuts as they are:

Answer	Result
Use Blendon keys	The default button: **Enter** picks it. Blendon answers the Blender keys in the Scene view, its context menu takes the right mouse button, and every Editor command it covers moves one modifier aside (Cycle Tool Modes to **Shift + G**, ...), like the Overview's **Make Editable Copy**; a key you tuned yourself stays as it is. On Unity's read-only **Default** profile Blendon first makes an editable copy named **Default copy** and switches to it without a second question. The right mouse button moves from the Editor's Scene view menu to Blendon's while Blendon runs, and goes back when the Editor closes. This is the recommended choice if you are coming from Blender.
Keep Unity keys	The Unity keyboard preset: Blendon's contested shortcuts move one modifier aside (grab to **Shift + G**, orbit to **Ctrl + Middle Mouse**, ...), and every Editor key keeps working in the Scene view too. On Unity's read-only **Default** profile the keys go into an editable copy, **Default copy**, made without a second question.
Skip	**Esc** and closing the window pick it too. Nothing is written to your shortcut profile and no copy is made. Blendon's keys answer in the Scene view, the Editor commands on the same keys everywhere else. The right mouse button stays the Editor's Scene view menu: on a profile Blendon could edit, its context menu starts switched off instead of taking the button, and switching it on later on the Context Menu page takes it. The tutorial still starts. Pick a preset whenever you like on the Overview page.

The question comes once per install in each project. An update of the package keeps your answer. On Windows, deleting Blendon's folder and importing it again is a new install, so the question comes again, unless the profile is already on the Unity keys: that is an answer of its own. On macOS and Linux an import into a project Blendon already ran in never asks again.

Both keyboard presets stay available on the Overview page of the settings window, shown below. See Keyboard presets for the full comparison, and for the right mouse button, which Blendon's context menu takes from the Editor's own Scene view menu. On Unity's read-only **Default** profile that write is not possible, so the right-click menu waits for an editable copy. Either preset answer to the first-run question makes one; after **Skip**, or if you switch back to **Default** later, the Overview page offers it.

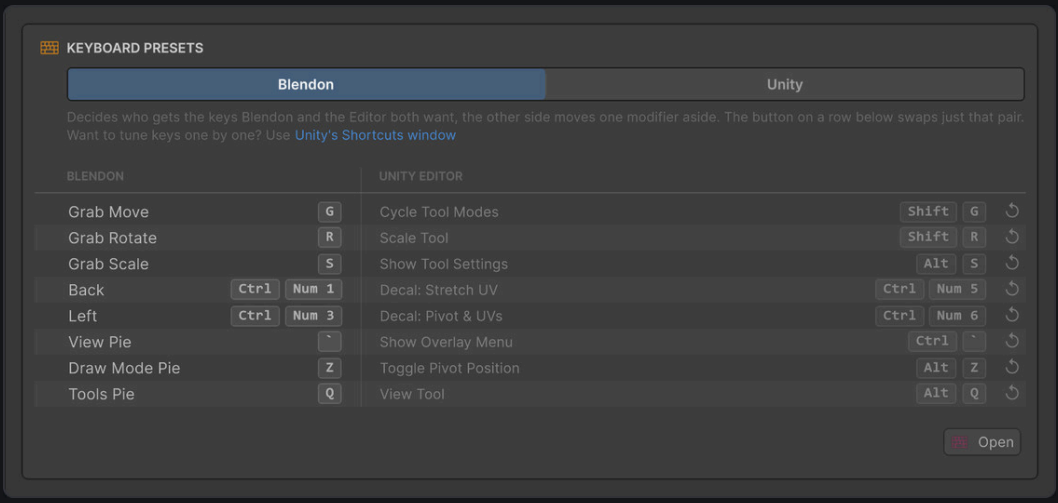


Figure 3.1 The Keyboard Presets panel on the Overview page after **Use Blendon keys**. The left column lists Blendon's shortcut and its key, the right column the Editor command and the key it moved to, one modifier aside. The button on the right swaps the pair: ↺ gives the Editor's command its key back and moves Blendon's aside, ↪ swaps them again.



Figure 3.2 Blendon preset: Blendon keeps the key, the Editor's command moves one modifier aside.

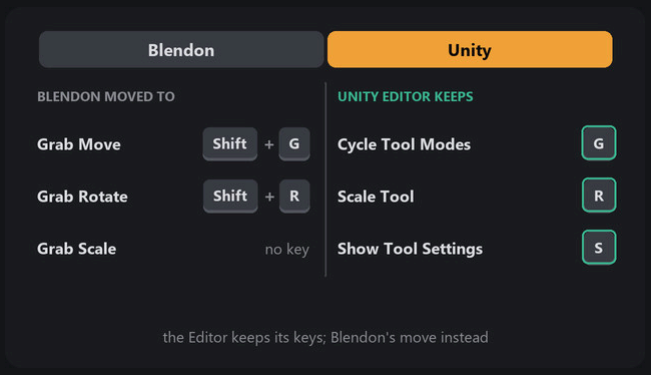


Figure 3.3 Unity preset: the Editor keeps its keys, and Blendon's move instead.

Every Blendon shortcut is an ordinary entry in **Edit -> Shortcuts**, under `Blendon/`, with the Blender key as its default. The folders follow the settings window's pages, as in `Blendon/Rotate/Grab` or `Blendon/Numpad Views/Front`. **Reset** there brings a Blendon key back to the Blender one.

3.3 Step 3 - Open the settings window

Open **Tools -> Blendon -> Settings**. The same menu holds **Copy Support Info**, for a bug report (see Reporting a problem), and **Prepare Uninstall...** (see Uninstalling).

This one window contains every setting Blendon has. The sidebar groups all features into **Get Started**, **Navigating**, **Transform Tools**, **Scene Tools** and **Keys & Menus**. The first time, the window opens on the Overview page, the **Blendon** entry at the top of the sidebar, which lists every feature with its current key

and an on/off toggle. After that it reopens on the page you last viewed.

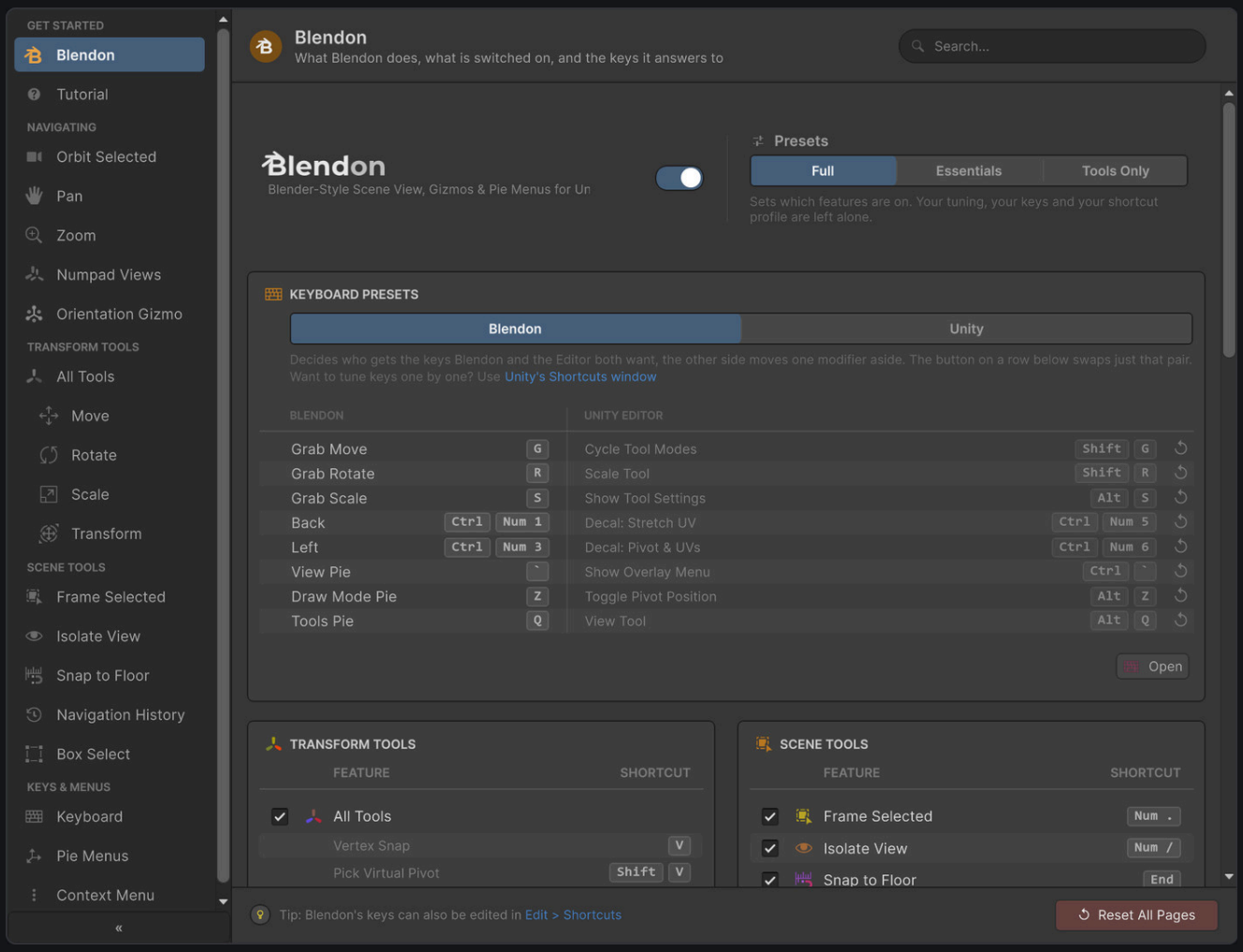


Figure 3.4 The settings window on its Overview page: the sidebar on the left, the master switch and feature presets at the top, and the Keyboard Presets panel below.

3.4 Step 4 - Verify the installation

Open any scene with at least one object in it and confirm the following. The keys below are the Blendon preset's; on the Unity preset orbit is **Ctrl + Middle Mouse** and grab is **Shift + G**. Click into the Scene view first: Blendon's keys answer while it has focus.

1. **Navigation.** Select an object, then hold the **Middle Mouse Button** in the Scene view and drag. The view should orbit around the selected object rather than around the Scene view's own pivot.
2. **Gizmos.** With the object still selected, press **W**. The Move gizmo should be drawn in the Blender style: three colored axis arrows, plane handles, a white screen-space ring and a center dot, as in Transform gizmos. Blendon's gizmos live in a tool context of their own, **Game Object (Blendon)**, which Blendon switches on by itself and **B** toggles; see The Blendon tool context.
3. **Grab.** Press **G** and move the mouse. The object should follow the cursor immediately. Press **Esc** or right-click to cancel, and the object returns to where it started.

If any of these does nothing, open **Tools -> Blendon -> Settings** and check that the feature's row on the Overview page is switched on and that its key is not highlighted as clashing with a Unity command. See Troubleshooting.

3.5 Step 5 - Work through the in-Editor tutorial

A small **Blendon Tutorial** badge sits in the bottom-right corner of the Scene view. Click it, and the settings window opens on the **Tutorial** page: a checklist of 56 tasks in eight chapters, in which you build a small scene and learn Blendon's gestures on the way. A task ticks off only once you actually perform the gesture, not by reading about it.

This is the fastest way to learn the plugin, and it is described in full in Scene tutorial. If you closed the badge, switch it back on from the Tutorial page of the settings window; your progress is kept. Answering **Keep Unity keys** in Step 2 switches the tutorial off once the Unity keys are written, since it teaches the Blender keys just declined; the switch at the top of the Tutorial page brings it back whenever you like.

3.6 Choosing a starting preset

The Overview page offers three feature presets, under **Presets**:

Preset	What it enables
Full	Every feature on. Your tuning, your keys and your shortcut profile are kept.
Essentials	Navigation and the transform gizmos, with their grab and vertex-snap keys. The numpad views, pie menus, context menu and scene tools stay off, and the Editor keeps the keys they would have taken.
Tools Only	Just the Move, Rotate, Scale and Transform gizmos, with their grab and vertex-snap keys. Everything else stays off, and the Editor keeps every key it would have taken.

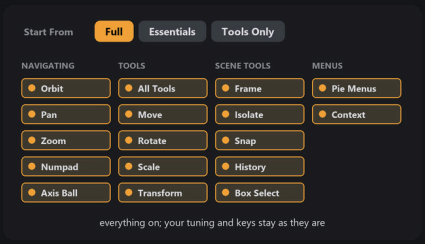


Figure 3.5 Full: every feature on, your tuning and keys kept.

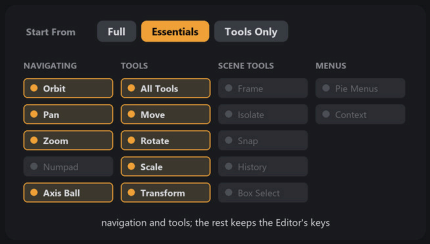


Figure 3.6 Essentials: navigation and the transform gizmos.

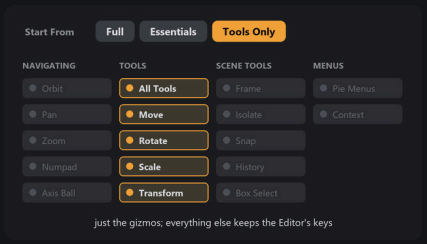


Figure 3.7 Tools Only: just the four gizmos.

Pick **Tools Only** or **Essentials** if you want to adopt Blendon gradually, and switch individual features on from the Overview page as you go. A preset only switches features on and off, and Blendon itself on if it is off; your tuning, your keys and your shortcut profile are left alone. The lit segment is the preset your switches match right now, so a fresh install shows **Full**, and flipping a single switch leaves none lit; clicking the lit one does nothing.

4 Navigation

Blendon moves the Scene view's camera the way Blender does: around the selection, anchored on the point under the cursor, and without ever leaving the mouse.

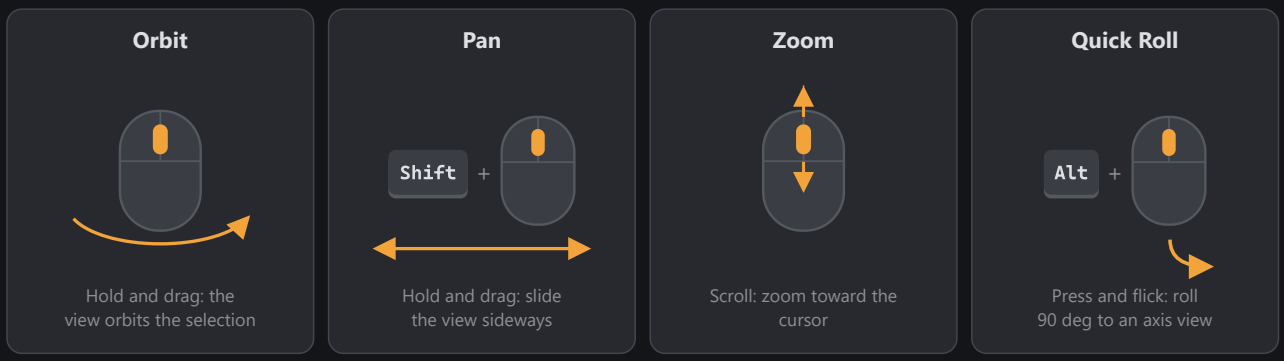


Figure 4.1 The four mouse gestures at a glance. Orbit, Quick Roll and Pan are rebindable, and the Unity keyboard preset moves them to Ctrl, Ctrl + Shift and Alt. Wheel zoom stays on the wheel.

Orbit, Quick Roll and Pan are drags, so their shortcut fields take mouse buttons only: any button, with or without **Shift**, **Ctrl** or **Alt**. The left button needs one of the three, since a bare left click selects. **Alt + Left Mouse**, say, orbits the way Unity's own orbit does. Unity's Shortcuts window also puts Orbit or Pan on a keyboard key: hold the key and drag with the middle mouse button.

A zoom, pan or orbit that starts while an animated view change is still flying, such as a numpad view, first lands it where it was heading, then acts, as in Blender.

4.1 Orbit Selected - Middle Mouse

Hold the **Middle Mouse Button** and drag to orbit the Scene view around the current selection. With multiple objects selected, the view orbits the point the gizmo sits on, as set by the Pivot Point mode. With a vertex settled by **Shift + V**, it orbits that vertex instead; see Vertex snapping. The numpad orbit steps and the Orientation Gizmo turn the view about the same point.

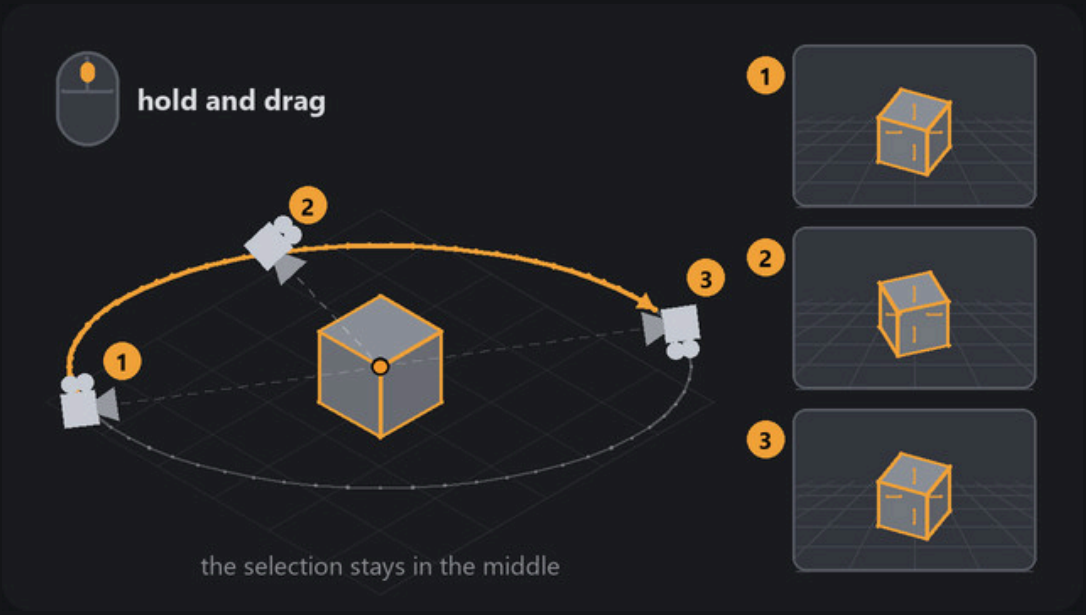


Figure 4.2 Hold the middle mouse button and drag. The camera circles the selection, which stays in the middle of the view.

- Separate horizontal and vertical sensitivity, in degrees per pixel.
- **Depth.** With nothing selected, orbit around the scene point under the cursor instead of the Scene view's own pivot. **Show Pivot Dot** marks that point while you orbit.
- **Snap view to world axis.** Hold the snap modifier (**Alt** by default) in the middle of an orbit to lock the view to the nearest world axis and switch to orthographic; release it to return to a free perspective orbit. The **Threshold** angle, 12 degrees by default, decides how close the view has to be before it snaps. When the orbit shortcut holds the snap modifier itself, as in **Alt + Left Mouse**, that first Alt belongs to the orbit: let go of it and press it again to snap.
- **Blendon Context Only.** Off by default. On, Orbit Selected and Quick Roll answer in the Blendon tool context only.
- **Right-click or Esc** while dragging cancels the orbit and restores the original view.
- **A click is still a click.** The orbit starts only once the drag passes 6 pixels, Unity's own click distance. A middle click without a drag does what Unity's **Scene View/Focus** does: the Scene view's pivot moves to the surface you clicked, and the view slides over to center it without zooming. This holds while the orbit sits on the middle button, Focus's own key; on any other key, Unity's Focus answers the middle click itself.

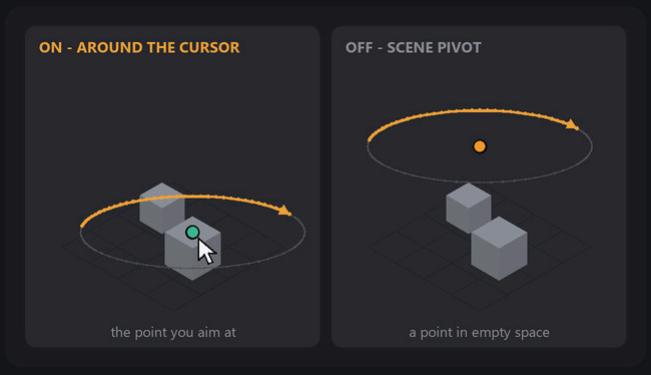


Figure 4.3 Depth: with nothing selected, the view turns around the point you aim at.

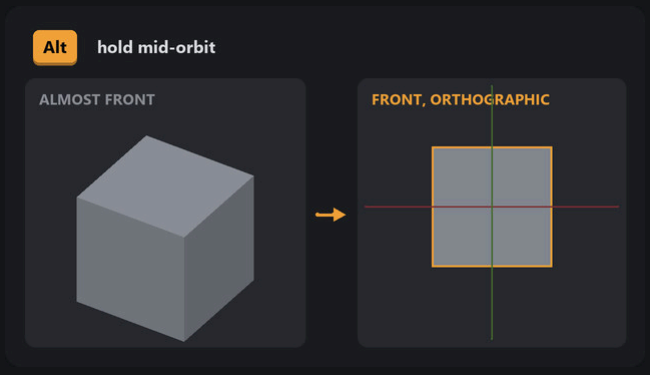


Figure 4.4 Hold Alt mid-orbit and the view snaps to the nearest axis, orthographic.

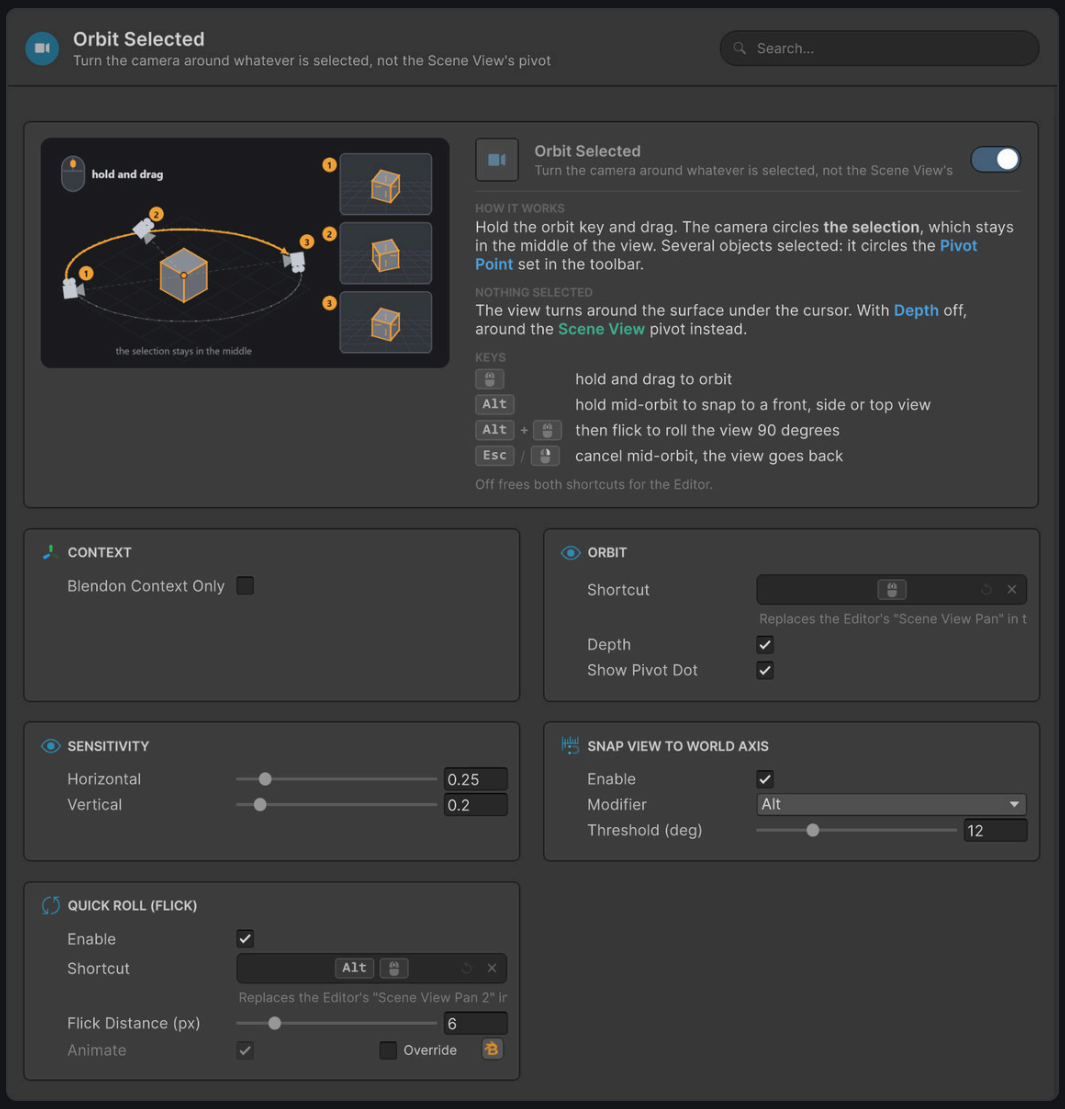


Figure 4.5 The Orbit Selected page. Its header card sums the gesture up, and Quick Roll lives here too, since it is the same middle-mouse gesture.

4.2 Quick Roll - Alt + Middle Mouse, flick

Hold **Alt**, press the **Middle Mouse Button**, then flick a short distance in any direction to roll the view 90 degrees that way and drop into orthographic. The view lands squarely on the nearest named axis view regardless of where the flick started.

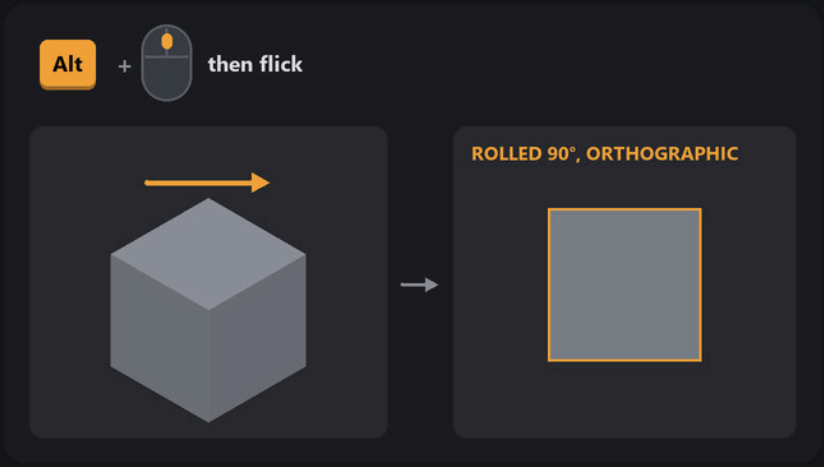


Figure 4.6 Alt, then a short flick of the middle mouse button: the view rolls 90 degrees onto an orthographic axis view.

- **Flick Distance.** How far the mouse must move before the roll commits, 6 pixels by default.
- **Animate.** The roll can be animated instead of cutting straight to the target.
- Quick Roll can be disabled on its own while the rest of Orbit Selected stays active.

4.3 Camera Pan - Shift + Middle Mouse

Pan the Scene view with **Shift + Middle Mouse Button**. It is an ordinary shortcut, `Blendon/Pan`, rebindable on the Pan page, and the Unity keyboard preset moves it to **Alt + Middle Mouse**.

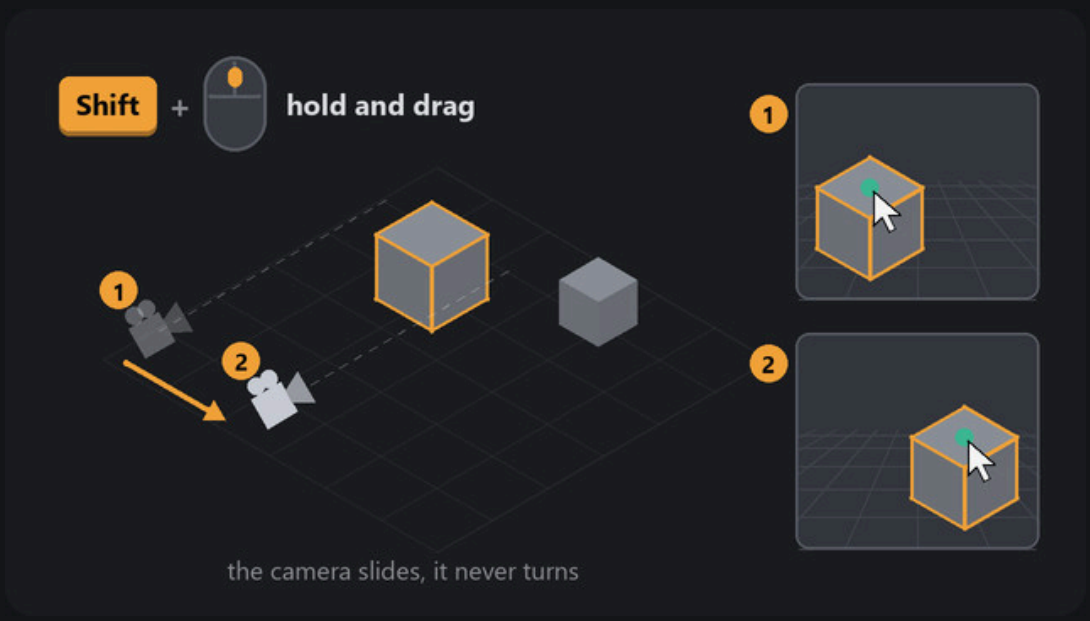


Figure 4.7 The camera slides sideways without turning, and the point you grabbed stays under the cursor.

- **Depth.** Keep the exact world point under the cursor anchored while panning.
- **Invert Drag.** Reverses the drag direction.
- **Cursor Wrap.** Lets a pan run past the edge of the view; see Defaults.
- **Blendon Context Only.** Off by default. On, the pan answers in the Blendon tool context only.
- **Right-click** while panning cancels the pan and restores the view, while **RMB cancels drag** is on.

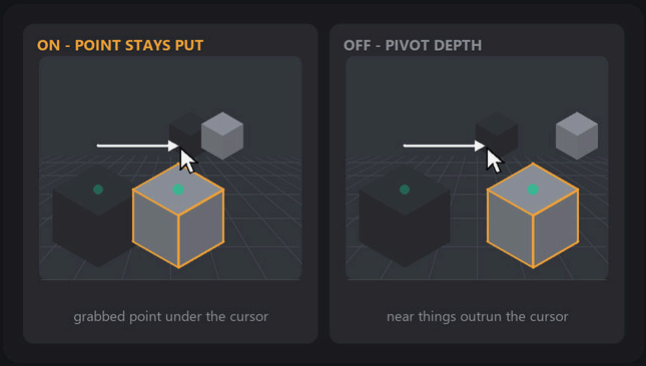


Figure 4.8 Depth on: the grabbed point stays exactly under the cursor, near or far.

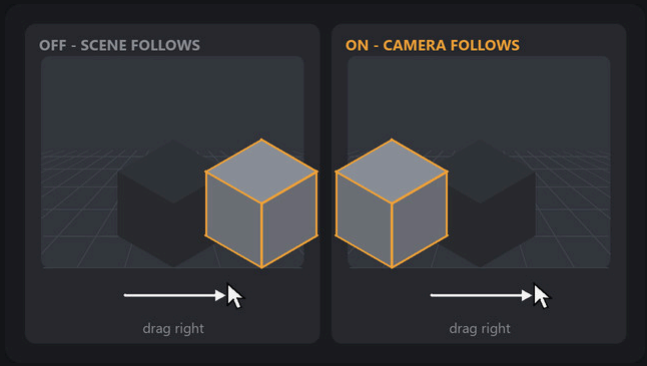


Figure 4.9 Invert Drag: the camera follows the cursor instead of the scene.

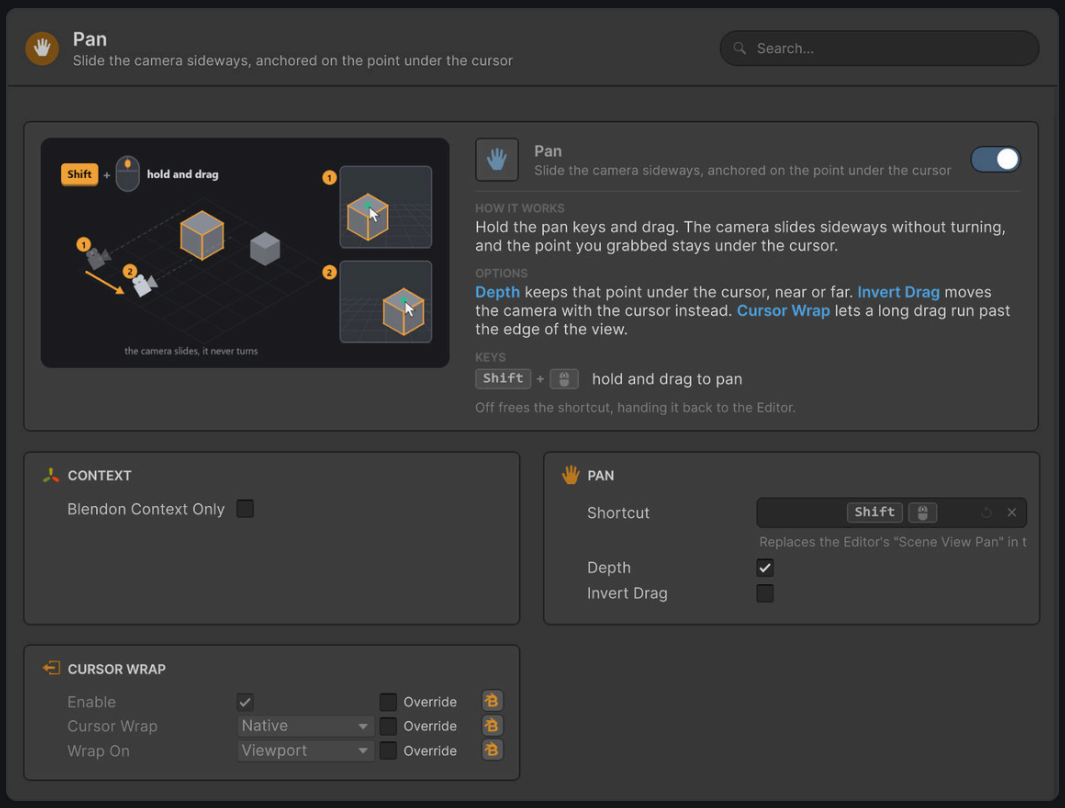


Figure 4.10 The Pan page. The Cursor Wrap panel reads the shared default until you tick one of its Override boxes.

4.4 Zoom - Scroll Wheel

Replaces Unity's scroll-wheel zoom with Blender's, mirroring Blender's *Preferences -> Navigation -> Zoom Method*:

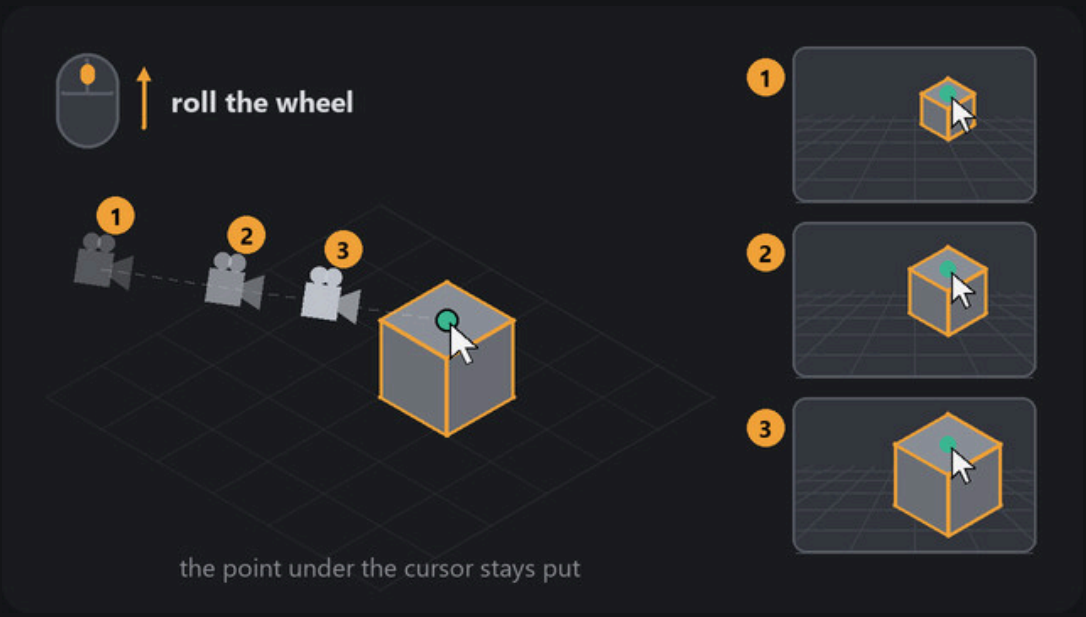


Figure 4.11 Roll the wheel over the Scene view. The point under the cursor stays put while the camera closes in.

Method	Behavior
Scale (default)	Stepwise multiplicative scale of the view size around the pivot.
Dolly	Moves the viewpoint toward the Zoom To point, or along the view axis with Zoom To on Center, keeping the view size constant.
Continue	Continuous and magnitude-sensitive: faster scrolls zoom proportionally more.

- **Zoom To.** The point that stays put on screen while the view zooms. **Cursor** (default) keeps the scene point under the cursor, **Selection** the selected object (several: the toolbar's Pivot Point), **Center** the middle of the view. Selection zooms to the middle when nothing selected is in view.
- **Blendon Context Only.** Off by default. On, the wheel and the keypad keys zoom Blendon's way in the Blendon tool context only, and the Editor's way everywhere else.
- **Sensitivity** is adjustable and **Invert Direction** reverses the wheel.
- **Keyboard Zoom. Numpad +** and **Numpad -** zoom in fixed steps, 20% of the view size by default, toward the Zoom To point (with Cursor, only while the cursor is over the view). A zoom-out step is the exact reverse of a zoom-in step. Keyboard zoom has its own switch, which hands both keys back to the Editor when it is off.

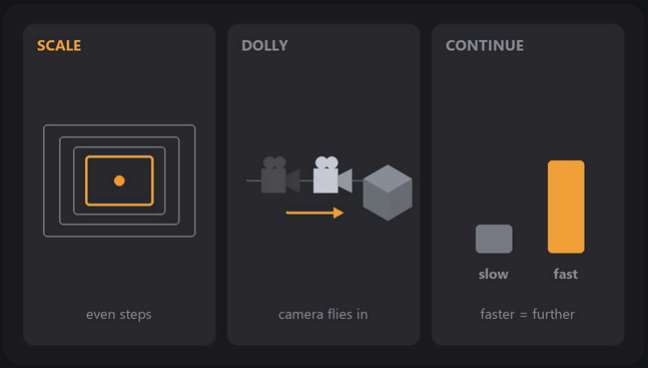


Figure 4.12 The three zoom methods: even steps, a camera that flies in, or speed-sensitive.

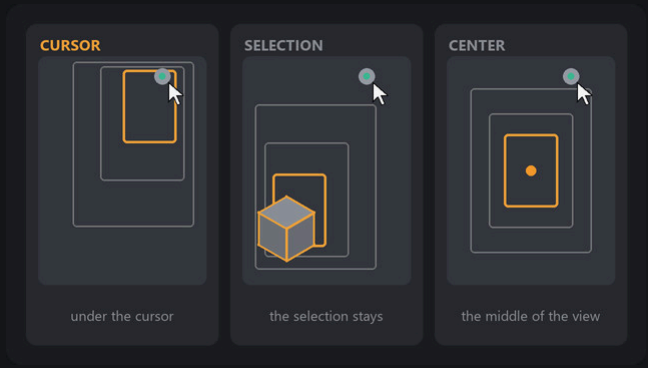


Figure 4.13 Zoom To keeps the cursor, the selection or the middle of the view where it is.

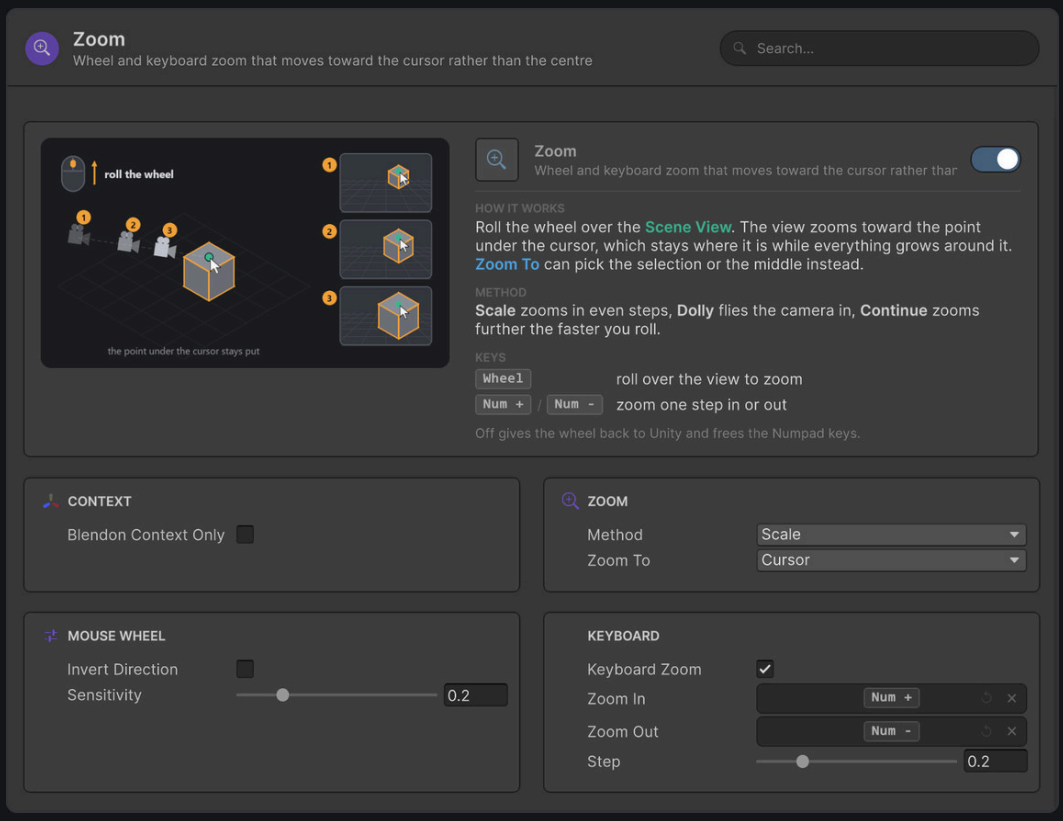


Figure 4.14 The Zoom page.

4.5 Orientation Gizmo

A Blender-style axis-ball overlay in the corner of the Scene view, in place of Unity's own scene gizmo.

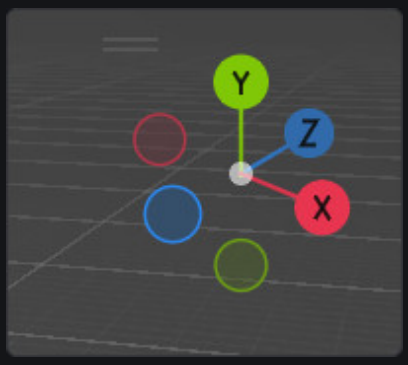


Figure 4.15 The Orientation Gizmo at rest.

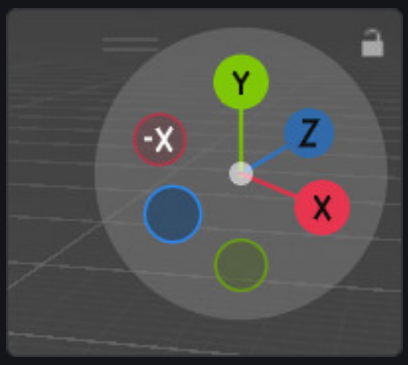


Figure 4.16 With the cursor over it: a backdrop, and the label of the negative handle under the cursor.

- Click an axis handle to snap the view to it, optionally animated instead of jumping. Clicking the handle the view already faces turns it around to the opposite side, so two clicks on Front give Back.
- **Drag to orbit.** Dragging inside the gizmo orbits the view with the Orbit Selected sensitivity. **Right-click** or **Esc** in the middle of the drag cancels it and restores the view.
- **MMB toggles projection.** Middle-click the gizmo to toggle perspective and orthographic, or use the **Center circle**, a dedicated button that does the same. It is on by default.
- X, Y and Z axis letters, with optional always-on **Negative letters** and a **View Name** label. Clicking the label also toggles perspective and orthographic.
- **Lock.** A padlock in the gizmo's corner shows while the cursor is over the gizmo. Click it to lock the Scene view's rotation, and click it again to unlock. While the view is locked, the padlock stays visible and the gizmo ignores clicks and drags.
- **Right-click menu. Free** turns the view to a three-quarter angle along its current heading. Below it are the six named views, a **Perspective** toggle, **Show Background** and **Settings**, which opens this page.
- Adjustable size, opacity, padding and line thickness, plus configurable **Hover Backdrop** and **Depth Fade** colors. **Show Background**, off by default, draws a dark panel behind the gizmo; the row on this page and the right-click menu switch the same setting.
- **Blendon Context Only.** Off by default. On, Blendon's gizmo shows in the Blendon tool context only, and Unity's own takes its corner everywhere else.
- **Hide Unity's gizmo.** On by default, so the two never overlap.

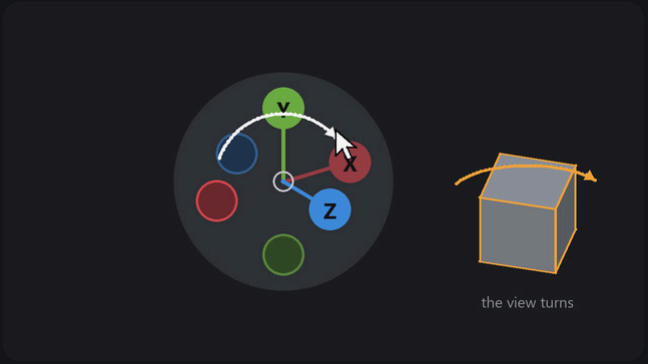


Figure 4.17 Drag inside the gizmo to orbit the view.

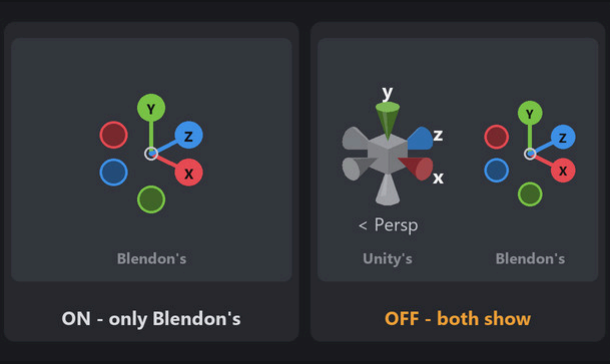


Figure 4.18 Hide Unity's gizmo: on by default, so only Blendon's is drawn.

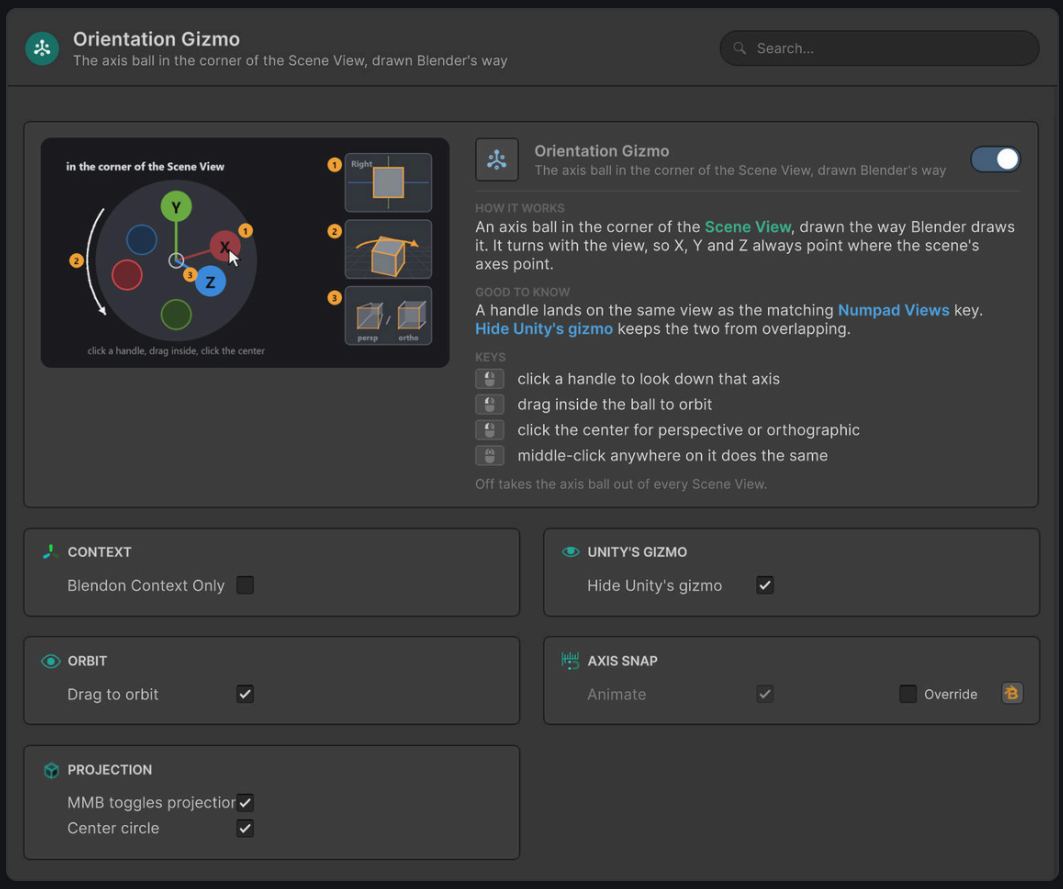


Figure 4.19 The Orientation Gizmo page.

4.6 Numpad Views

Blender's numpad viewport navigation, rebuilt on Unity's Shortcut Manager. NumLock must be on; the settings page warns inline when it is off.

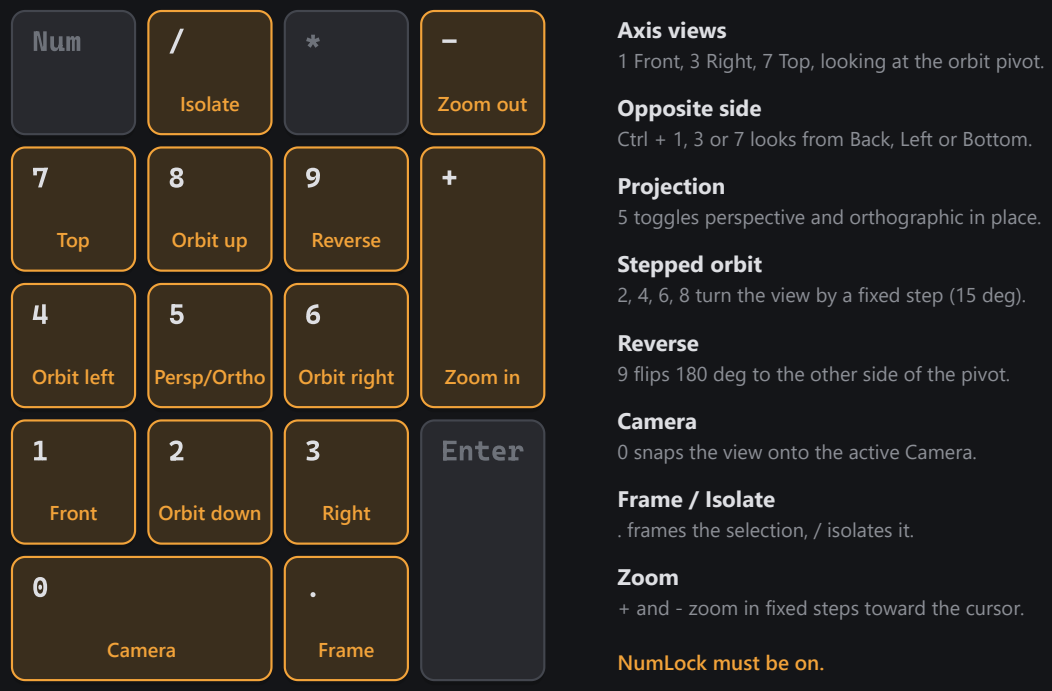


Figure 4.20 Every key of the numeric keypad, as Blendon binds it by default.

- **Blendon Context Only.** Off by default. On, the numpad keys answer in the Blendon tool context only. Zoom, Frame Selected and Isolate View follow their own pages.
- **Axis views. Numpad 1**, **3** and **7** look at the orbit pivot from the front, the right and the top. Holding **Ctrl** with the same key looks from the opposite side. Pressing the same key again goes back to the perspective view you came from. This table is shared with the Orientation Gizmo, so a numpad press and a handle click land on the same pose. In a URP or HDRP project the Unity keyboard preset moves **Back** to **Ctrl + Numpad 5** and **Left** to **Ctrl + Numpad 6**, away from the decal handles.
- **Toggle projection. Numpad 5** switches between perspective and orthographic in place, the same toggle as the gizmo's center circle.
- **Orbit steps. Numpad 2**, **4**, **6** and **8** turn the view a fixed number of degrees around the pivot, 15 by default. With Orbit Selected enabled, the view turns around the selection instead.
- **Reverse view. Numpad 9** flips 180 degrees to the opposite side of the pivot.
- **Align to camera. Numpad 0** snaps the Scene view onto the active Camera component: the selected camera first, then `Camera.main`, then any enabled camera. **Match Field of View** optionally copies the camera's field of view as well, and gives your own back once the view turns away from the camera.
- The page also shows the keyboard zoom keys and the Frame Selected and Isolate View keys. They keep their own bindings: changing one here changes it on its own page, with no duplicate shortcut.
- Every jump can be animated instead of cut to.

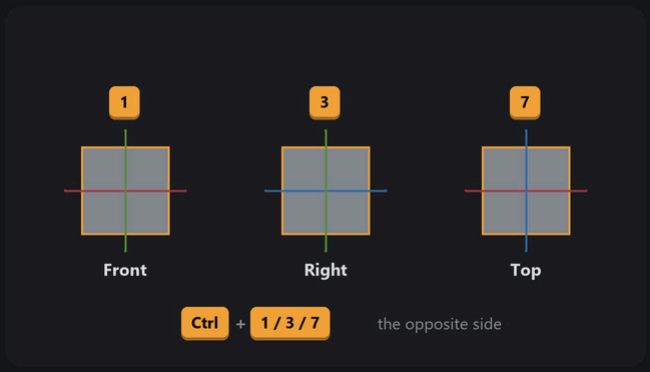


Figure 4.21 Numpad 1, 3 and 7 look from the front, right and top; with Ctrl, from the opposite side.

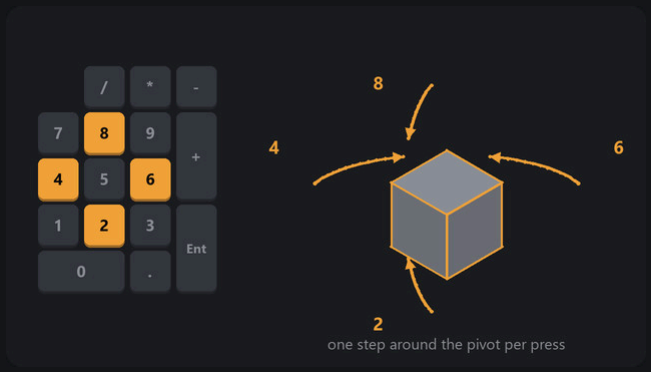


Figure 4.22 Numpad 2, 4, 6 and 8 turn the view one step around the pivot per press.

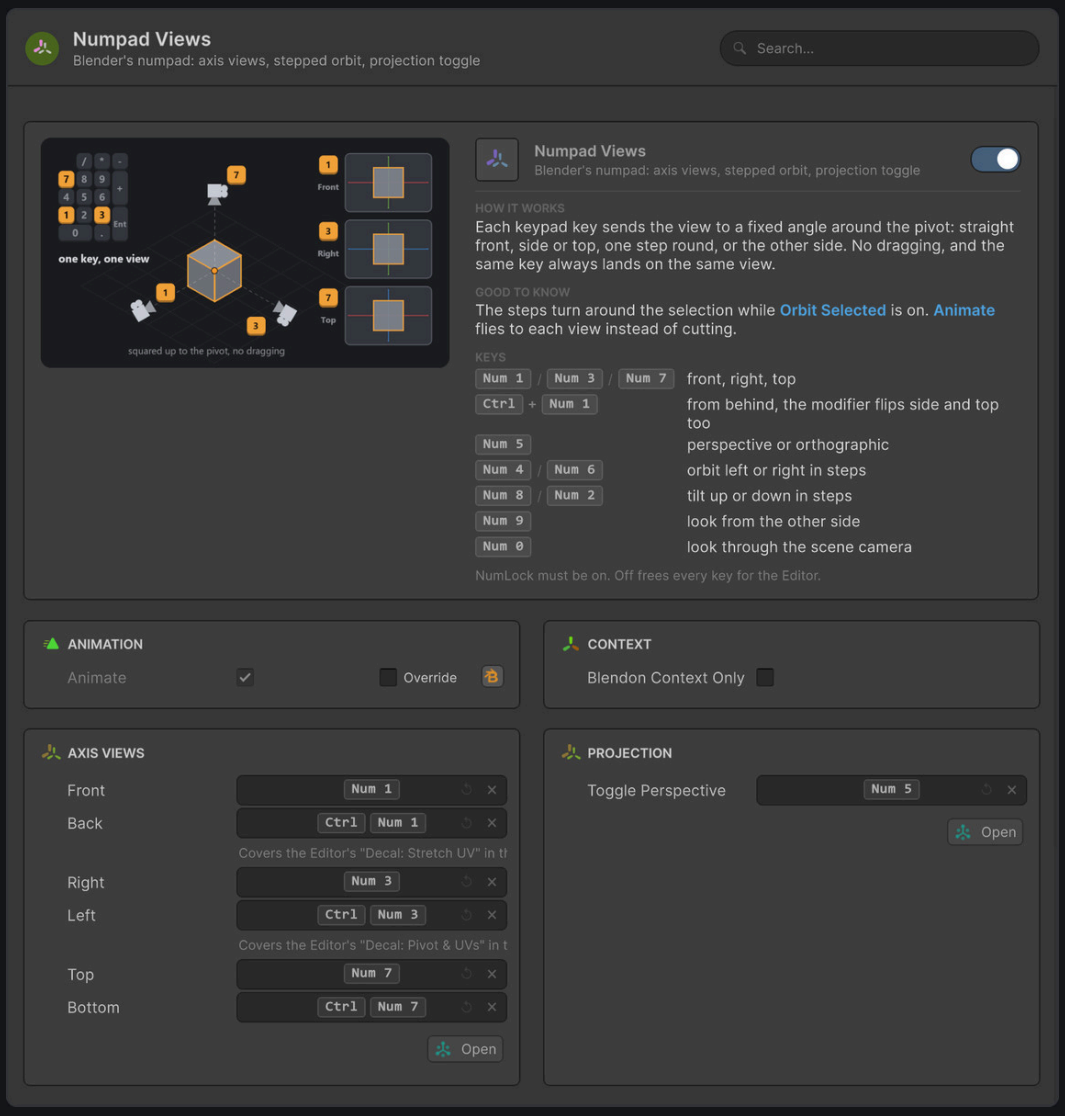


Figure 4.23 The Numpad Views page. Each binding can be changed here or in Unity's Shortcuts window.

5 Transform gizmos

Blendon replaces Unity's **Move (W)**, **Rotate (E)**, **Scale (R)** and **Transform (Y)** gizmos with Blender-style ones. Each can be enabled or disabled independently, and a disabled tool falls back to Unity's own gizmo. Move, Rotate and Scale can also be triggered straight from the keyboard without clicking a handle; see Grab.

In the Scene view **R** is Rotate Grab on the Blendon keyboard preset, which moves the Scale tool to **Shift + R**. The Unity preset leaves **R** to the Scale tool and moves Rotate Grab to **Shift + R** instead, and one click in the Overview's table swaps the pair either way; see Keyboard presets.

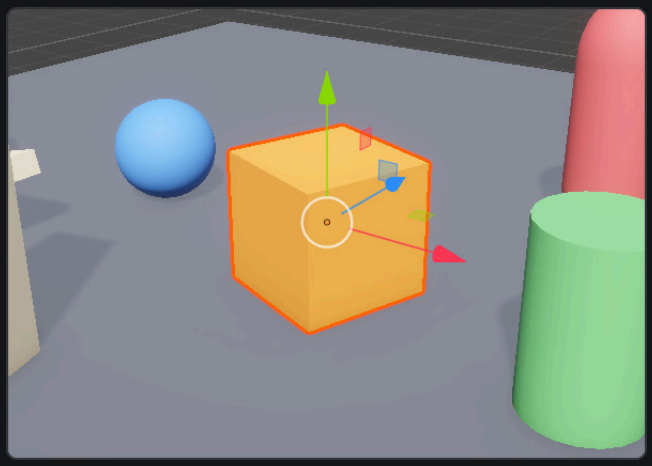


Figure 5.1 Move (W): axis arrows, plane handles, a white screen ring and a center dot.

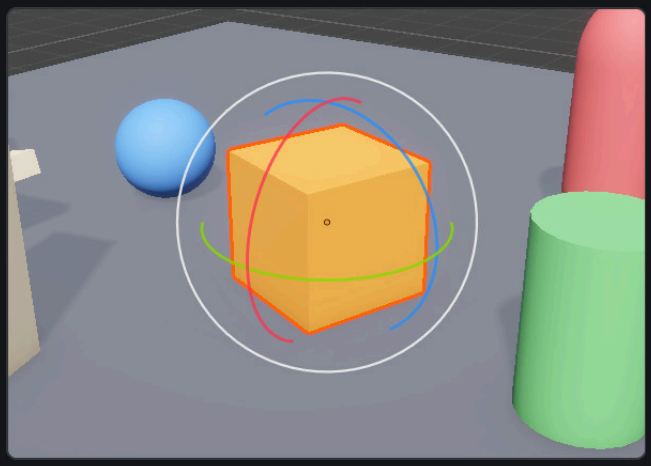


Figure 5.2 Rotate (E): axis rings, an outer view ring and a shaded trackball.

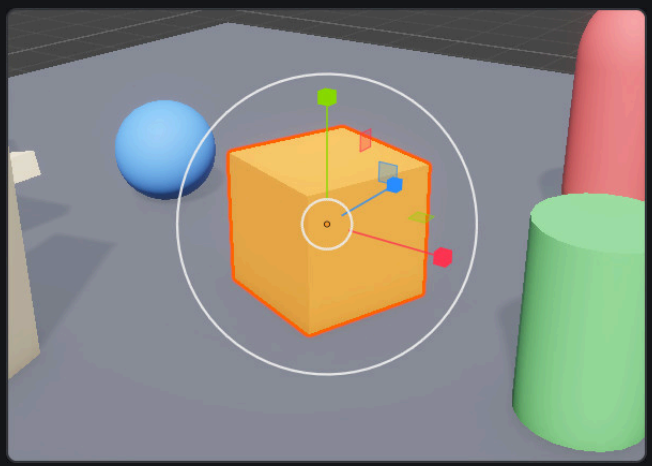


Figure 5.3 Scale (R): box-tipped axes, plane handles and a uniform-scale ring.

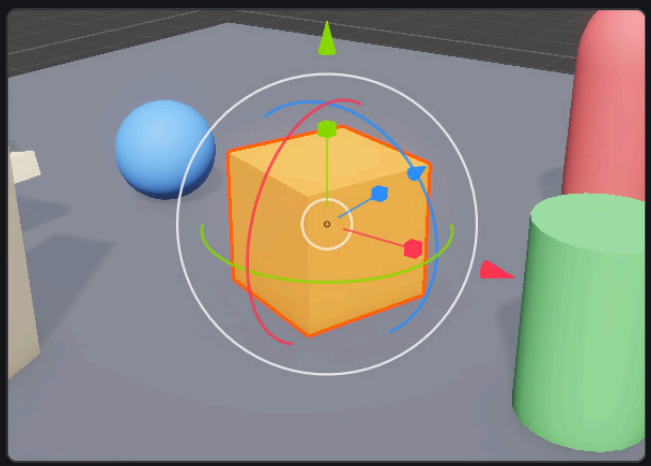


Figure 5.4 Transform (Y): all three nested into one gizmo.

All four gizmos:

- Honor Unity's **Local/Global** toggle, and Blendon's Pivot Point modes, two of which are Unity's own **Center** and **Pivot**.
- Support multi-object selection, and are fully undoable.

- Have their own vertex snapping (hold **V**) and work with Unity's grid snapping (hold **Ctrl**).
- Support **right-click to cancel** an in-progress drag, restoring the pre-drag transforms.
- Show a live readout of the exact change while dragging, and accept typed values; see Modal numeric input.

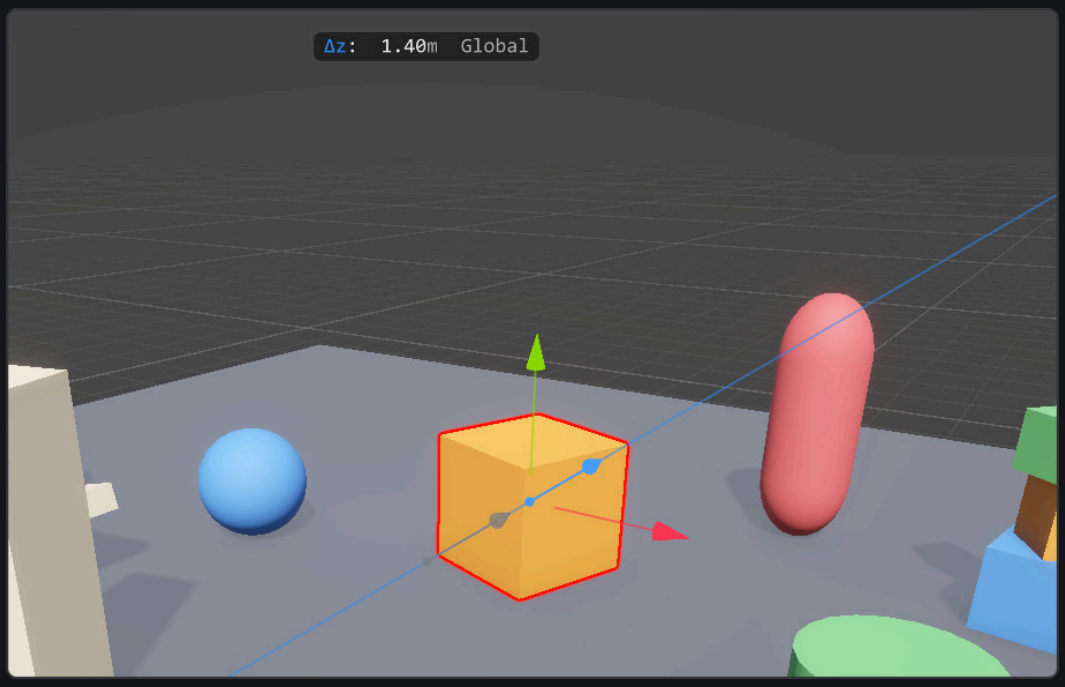


Figure 5.5 A Move drag along Z: the constraint line runs the length of the axis, a ghost dot marks where the drag started, and the readout at the top shows the exact distance.

5.1 The Blendon tool context

Unity's Scene view tools belong to a *tool context*. The built-in one is **Game Object**, and packages such as Splines and ProBuilder bring their own. Blendon adds one more, **Game Object (Blendon)**. In it, the Move, Rotate, Scale and Transform tools are Blendon's gizmos, and every other tool, such as Rect and View, is Unity's own.

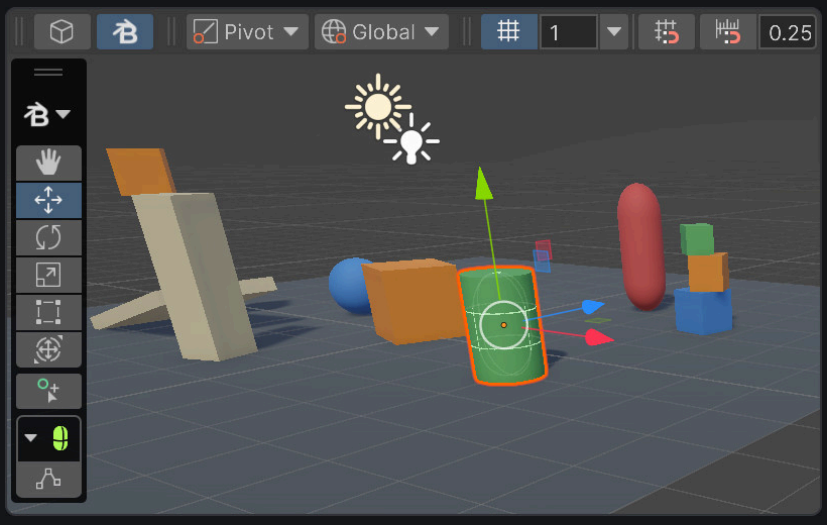


Figure 5.6 The Blendon tool context in the Tool Contexts overlay: the B switch, active, next to Game Object.

- **Where to find it.** Where the Scene view has a **Tool Contexts** overlay, as Unity 6.4 and newer do, the Blendon tool context is the switch with Blendon's **B** logo, next to Game Object. On Unity 6.0, which has no such overlay, it is an entry in the menu of the tool context button in the **Tools** overlay.
- **It switches itself on.** When the Editor starts with Blendon and at least one of the four gizmos switched on, Blendon moves the Scene view to its context. A script reload keeps it there.
- **Toggle Blendon key. B** switches between Blendon's context and the one you were in before, Game Object or a package's, such as ProBuilder's. When that context is no longer available, for example its object is not selected any more, or none is known yet, **B** goes to Game Object. The key works in every window. Change it on the Overview page, under **Tool Context**.
- **Unity's gizmos for a while.** Pick **Game Object** by hand, in the overlay or with **B**, and you get Unity's own gizmos until the Editor restarts, or until you pick **Game Object (Blendon)** again. Switching the gizmos or the master switch back on, after all of them were off, also brings the Blendon context back.
- **Esc keeps it.** In Unity, **Esc** in the Scene view drops you out of a tool context to Game Object. In Blendon's it does not: **Esc** still cancels a grab, a typed value or a menu, and a tool of its own, such as a Terrain brush, still steps back on it, but the context stays. This holds while Blendon draws a gizmo; with the master switch or all four gizmos off, **Esc** works the Unity way even in the Blendon context. Switch **Esc Keeps Context** off on the Overview page, under **Tool Context**, for Unity's way; leaving by **Esc** then counts as picking Game Object by hand.
- **Outside it, Unity's keys.** In Game Object and in a package's context, the keys that cover an Editor command in the Scene view step aside, and the Editor's command answers instead: the grabs on **G**, **R** and **S**, vertex snapping on **V** and **Shift + V**, and the Draw Mode and Tools pies while they sit on **Z** and **Q**. A drag draws the Editor's own selection box and a click picks the way the Editor does. Leaving the context ends a grab that is still running. Navigation, the other pie menus, the context menu, the hierarchy keys, Snap to Floor, Frame Selected, Isolate View and Navigation History work in every context, unless their page has **Blendon Context Only** on.
- **Blendon Context Only.** Each of those tools has this row on its page, off by default. On, the tool answers in the Blendon context only, and everywhere else it is plain Unity, as if the tool were switched off: Orbit Selected with Quick Roll, Camera Pan, Zoom with its wheel and keys, Numpad Views, Snap to Floor, Frame Selected, Isolate View, each half of Navigation History, the pie menus with Toggle Wireframe, and the hierarchy keys on the Box Select page. Two do it their own way: the Orientation Gizmo gives its corner to Unity's own gizmo, and the context menu opens the Editor's own menu, so the right button never changes hands. With all four gizmos off, Blendon does not move to its context by itself, so these tools only answer after **B**.
- **One gizmo off.** A switched-off gizmo leaves Unity's own tool in its slot, and the other three stay Blendon's.
- **All gizmos off.** Switching off all four gizmos, or the master switch, moves the Scene view to Game Object. Box Select and Click Selects Parent follow the context, so they rest as well.
- **Splines, ProBuilder and other packages.** Their contexts work exactly as they do without Blendon, with their own tools and the Editor's keys. Blendon's gizmos are not drawn there, on purpose.
- **Back from a package.** When Unity drops back to Game Object on its own, for example after you deselect a spline or press **Esc** in ProBuilder, Blendon takes its context back one frame later. It does the

same when you click Game Object yourself while a package's context is active. Unity then picks the tool by itself, Move or Rect, just as it does without Blendon.

5.2 Shared gizmo settings

One settings page, **All Tools** in the sidebar, drives everything the gizmos have in common.

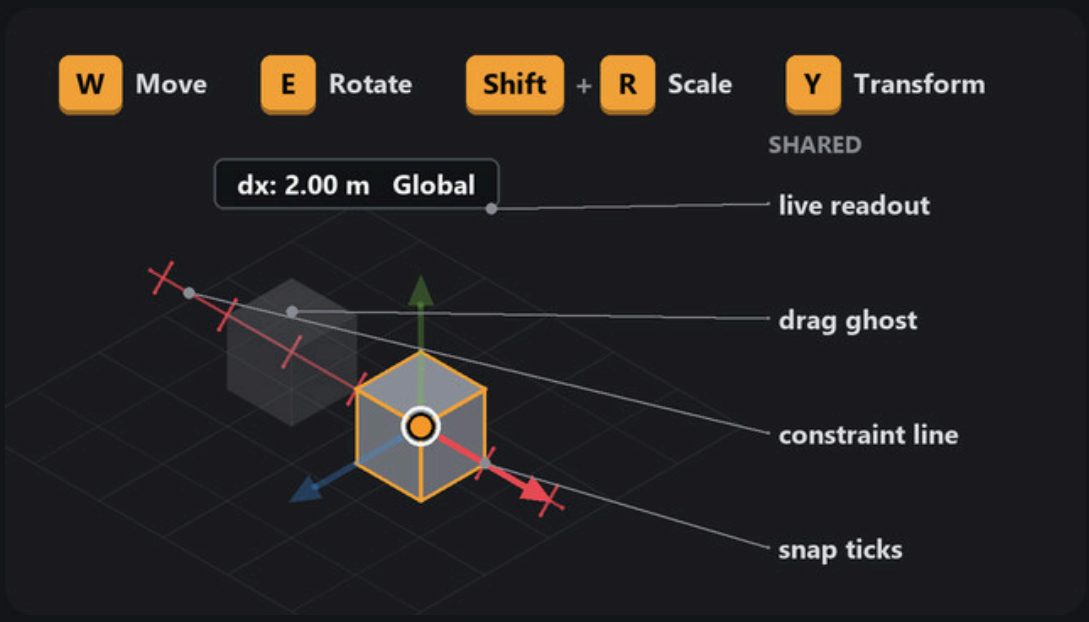


Figure 5.7 What every drag shares, whichever tool runs it: the live readout, the drag ghost, the constraint line and the snap ticks.

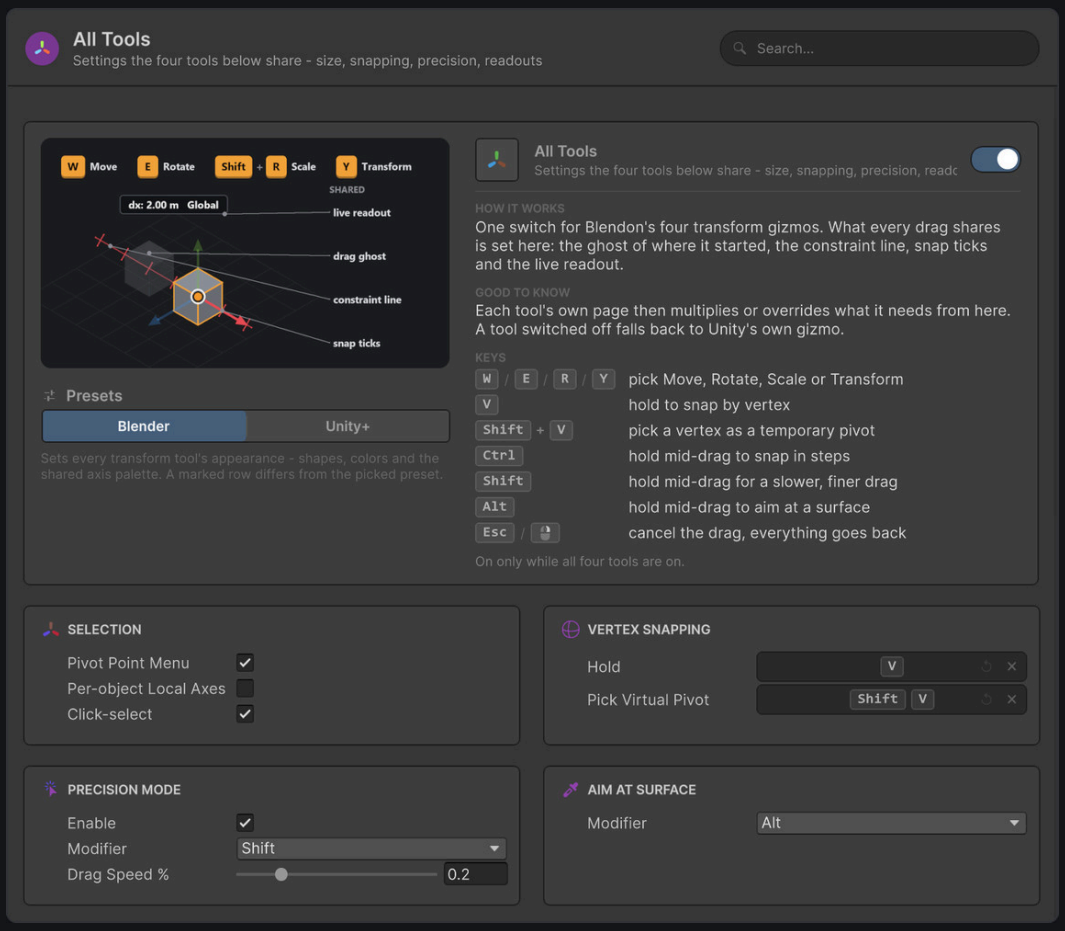


Figure 5.8 The All Tools page. Each tool's own page then multiplies or overrides what it needs from here.

- **Presets.** Picks the **Blender** or **Unity+** look (shapes, sizes, colors and the shared axis palette) for every transform gizmo at once, from a shipped, read-only preset file. This control appears on All Tools and on the Move, Rotate, Scale and Transform pages alike, since one click reaches all of them regardless of where it is clicked. The picked preset is the default of every setting it covers: a row is marked as changed only where it differs from that preset, and **Reset to Defaults** and **Revert to Default** go back to its values. While any value differs, the picked preset carries a dot and the note under it counts the changes; click the picked preset again to put them back. Picking the other preset applies it in full and drops the changes.
- **Resolution, scale, opacity and color contrast.** Global multipliers. Each gizmo then has its own layout settings on top of these.
- **Pivot Point Menu.** Gives the Pivot Point dropdown of Blendon's tools four of Blender's pivot modes. Off, it offers only Unity's Center and Pivot; see Pivot Point.
- **Per-object Local Axes.** Off by default, which is Unity's way: in Local space every selected object follows the active object's axes. On, it matches Blender: each object moves, turns and scales along its own axes, still about the shared pivot point.
- **Click-select.** A click on a handle, with no drag, selects what is under the cursor the way the Editor's own click does: a prefab root or a `[SelectionBase]` parent stands for its parts and a second click enters the part, **Shift** adds and **Ctrl** (**Cmd** on macOS) toggles.
- **Fade out angle.** Axis handles fade as they approach a grazing angle to the camera.

- **Axis Colors** and the outline color, shared plugin-wide with the Overview page, and the **Center Dot**'s size, outline thickness and color.
- **Drag ghosts.** A translucent marker of the drag-start state, with a configurable tint.
- **Constraint line.** A full-extent line along the dragged axis or plane, with adjustable thickness, opacity and contrast.
- **Snap Ticks.** While Ctrl-snapping, tick marks are drawn along the dragged handle at every snap increment. Length and gap are configurable.
- **Live Readout.** A Blender-style info box showing the exact delta while dragging, for example `Δz: 1.40m Global`. **Snap to Cursor** pins it next to the cursor instead of the top of the view.
- **Axis labels.** An optional X, Y or Z label near the dragged handle's tip.
- **Precision Mode.** Hold the precision modifier (**Shift** by default) to slow the drag down (20 percent speed by default) and halve the snap increment. With precision mode switched off, the modifier does neither.
- **Aim at Surface** (**Alt** by default). Drives Move's surface snap and Rotate's look-at, in a handle drag and in the **G** and **R** grabs, until the Move or Rotate page picks a key of its own.
- **Plane Lock Modifier** (**Shift** by default), under **Numeric Input**. Held with X, Y or Z, it locks a plane instead of an axis; see Modal numeric input.
- **Vertex Snapping.** The **Hold** and **Pick Virtual Pivot** keys; see Vertex snapping.
- **Cursor Wrap.** Wraps the cursor to the opposite edge during gizmo handle drags and the G, R and S grabs, so that they can continue indefinitely. It follows the Overview page's default until you tick **Override**; camera pan has its own on the Pan page. See Defaults.

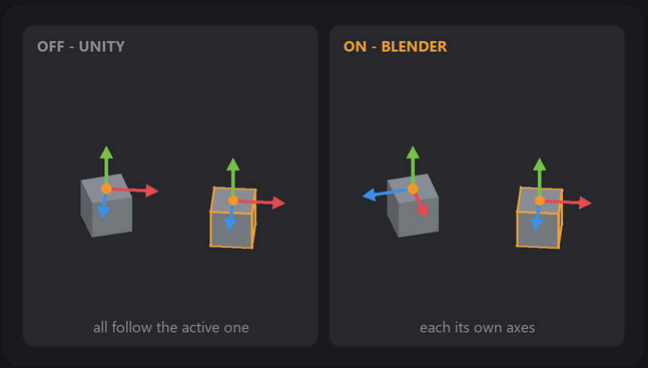


Figure 5.9 Per-object Local Axes: off, every object follows the active one's axes; on, each keeps its own.

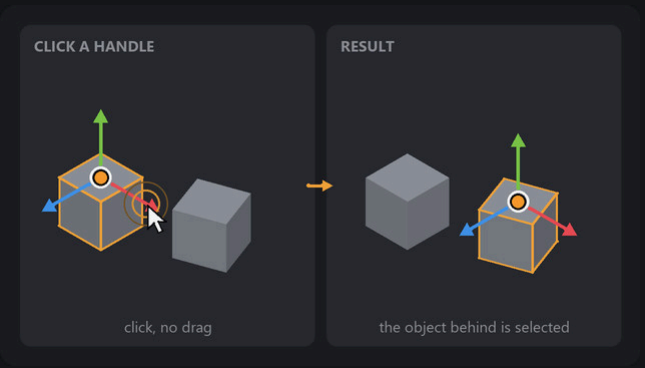


Figure 5.10 Click-select: a plain click on a handle selects the object behind it.



Figure 5.11 The drag ghost marks where the drag started.

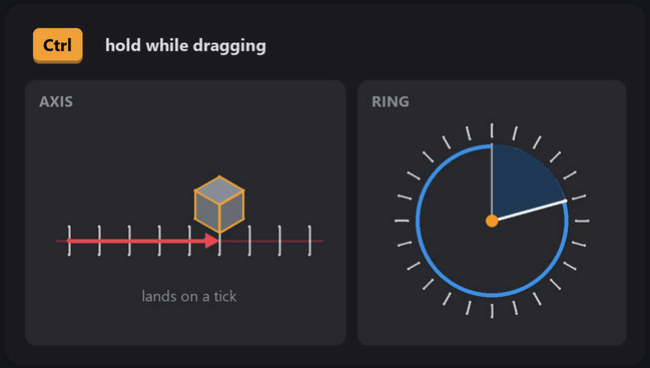


Figure 5.12 Snap Ticks: while Ctrl-snapping, ticks mark every increment along the handle.

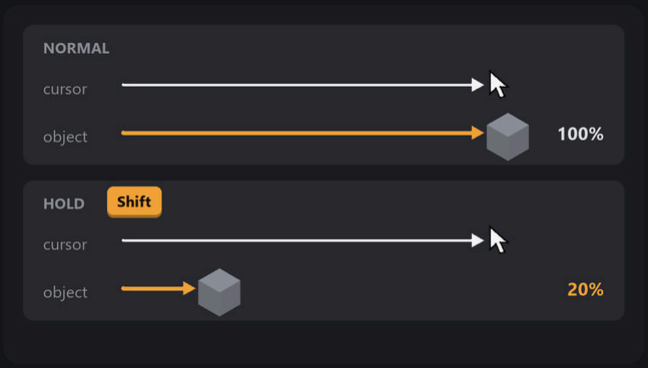


Figure 5.13 Precision Mode: hold Shift and the object moves a fifth as far as the cursor.

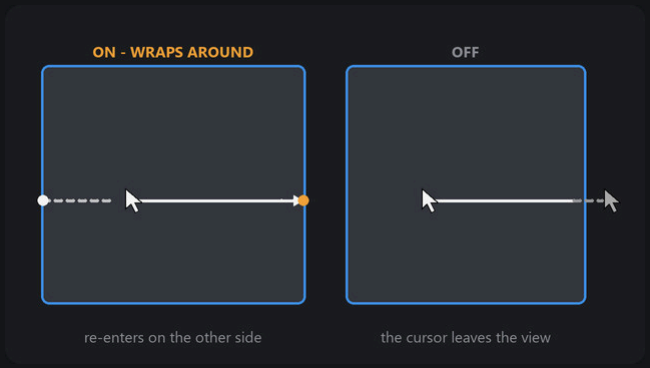


Figure 5.14 Cursor Wrap: a drag that reaches the edge carries on from the other side.

5.3 Pivot Point

Blender's pivot point decides what two or more selected objects turn and scale around. While one of Blendon's tools is active, the Tool Settings overlay shows Blendon's **Pivot Point** dropdown where Unity's own **Pivot/Center** dropdown would be, and Unity's Global/Local dropdown stays beside it. It looks and works like Unity's: click it and pick a mode.

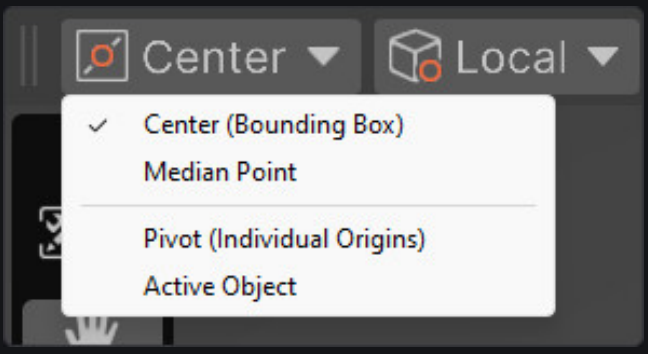


Figure 5.15 Blendon's Pivot Point dropdown in the Tool Settings overlay, with Pivot Point Menu on. The button shows the active mode, here Center (Bounding Box).

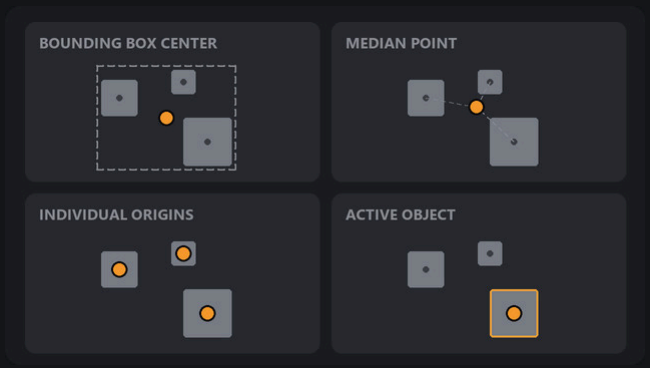


Figure 5.16 The four modes at a glance. The orange dot is where the selection turns and scales.

The four modes sit on the two sides of Unity's own Pivot/Center toggle, so the two never disagree. Picking a mode flips Unity's toggle to its side, and picking Pivot or Center in Unity's own dropdown, which the Rect tool and the Game Object context show, brings back the mode last used on that side.

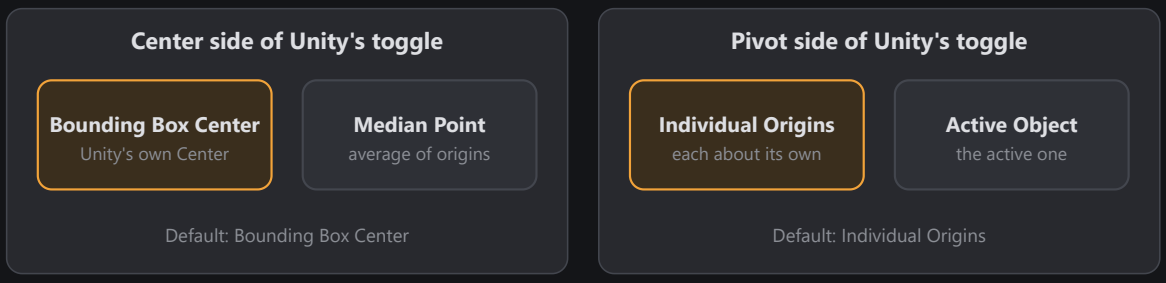


Figure 5.17 Which of Unity's two toggle states each Blender mode belongs to, and where each side starts.

Mode	Button label	Where the selection turns and scales
Center (Bounding Box)	Center	The center of the box around the whole selection. This is exactly Unity's own Center.
Median Point	Median	The average of the objects' origins, Blender's default.
Pivot (Individual Origins)	Pivot	Each object about its own origin, and in Local space along its own axes too.
Active Object	Active	The active object's origin. The rest of the selection turns and scales around it.

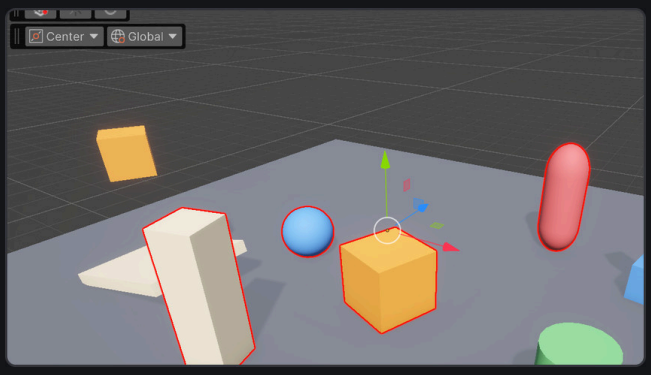


Figure 5.18 Bounding Box Center: the gizmo sits in the middle of the selection's bounds.

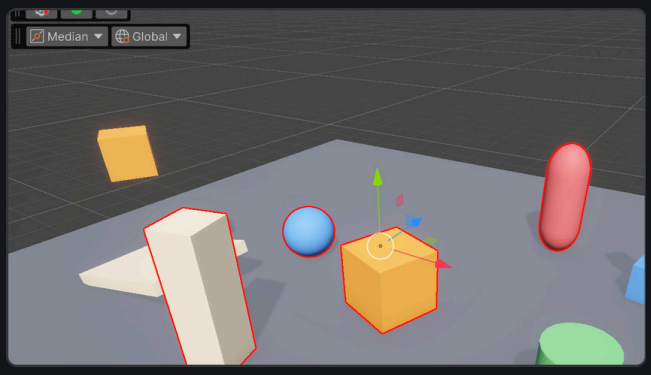


Figure 5.19 Median Point: the average of the four origins, pulled toward the cluster.

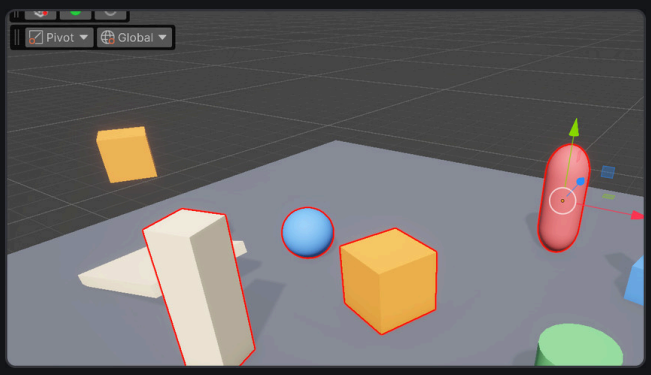


Figure 5.20 Individual Origins: every object turns about its own origin.

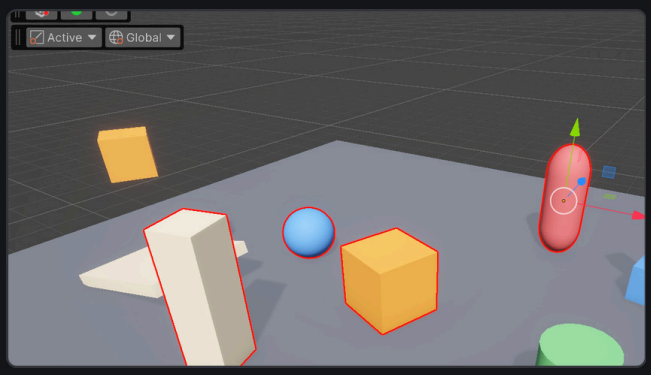


Figure 5.21 Active Object: the gizmo sits on the active object, the capsule.

- **Global/Local only changes the axes, never the pivot.** Individual Origins works in Global space too, and applies to screen-ring and trackball rotation as well.
- **Scaling about a shared pivot spreads positions.** With Bounding Box Center, Median Point or Active Object, a scale moves the objects away from the pivot as well as resizing them, and each object's scale is worked out along its own axes.
- **A single selection follows the toolbar alone**, as Unity's gizmos do.
- The defaults are **Bounding Box Center** on the Center side and **Individual Origins** on the Pivot side, which is Unity's own behavior until you pick something else.
- **Pivot Point Menu** on the All Tools page switches the four modes off. The dropdown then offers only Center and Pivot, which are Unity's own two. The Pivot Point pie on the **.** key lists all four modes; with Pivot Point Menu off, or outside the Blendon tool context, only Center and Pivot can be picked.
- **Only with Blendon's tools.** The Rect tool, a tool whose Blendon gizmo is switched off, and every other tool context keep Unity's own dropdown. On Unity 6.4 and newer, pivot modes that other packages add to that dropdown are therefore not offered while one of Blendon's tools is active.

5.4 Move Tool - W

Three colored axis arrows, two-axis plane handles, a white screen-space ring for free dragging, and a center dot.

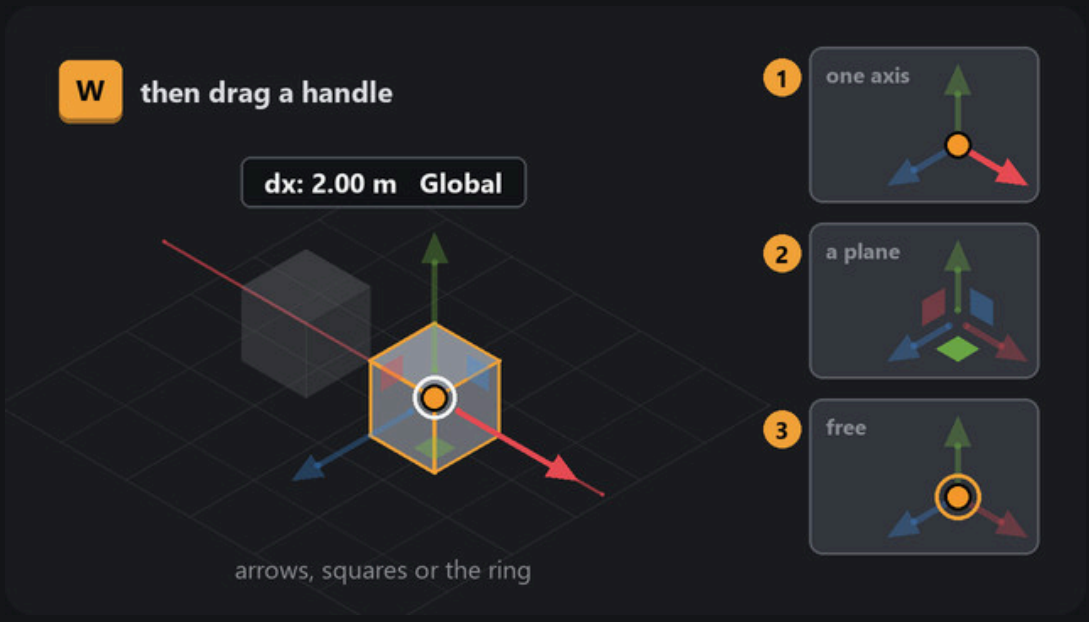


Figure 5.22 Press W and drag a handle: an arrow moves along one axis, a square across a plane, the ring freely on screen.

- Adjustable axis length, thickness and offset, and arrow-head size, with flat 2D or 3D arrow tips.
- Optional plane handles, with configurable shape, size, offset and outline.
- Optional screen-move ring with an adjustable radius.
- **Surface Snap.** Hold the aim modifier (**Alt**) while dragging to drop the selection onto the surface under the cursor, for placing objects on a floor, a wall or another object. A drag that is already running on an

axis or plane keeps its constraint and only snaps as far along it as it can reach. Let go of **Alt** and the selection comes back under the cursor, as in a **G** grab, and increment snapping lands on its grid steps again.

- **Grab.** The **G** key; see Grab.



Figure 5.23 Surface Snap: hold Alt while dragging and the object lands on the surface under the cursor.

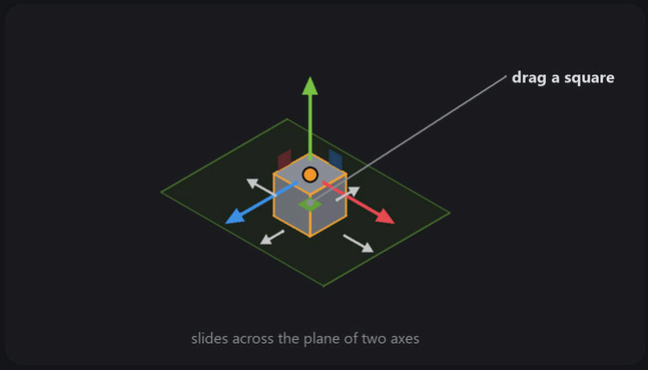


Figure 5.24 A plane handle slides the object across the plane of two axes.

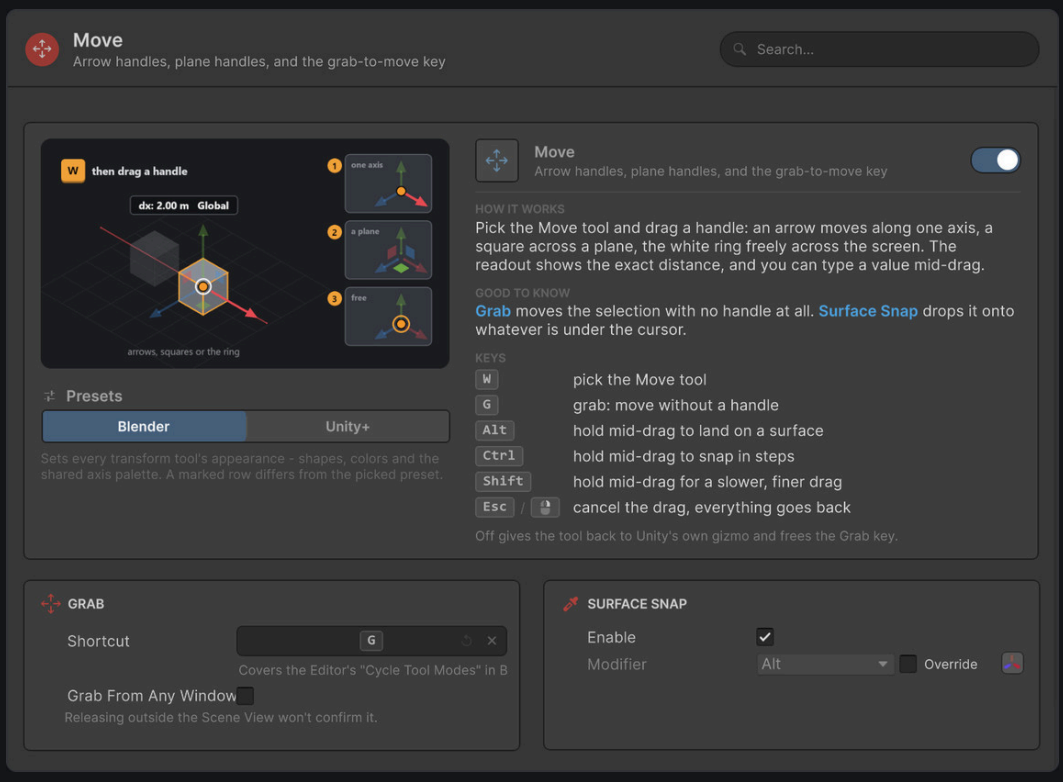


Figure 5.25 The Move page: the header card, the appearance presets, and the Grab and Surface Snap panels.

5.5 Rotate Tool - E

Three axis rings, an outer screen-space ring that rotates around the view axis, and an optional shaded trackball sphere for free rotation.

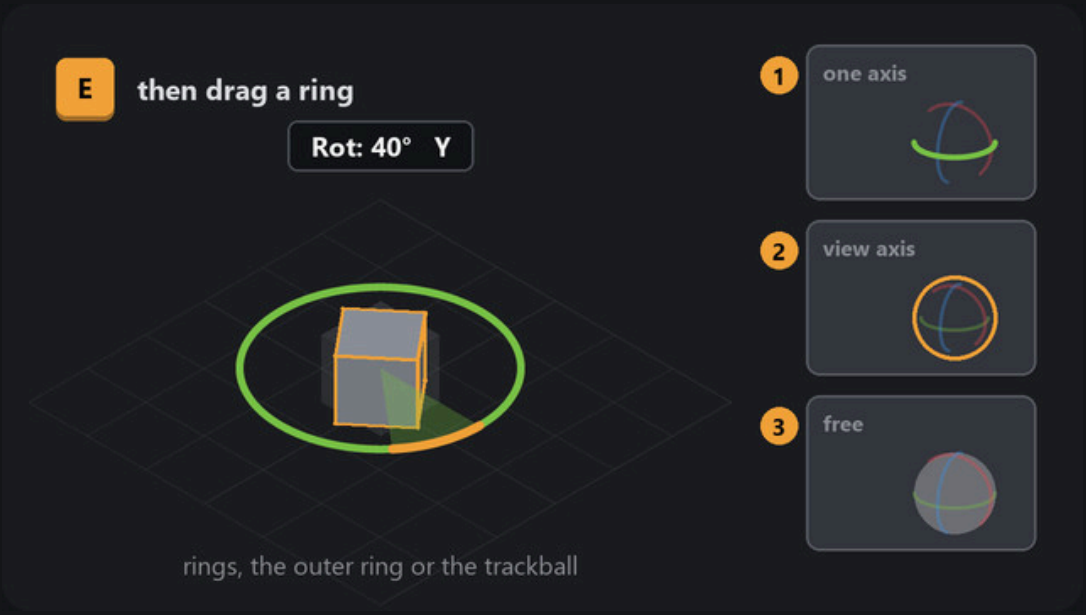


Figure 5.26 Press E and drag a ring: a colored ring turns around one axis, the outer ring around the view, the trackball freely.

- **Rotate To Cursor.** A Blender-style arc drag where the angle follows the cursor around the pivot. Turn it off for Unity's native behavior, which reads raw mouse movement along one screen axis.
- **Look At.** Hold the aim modifier (**Alt**) while dragging, or during the **R** grab, to aim the selection at the surface under the cursor, which is useful for orienting an object toward a camera or a player. The trackball turns the whole object; a single axis ring turns it only as far as that axis allows. The **R** grab aims the same way about its axis: the view axis, or the one **X**, **Y** or **Z** picked. It draws the aim line, and an axis key pressed while aiming aims about the new axis at once. Let go of the key and the turn follows the cursor again from the aimed angle. Typed digits win over the aim. Under Individual Origins, or in Local space with **Per-object Local Axes**, each object aims on its own and keeps that aim when the key is let go, then all turn on by the same angle.
- **Depth Gradient.** Shades the dragged ring from the axis color on the near side to the constraint color on the far side, to suggest depth.
- **Angle arc.** A swept pie showing the rotation delta, with adjustable opacity and a **Dynamic Opacity** mode so that overlapping 360-degree laps do not wash out to white.
- Adjustable ring length and thickness, and trackball radius and opacity.

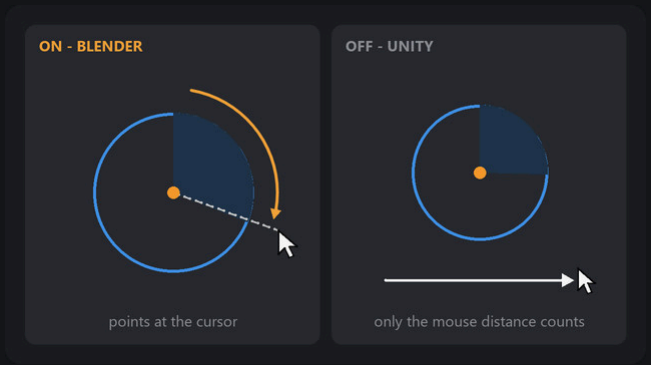


Figure 5.27 Rotate To Cursor: the angle follows the cursor around the pivot, as in Blender.



Figure 5.28 Look At: hold Alt and the selection turns to face the point under the cursor.



Figure 5.29 The trackball spins the object freely in any direction.



Figure 5.30 Dynamic Opacity keeps the angle arc even over several full turns.

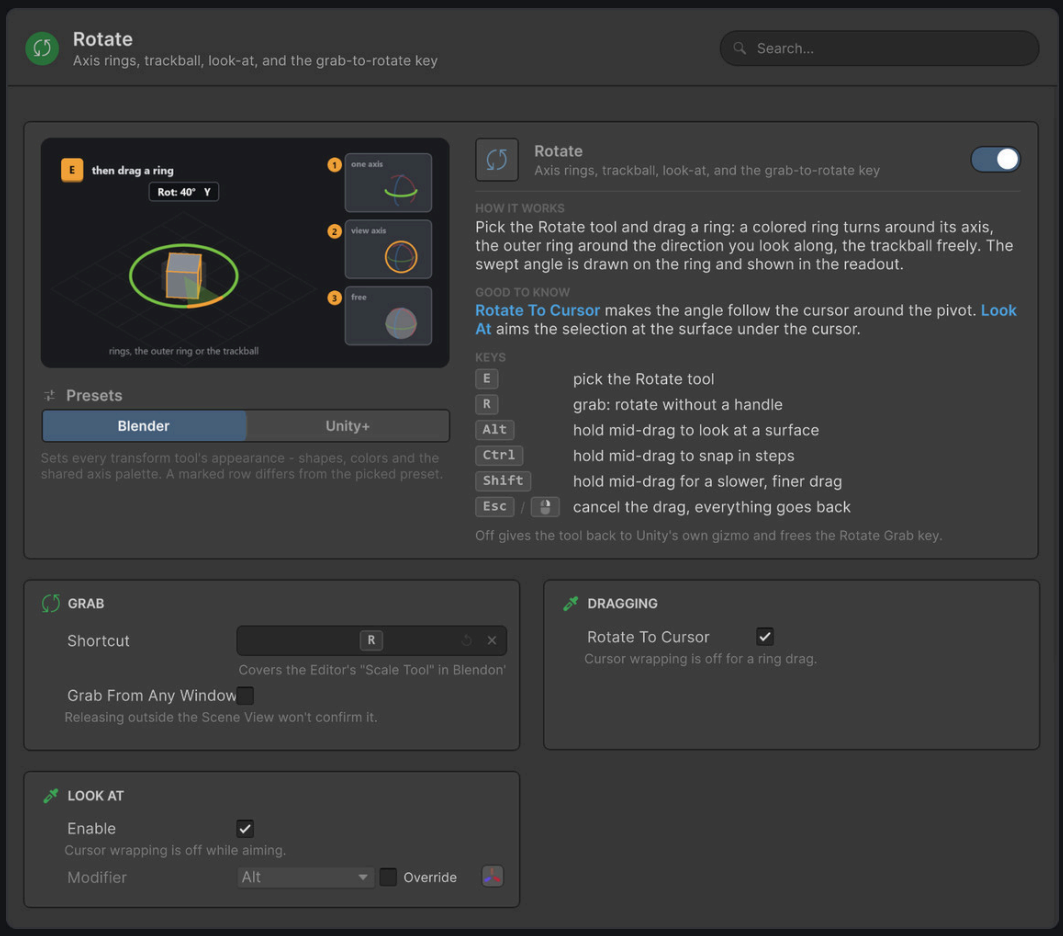


Figure 5.31 The Rotate page.

5.6 Scale Tool - R

Box-tipped axis handles, plane handles, an inner uniform-scale circle and an outer pick ring.

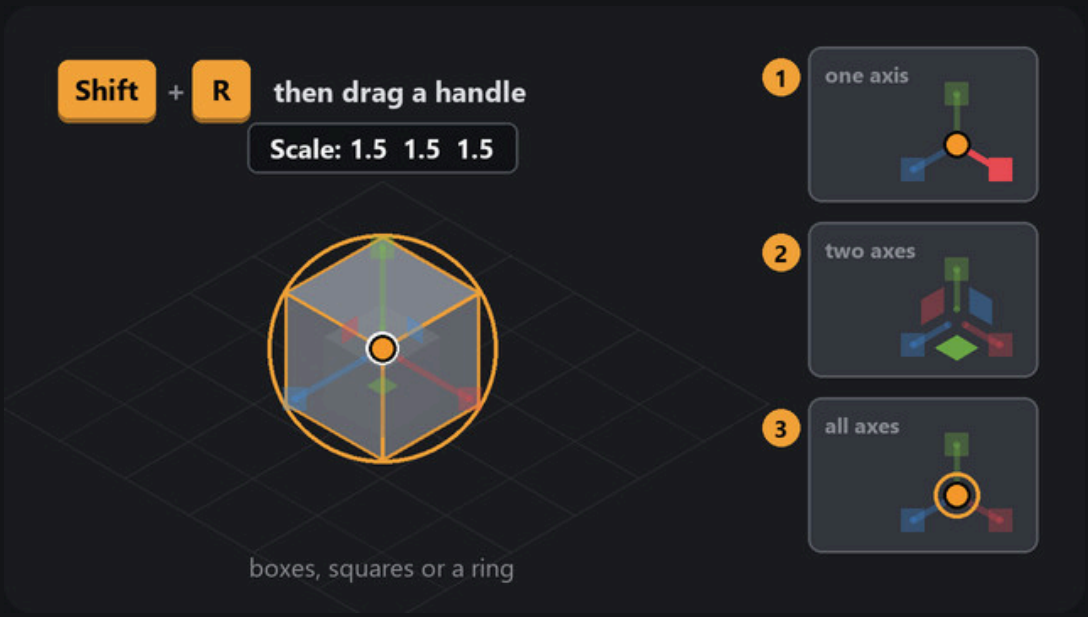


Figure 5.32 Pick the Scale tool and drag a handle: a box scales along one axis, a square along two, a ring along all three.

- **Force Local Orientation.** Unity-native behavior: always draw and scale along the object's local axes, ignoring the handle-rotation toggle. Off by default, which gives Blender-style Global and Local behavior that approximates scaling along the world axis you drag on a rotated object.
- Adjustable axis length, thickness and offset, and box size, with flat or 3D box tips.
- Optional plane handles and screen rings, with configurable inner and outer radii.

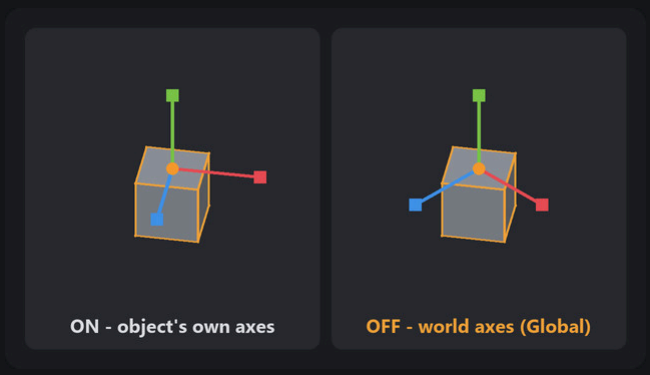


Figure 5.33 Force Local Orientation: on, always the object's own axes; off, the world axes in Global.



Figure 5.34 The outer ring: drag anywhere inside it, away from the handles, to scale evenly.

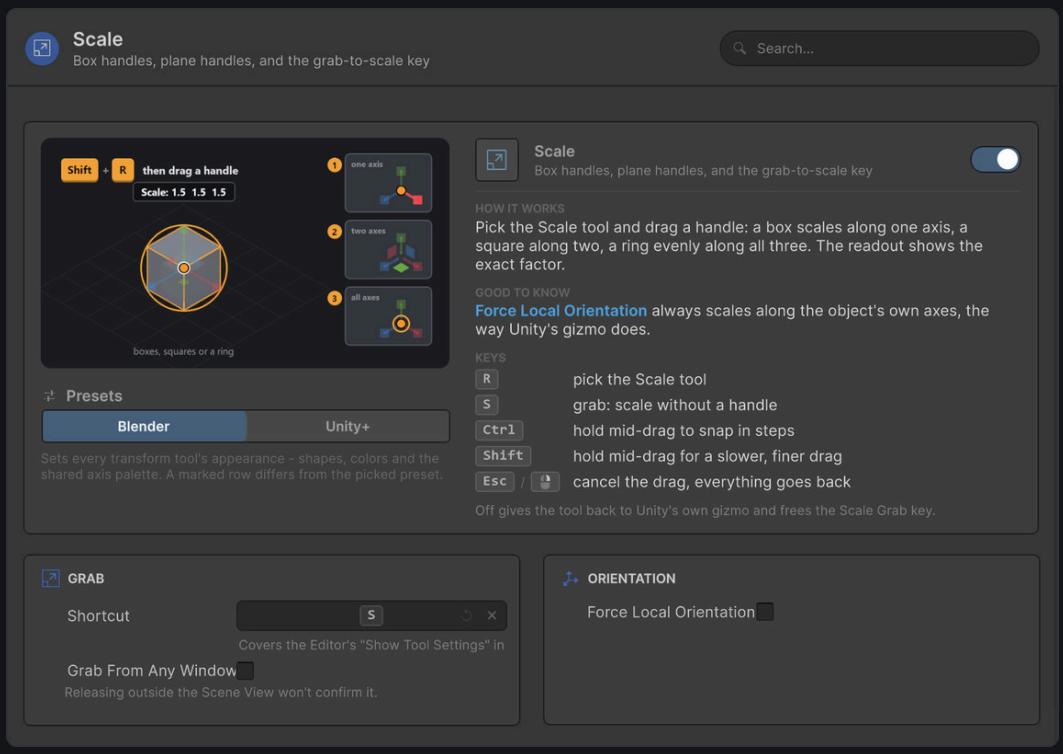


Figure 5.35 The Scale page.

5.7 Transform Tool - Y

All three gizmos composed into one and nested so that they do not fight each other: arrows outermost, rotate rings in the middle, scale boxes innermost.

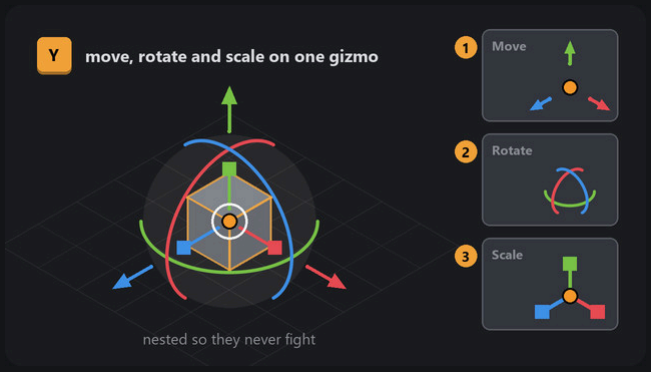


Figure 5.36 Press Y: move, rotate and scale on one gizmo.

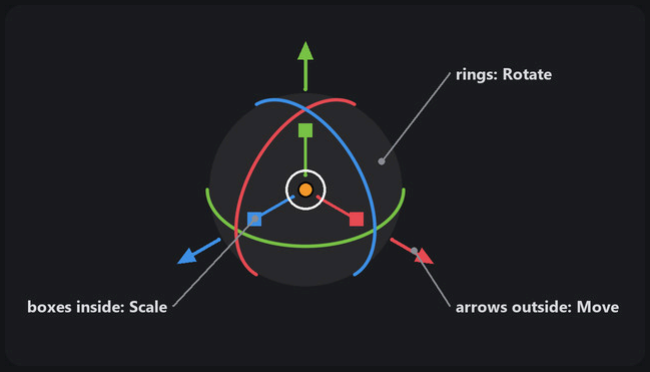


Figure 5.37 Arrows outside move, rings rotate, boxes inside scale.

- Each sub-gizmo has its own layout settings: length, offset, thickness, head size and screen rings, plus the rotate part's ring radius, trackball and ring color. Tuning a standalone tool never disturbs the nesting.
- Scale's outer pick ring is hard-disabled, since its click-anywhere fallback would swallow the other handles. Plane squares are off by default and can be switched on.
- The rotate trackball and the free-move screen ring are shown by default; either can be switched off.
- Grab keys, flat tips, surface snap, look-at, rotate-to-cursor, depth gradient, angle arc and force-local always follow the standalone tools' own values rather than storing duplicates. The page's **Reset to**

Defaults puts back only those it shows, such as the flat tips and the angle arc. Surface Snap, Look At, Grab From Any Window and the rest stay as the Move, Rotate and Scale pages have them.

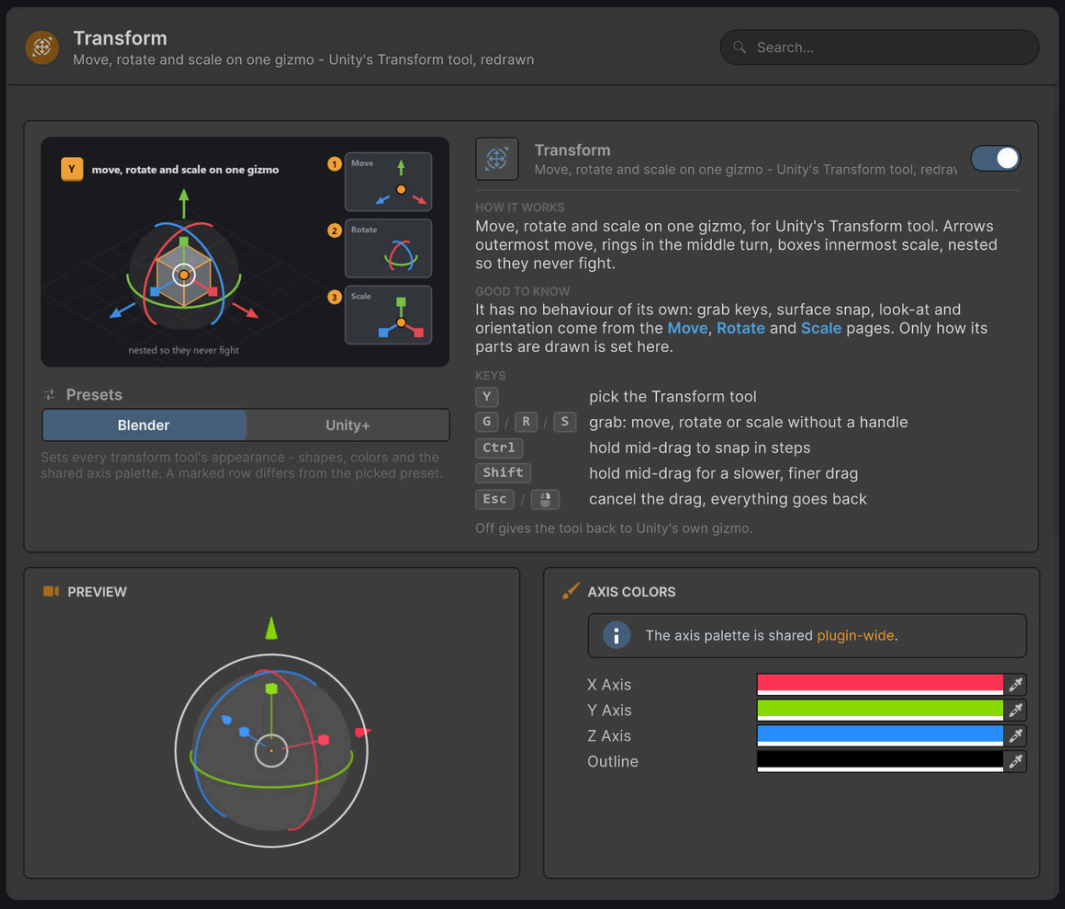


Figure 5.38 The Transform page. It has no behavior of its own, only how its parts are drawn.

5.8 Grab - G / R / S

Blender's modal grab. Press **G**, **R** or **S** anywhere in the Scene view, in the Blendon tool context, and the selection follows the cursor immediately, with no handle to click, and whichever tool is currently active in the toolbar stays active. Confirm with a click or **Enter**, and cancel with a right-click or **Esc**, exactly as you would a handle drag.

- **Grab (G)** moves the selection across a camera-facing plane through the pivot.
- **Rotate Grab (R)** spins it around the view axis.
- **Scale Grab (S)** scales it uniformly by the cursor's distance from the pivot. Under Individual Origins each object scales about its own origin, and the distance is measured from the active object, where the gizmo sits.



Figure 5.39 Three complete grabs, typed without touching the mouse.

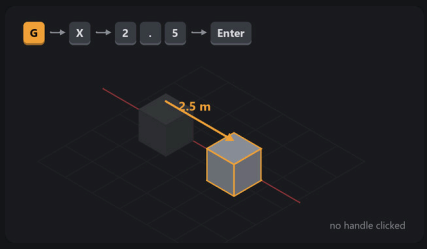


Figure 5.40 G, X, 2.5, Enter: moved 2.5 m along X, no handle clicked.

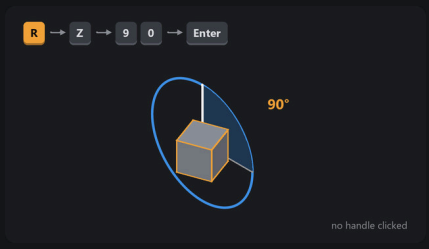


Figure 5.41 R, Z, 90, Enter: turned 90 degrees about Z.

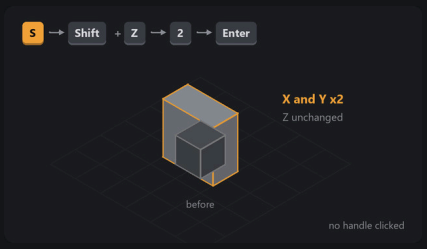


Figure 5.42 S, Shift + Z, 2, Enter: doubled in X and Y, Z unchanged.

A grab runs through the same modal machinery as a handle drag:

- **X**, **Y** and **Z** constrain it to an axis. **Shift** with an axis key locks a plane on Move and Scale, and turns about the excluded axis on Rotate. A third press of the same key drops the constraint again.
- Typed digits enter an exact value: a distance for **G**, degrees about the view axis for an unconstrained **R**, and a uniform factor for an unconstrained **S**.
- The precision modifier slows it down, and grid, rotation and scale snapping apply as usual.
- Grabs respect the shared Pivot Point. Hold the aim modifier and Move's grab honors **Surface Snap**, and Rotate's grab **Look At**: it turns the selection about the grab's axis toward the surface under the cursor; see Rotate Tool.
- Every gizmo handle except the center dot is hidden while a grab is running. When the active tool is Unity's own, because its Blendon gizmo is switched off, its handles stay visible meanwhile.
- In Game Object and in a package's context, **G**, **R** and **S** are the Editor's own keys, and leaving the Blendon context ends a grab that is still running.
- Pressing another grab key in the middle of a grab switches to that grab. The first one is undone, and the new one starts from where the selection stood before. Pressing the same key again does nothing.
- **G**, **R** or **S** pressed during a handle drag takes the drag over the same way, unless the drag is already that kind of operation, such as **G** on a Move arrow. The grab then confirms when you let go of the mouse button. Let go outside the Scene view and the grab stays open until you confirm it with a click or **Enter**.
- Each key is rebindable from its own tool's settings page (Move, Rotate or Scale), next to that tool's other options. There is no separate on/off switch: a grab is enabled together with its tool.

- **Grab From Any Window**, off by default, sits on each of those pages too. On, the key starts its grab from the Hierarchy, the Inspector or any other window: the Scene view under the cursor, or the last one used, comes to the front and the grab runs there, still in the Blendon tool context only.
- **R** is also Unity's own Scale-tool hotkey. The Blendon keyboard preset moves the Scale tool to **Shift + R**. While the two still share **R**, Rotate Grab answers it in the Scene view and the Scale tool everywhere else, unless Rotate's **Grab From Any Window** is on. The Unity preset moves Rotate Grab to **Shift + R** instead; see Keyboard presets.

5.9 Vertex snapping - V / Shift + V

Blendon has its own vertex-snap mode on its own two keys. It applies to every gizmo handle and to the G, R and S grabs alike. The gizmo swaps to a square snap handle, small markers show the selection's vertices it can be picked up by, and the mesh edges leaving the vertex it currently sits on are traced out so that the chosen point stands apart from the rest. During a drag, the object it is pulled onto marks its vertices too.

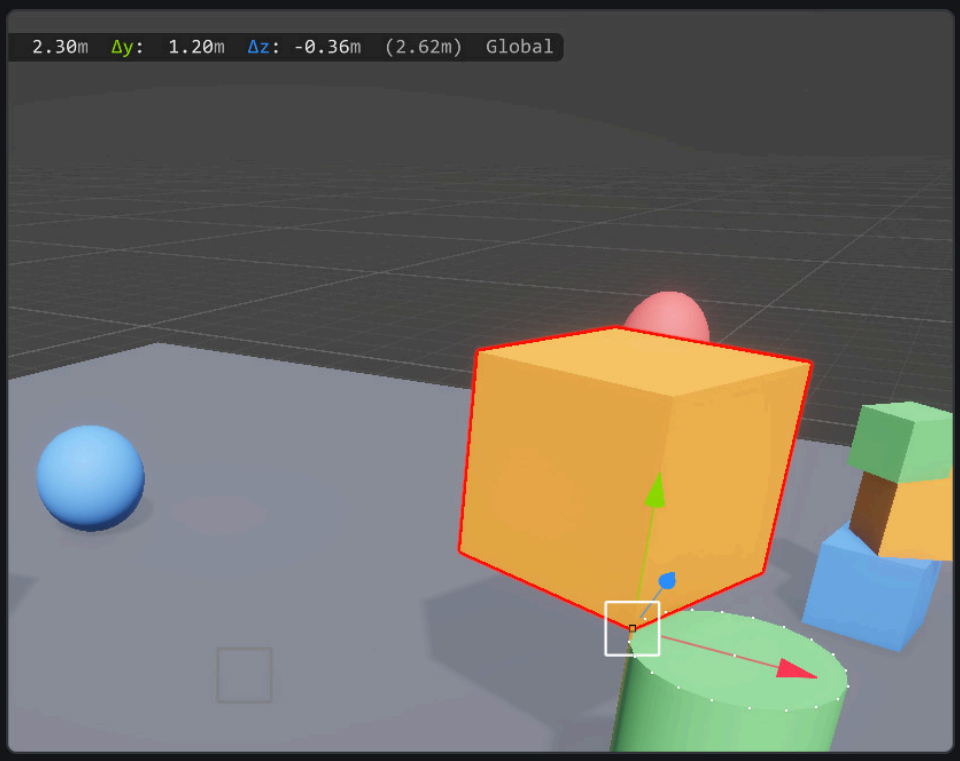


Figure 5.43 Hold V and drag: the crate's corner is carried onto a vertex on the barrel's rim. The small square at the bottom left marks where the drag started.

Hold V to snap for as long as the key is down. The selection is picked up by the vertex nearest the cursor and dropped onto the nearest vertex of another object.

Shift + V (Pick Virtual Pivot) opens a mode that stays on. It begins by ringing the vertex nearest the cursor; click to settle on it, and from then on drags run from that point instead of from the selection's own pivot, until the mode is switched off.

Both keys are Blendon's in the Blendon tool context. In Game Object and in a package's context they are the Editor's own vertex snapping again.

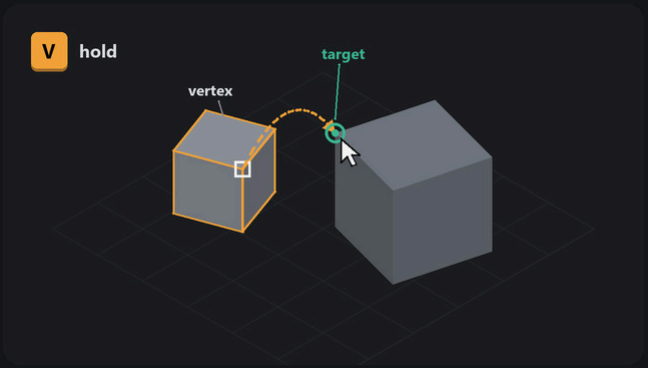


Figure 5.44 Hold V: the vertex under the cursor snaps onto a vertex of another object.

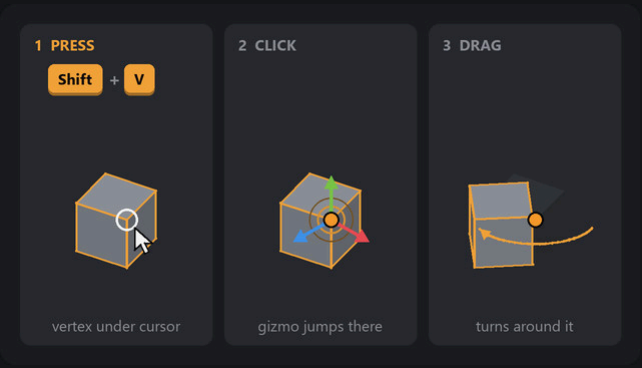


Figure 5.45 Shift + V: pick a vertex, and the gizmo moves there as a temporary pivot.

- Every entry starts the choice over, so the mode never silently reuses the vertex you picked last time.
- While a vertex is settled, Orbit Selected, the numpad orbit steps and the Orientation Gizmo turn the view about it.
- **Right-click** or **Esc** leaves the mode. A click past the gizmo drops just the settled vertex and reopens the choice, as does changing the selection.
- **Holding V** with nothing being manipulated reopens the choice for as long as the key is down. Release it over nothing and the old pick comes back.
- **Pressing V during a drag** toggles the pull toward other objects' vertices. Tapped, it toggles; held past a quarter of a second, it acts as a clutch and releasing it turns the pull back off.
- **On an axis or a plane**, from its handle or after **X**, **Y** or **Z**, the pull lands on the point of that axis or plane nearest the target vertex, along the gizmo's own axes, in Local as in Global.
- While that pull is on, grid snapping and the precision modifier stand down: a drag that ends on one exact point leaves nothing for them to refine. Without the pull, a drag from a settled vertex snaps and slows down as usual.
- **Alt** (surface snap) suspends the pull for as long as it is driving the drag. The surface under the cursor and the vertex under it answer the same question, and Alt says which one you meant.

Enter makes a settled vertex permanent. Unity cannot move an object's own pivot, so Blendon creates an empty on that vertex and parents the selection under it. The empty is selected and the mode closes: from then on the group turns and scales around that vertex with nothing left running.

- The empty takes the active object's rotation, so a Local-space rotation of the group still turns the way the object does. It is named after the object, as in `Crate Pivot`, or `Pivot Group` when several objects are selected.
- It takes the objects' place in the Hierarchy. When they come from different parents, it goes to the scene root instead. Only the topmost selected objects move under it, so a selected child stays under its own parent.
- Creating the empty and parenting the selection under it is a single undo step.
- Enter groups only while nothing is being manipulated. During a grab or a typed value it confirms the transform as usual.

- An object inside a prefab instance, other than its outermost root, cannot be reparented. Neither can the root of the prefab being edited. Blendon then leaves the selection as it is and explains why in the Console.

Skinned meshes are included. Their vertices are read from the posed mesh rather than from the bind pose, so a character snaps where it actually stands.

Engaging the mode never re-orients the gizmo. Only Scale's **Force Local Orientation** overrides Unity's handle-rotation toggle.

Both keys are rebindable from the All Tools page or from the Keyboard page.

6 Modal numeric input

While any gizmo drag or keyboard grab is active, you can type an exact value, exactly as you would in Blender.

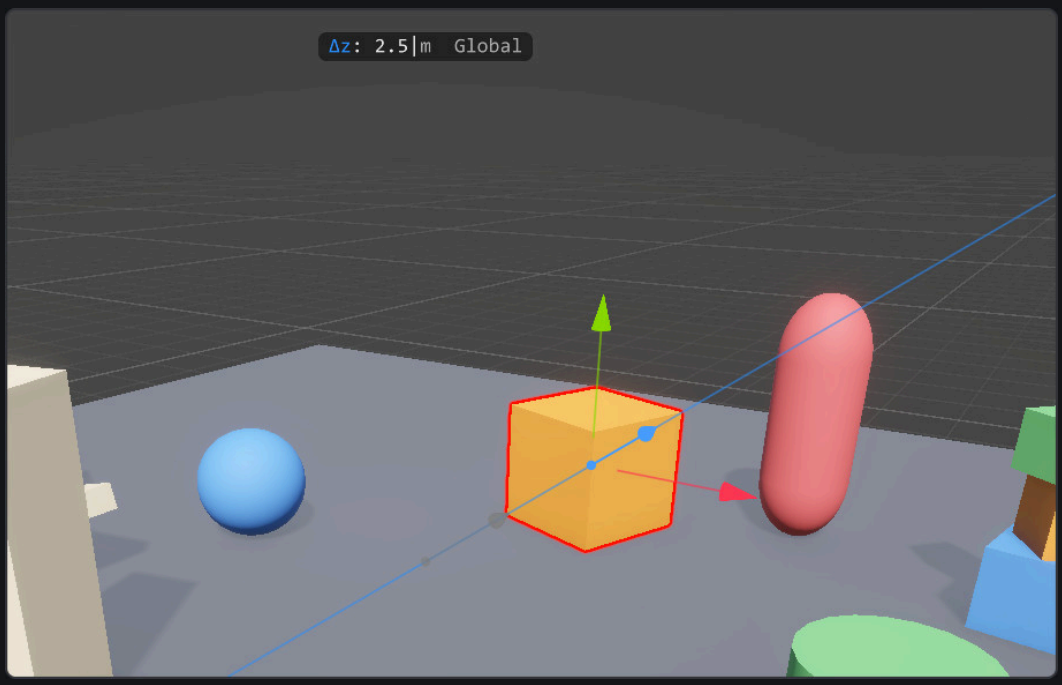


Figure 6.1 Typing 2.5 in the middle of a Move drag along Z. The mouse stops driving the drag, and the readout shows the value being typed, with its caret.

Key	Effect
`0` - `9`, `.`	Type the value. The first digit commits to numeric mode, and the mouse stops driving the transform. A comma types the decimal point too.
X / **Y** / **Z**	Constrain to that axis. Whatever you have already typed transfers to the new axis.
The same axis key again	Switch to the other orientation, as in Blender; see the figure below.

Key	Effect
Shift + **X** / **Y** / **Z**	Lock the opposite plane, excluding that axis instead of isolating it. It counts as the same letter for the orientation cycle, and on a rotation it turns about the excluded axis. The modifier is configurable.
`-`	Toggle the sign.
`/`	Toggle the reciprocal, `1/x`.
Backspace	Delete a character. The press that deletes the last one exits numeric mode and hands the drag back to the mouse at once.
Enter or **left-click**	Confirm.
Esc or **right-click**	Cancel and restore the pre-drag transform.

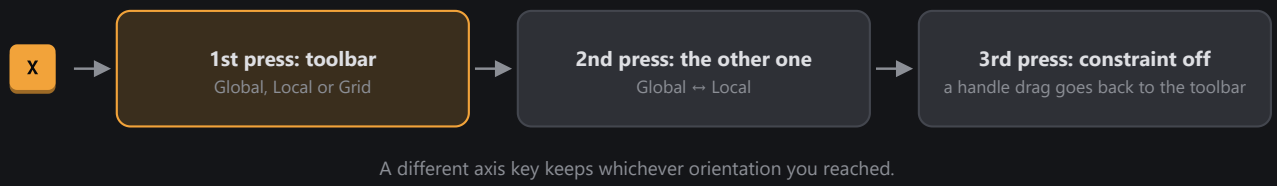


Figure 6.2 Pressing the same axis key repeatedly. The first press uses the toolbar's orientation, the second switches to the other one, and the third drops the constraint on a grab.

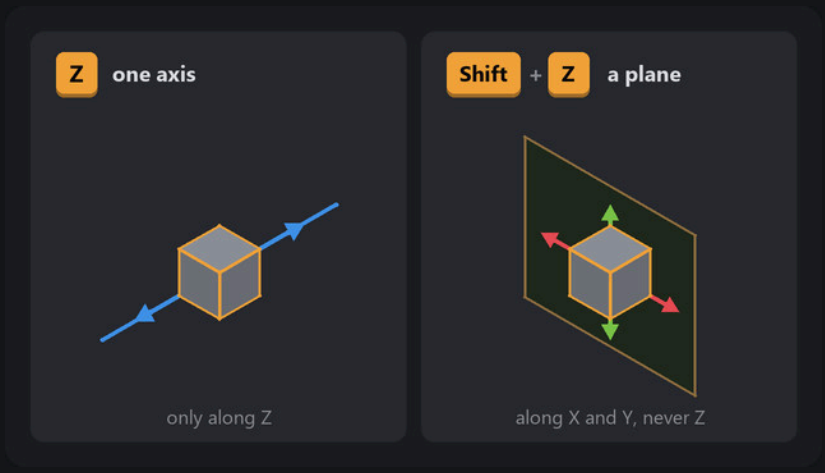


Figure 6.3 Z alone constrains to the Z axis. Shift + Z locks the plane of the other two, so the object slides along X and Y and never Z.

Pressing an axis key before typing anything does not enter numeric mode. The mouse keeps driving the drag, simply re-projected onto the newly selected axis or plane. The precision modifier and snapping keep working after an axis key, on a rotation ring as anywhere else. A typed value is applied exactly as typed: snapping never rounds it. On a plane handle it applies to both of the plane's axes: Move slides that distance along each, and Scale scales both by the typed factor, as **Shift** with an axis key does in a grab.

7 Scene utilities

7.1 Snap to Floor - End

Drops the selected objects onto whatever is below them. Blendon looks along the chosen direction for the first surface that is drawn in the scene, so a floor needs no collider. The object itself and its children are skipped. Each object falls straight down until its base meets something, even under one corner: Blendon probes the middle and the four corners of its footprint, so an object whose pivot sits outside its body, or one hanging over a table edge, lands on the table. What it draws, its children included, comes to rest on the surface; an object that draws nothing, such as an empty, lands with its pivot on it. The operation is fully undoable.

With nothing below, an object lands on the Terrain under it, or on the ground plane at Y = 0 when there is no terrain. An object buried in terrain comes back up onto it.

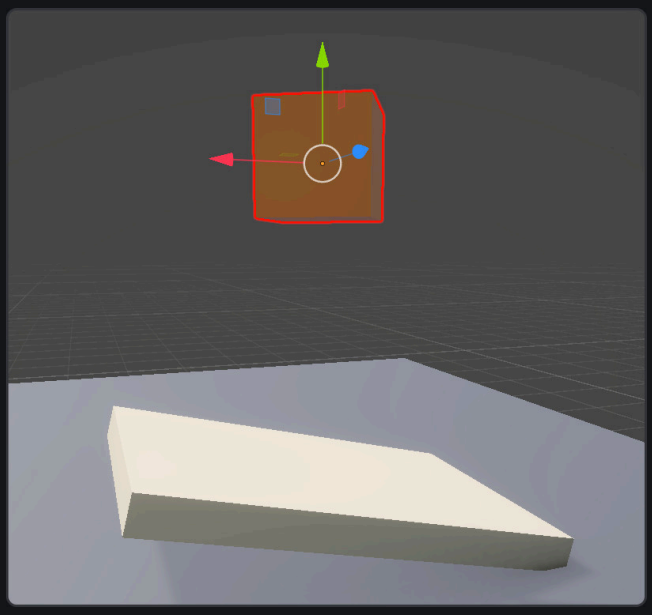


Figure 7.1 Before: a crate floating above a sloped ramp.

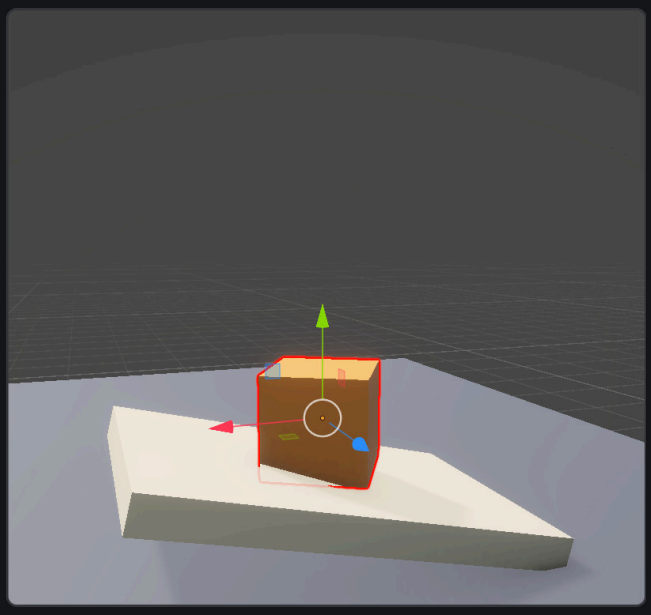


Figure 7.2 After pressing End: the crate rests on the ramp.

- **Blendon Context Only.** Off by default. On, the key answers in the Blendon tool context only.
- **Direction.** Down onto floors, or Up onto ceilings.
- **Surface Layers.** Only snap onto surfaces on the chosen layers.
- **Align to Surface.** Also rotate each object so that its up axis matches the surface slope.
- **Surface Offset.** Leave a gap between the object's base and the surface.
- An optional bounce animation, with an adjustable duration.



Figure 7.3 Align to Surface: the object lies flush with the slope instead of staying upright.

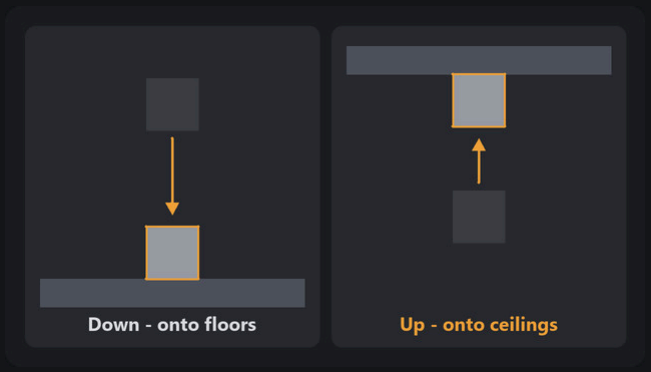


Figure 7.4 Direction: down onto floors, or up onto ceilings.

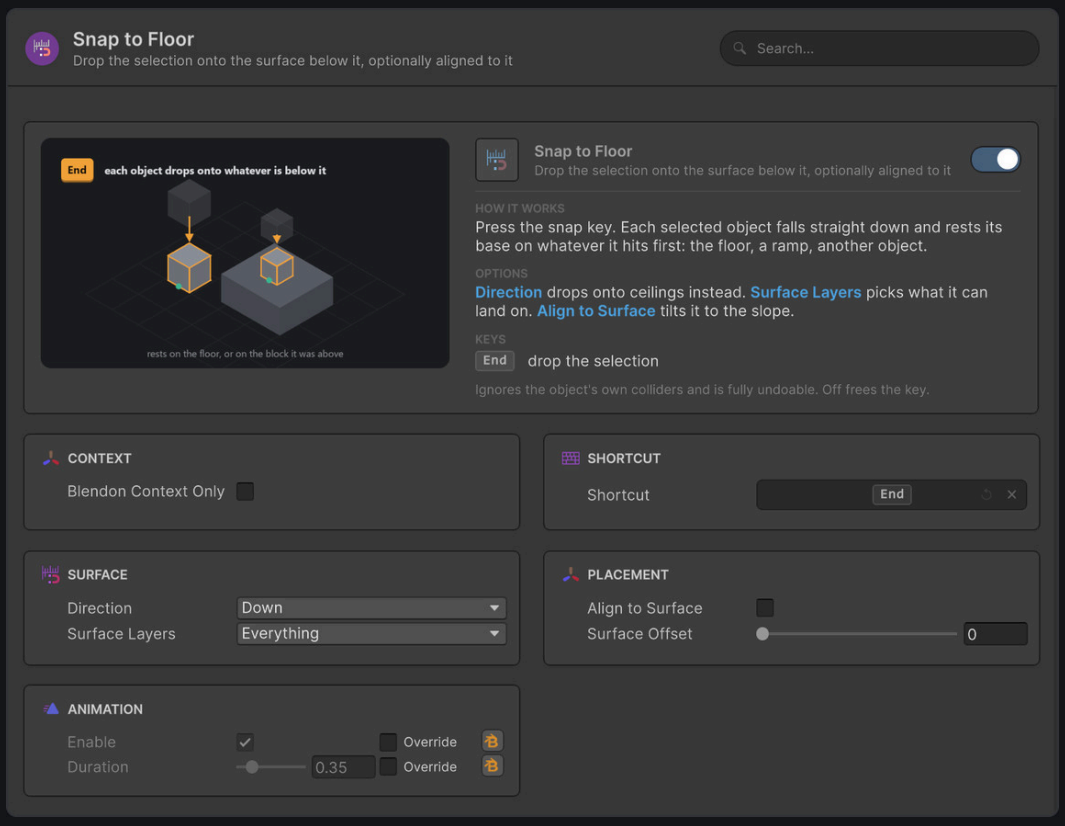


Figure 7.5 The Snap to Floor page.

7.2 Frame Selected - Numpad .

A cycling frame shortcut. Pressing it repeatedly steps through four steps and then starts over: **Selection Center** turns the camera to the middle of the selection, **Active Pivot** to the active object's pivot, **Zoom In** runs Unity's own Frame Selected for a close-up, and **Back to Start** returns to the view you started from. The Sequence table on the page sets which of these steps the key walks through and in what order. Moving the camera yourself between presses starts the cycle over. A step that would leave the view where it already is gets skipped, so every press visibly does something.

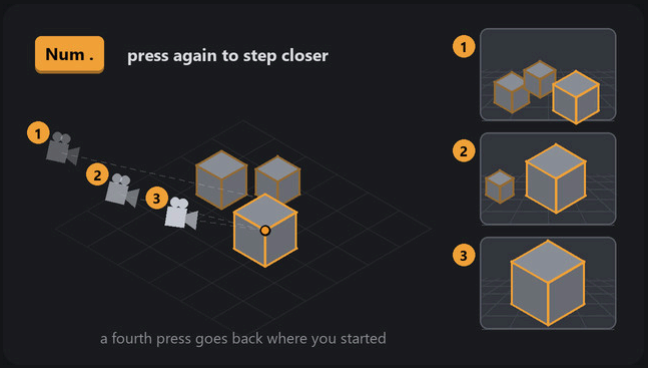


Figure 7.6 Each press steps the camera closer; a fourth press goes back where you started.

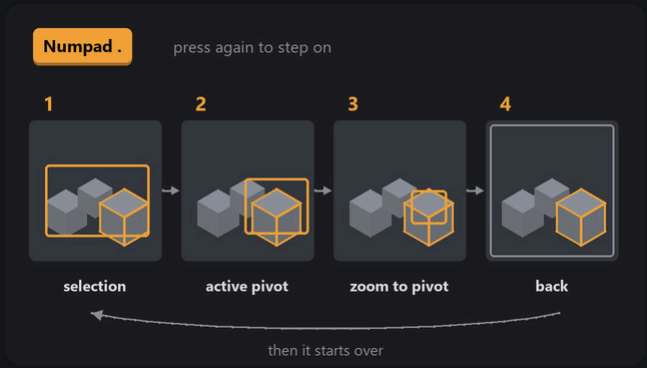


Figure 7.7 The four steps of the cycle, then it starts over.

The camera fly-to can be animated instead of snapping, with an adjustable duration. **Blendon Context Only**, off by default, keeps the key to the Blendon tool context.

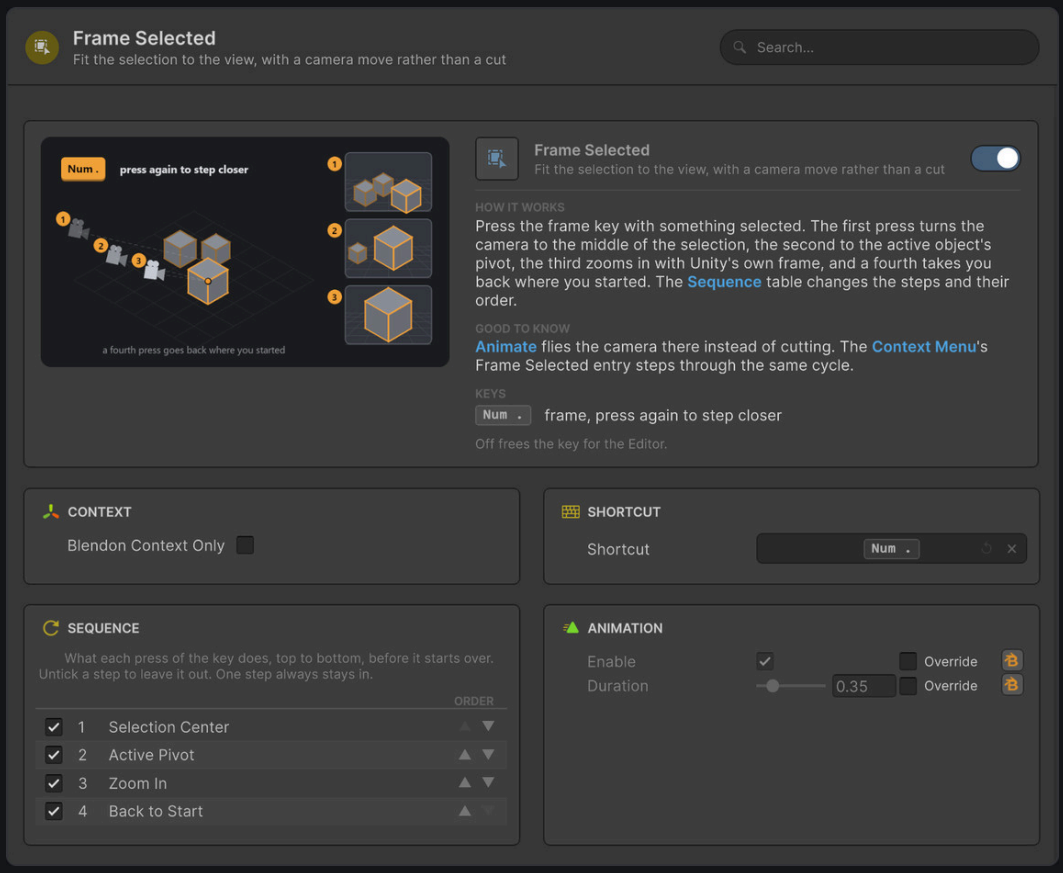


Figure 7.8 The Frame Selected page.

7.3 Isolate View - Numpad /

Blender-style local view. Blendon hides everything except the selection using Unity's Scene Visibility, and restores the scene when you press the key again.

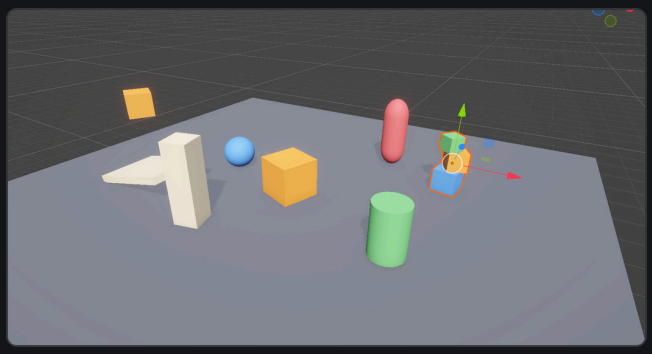


Figure 7.9 Before: the whole scene, with the stack of boxes selected.

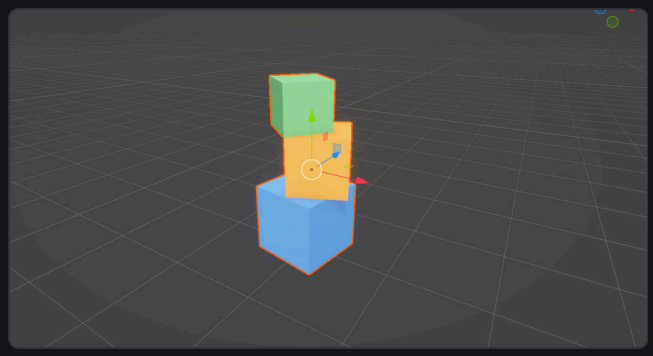


Figure 7.10 After Numpad /: only the stack remains, until the key is pressed again.

- **Blendon Context Only.** Off by default. On, the key answers in the Blendon tool context only. An isolation started there stays after you leave, and Unity's own isolation bar ends it.
- **Include Children.** Isolate the whole hierarchy under each selected object. On by default.
- **Frame On Isolate.** Frame the selection when isolating. Off by default.

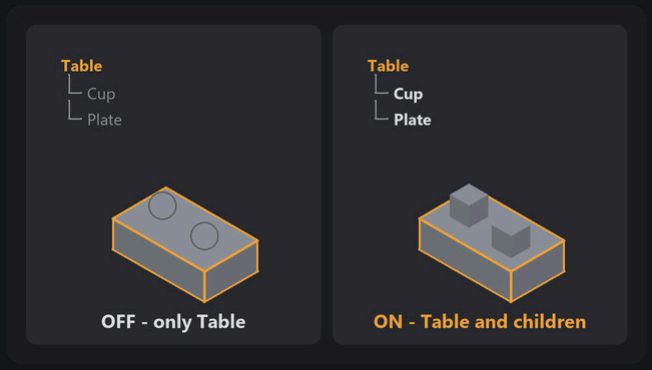


Figure 7.11 Include Children: on, the table keeps its cup and plate.

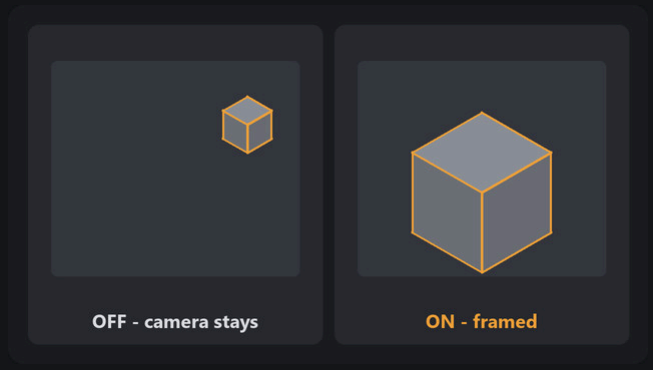


Figure 7.12 Frame On Isolate: on, the camera frames what is left.

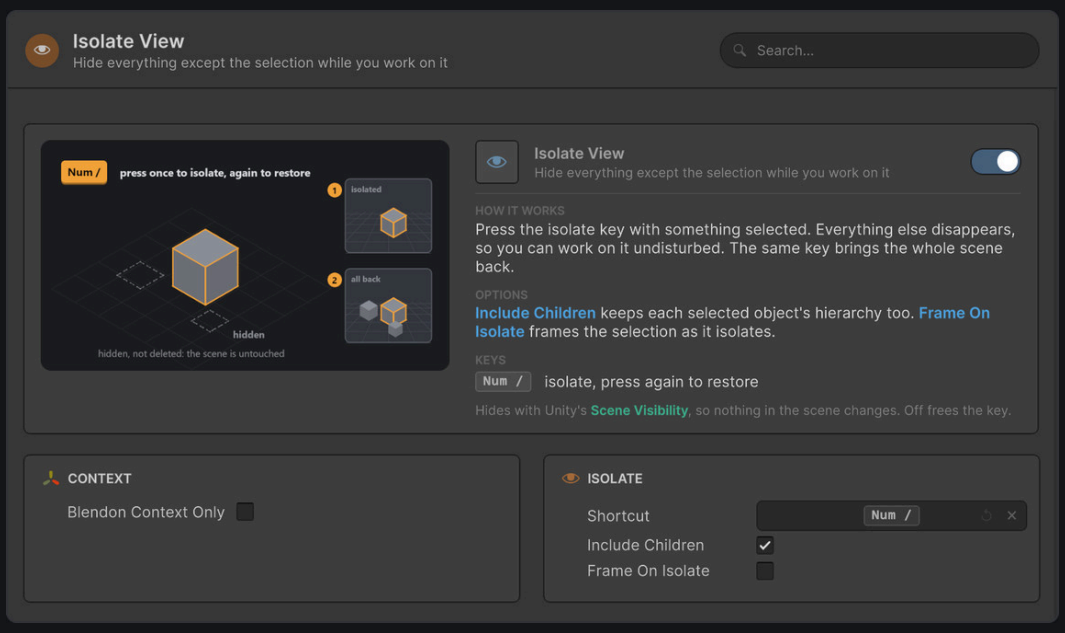


Figure 7.13 The Isolate View page.

7.4 Navigation History

Browser-style back and forward for both what you had selected and where you were looking, keeping up to 100 entries each.

- **Selection History.** Jump back and forward through previous selections. The default binding is **Mouse 3 / Mouse 4**, the side buttons on most mice. Only objects in the scene are recorded, so a click on an asset in the Project window adds no step, and the history starts over when a scene is opened, or a new one created, other than additively.
- **View History.** Undo and redo Scene view camera positions. The default binding is **Shift + Mouse 3 / Shift + Mouse 4**, optionally animated.
- **Blendon Context Only.** Each half has its own, off by default. On, its keys answer in the Blendon tool context only. Both halves keep recording everywhere.



Figure 7.14 Selection History: the mouse's side buttons step back and forward through what you had selected.

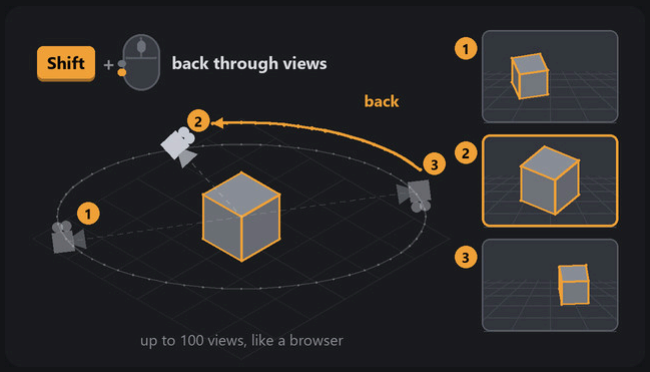


Figure 7.15 View History: with Shift, the same buttons step through camera views.

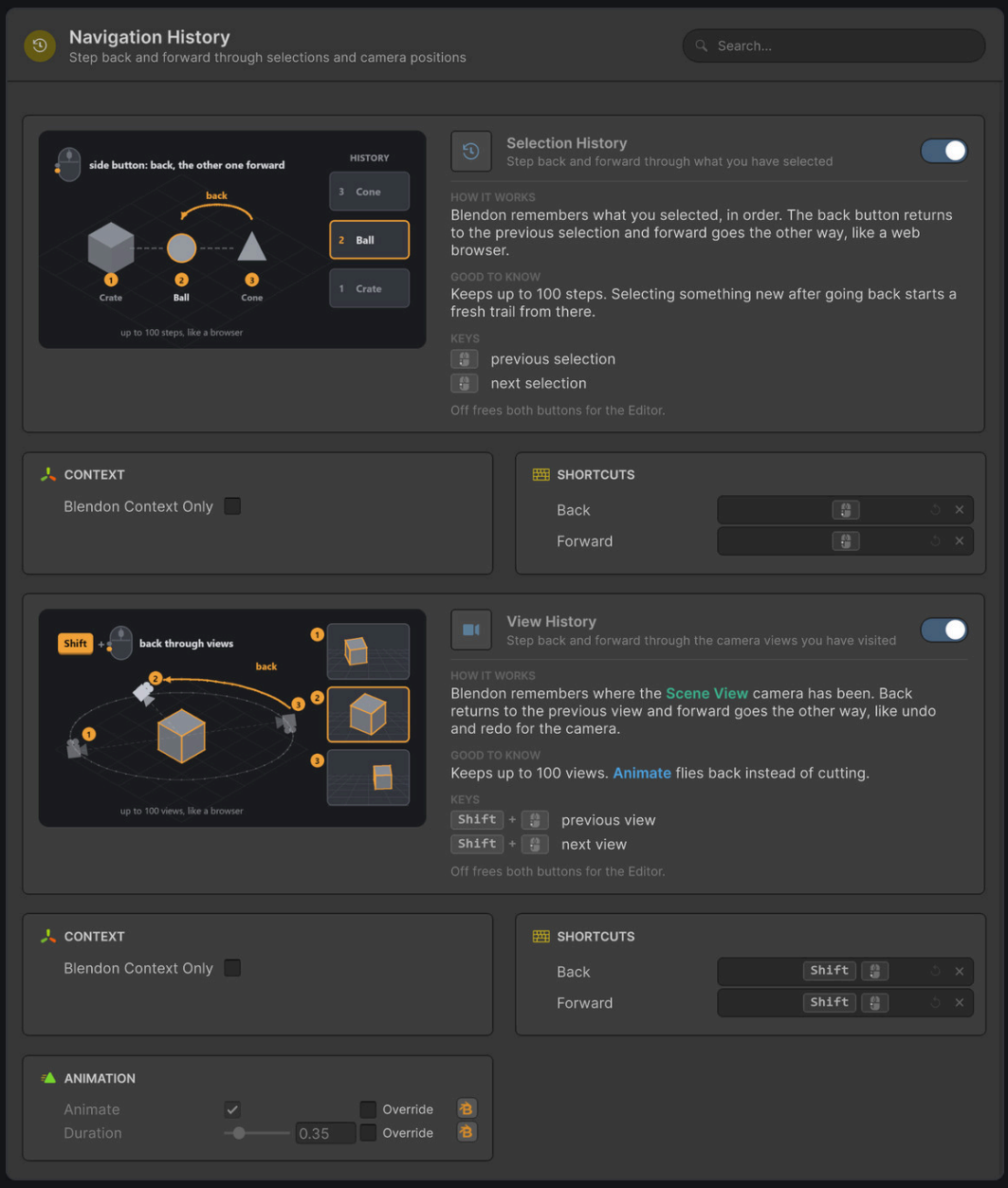


Figure 7.16 The Navigation History page. Each half can be switched off on its own.

7.5 Box Select - Left Mouse drag

Blender's box select. A box dragged over the Scene view selects every object it touches, not only the ones that fit wholly inside it as the Editor's own box does. Box Select is on by default; switch it off from **Scene Tools -> Box Select** or from the Overview page to hand the drag back to the Editor's own box.

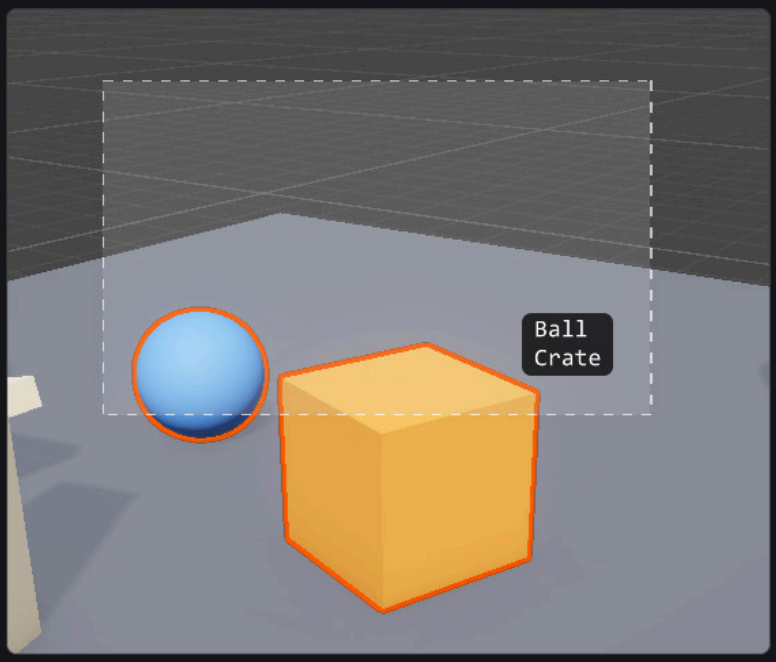


Figure 7.17 Only the top of the ball and a corner of the crate are inside the box, yet both are marked for selection. The readout at the box's corner names them.

- **Exact, not approximate.** The box is tested against each object's real triangles, sprite shape or line path, so a long diagonal object is selected only where the box actually crosses it, never just because its bounding box reaches in.
- **Visible Only.** On by default: the box only selects what you can actually see inside it, like Blender with X-Ray off. Press **C**, also in the middle of a drag, to switch it and select objects hidden behind others too. Sprites, lines and trails are always tested whole.
- **Add, remove and combine.** Hold **Shift** to add to the selection and **Ctrl** (**Cmd** on macOS) to remove from it. With both held, **Shift + Ctrl** does what its setting says: **Difference** (default) flips everything under the box, **Intersect** keeps only what was selected and is also under the box. These follow the Editor's own *Add Box Select* and *Invert Box Select* shortcuts, so rebinding those in Unity's Shortcuts window rebinds them here as well.
- **Move Box.** Hold **Space** while dragging to move the whole box instead of resizing it. Letting go carries on resizing from where the box now is.
- **While Dragging. Highlight** (default) marks the objects the release will select, with an outline or a fill. **Live** selects them for real as you go, Hierarchy and Inspector included. **On Release** shows nothing until the button comes up.
- **Cursor Readout.** Names the objects the release changes, or counts them, or both.
- **Fully Enclosed.** Off by default. On, only objects that fit entirely inside the box are selected, as in the Editor.
- **Select Prefab Roots** and **Respect Selection Base.** Touching any part of a prefab instance selects the instance, and touching part of an object under a parent marked `[SelectionBase]` selects that parent, the way a click does.
- **The Scene view's picking rules still apply.** Hidden objects, objects with picking disabled, locked layers and the Prefab Mode stage are respected, and in a LOD group only the level on screen counts.

- **Only in the Blendon tool context.** In Game Object and in a package's context, such as Splines, a drag draws the Editor's own box. Switching off all four gizmos leaves the context, so it hands the box back too. The hierarchy keys work in every context.
- **Everything else stays the Editor's.** Handles, Terrain painting, Alt-orbit and a tool that takes the press for itself, such as ProBuilder's Create Shape, keep the box away exactly as they do without Blendon. Terrain, particle systems, UI and gizmo icons (lights, cameras) are selected when the box contains them wholly, as in the Editor.

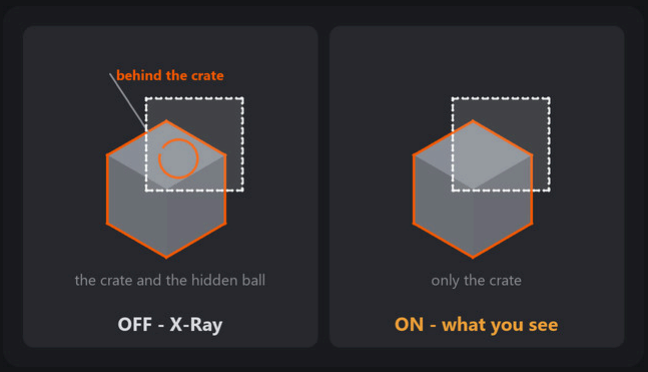


Figure 7.18 Visible Only: on, only what you can see; off, hidden objects behind others too.

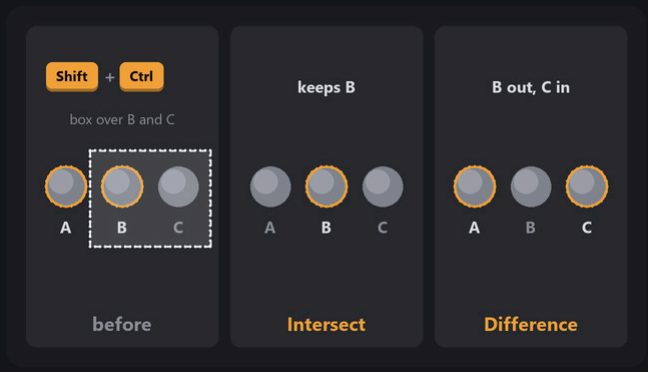


Figure 7.19 Shift + Ctrl: Difference flips what the box touches, Intersect keeps the overlap.

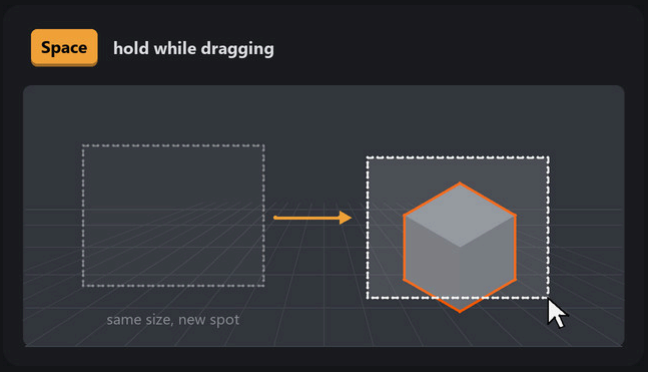


Figure 7.20 Hold Space while dragging to move the whole box.

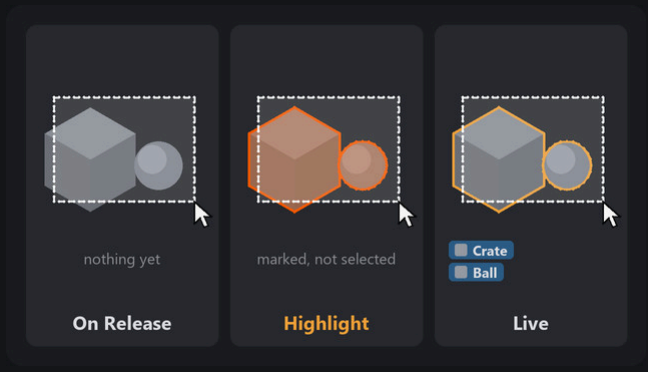


Figure 7.21 While Dragging: nothing until release, a highlight, or a live selection.



Figure 7.22 Fully Enclosed: off, touching is enough; on, only whole objects.

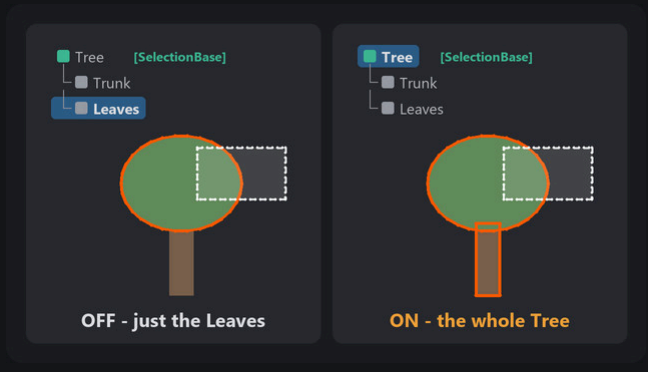


Figure 7.23 Respect Selection Base: touching the leaves selects the whole tree.

Each box is a single undo step. **Esc** or a right-click cancels a box mid-drag.

7.5.1 Click Selects Parent

On the same page, and on by default. Clicking the selected object again selects its parent, and each click after that the next parent up. Past the topmost one it goes to the object right under the cursor, so every level can be reached by clicking, including the parts under a `[SelectionBase]` or a prefab root. Clicks with **Shift**, **Ctrl**, **Alt** or **Cmd** are left as they are, and so is every click outside the Blendon tool context. A click a tool takes for itself, such as one of ProBuilder's Create Shape, keeps the selection too: only a click that makes the Editor pick an object climbs.

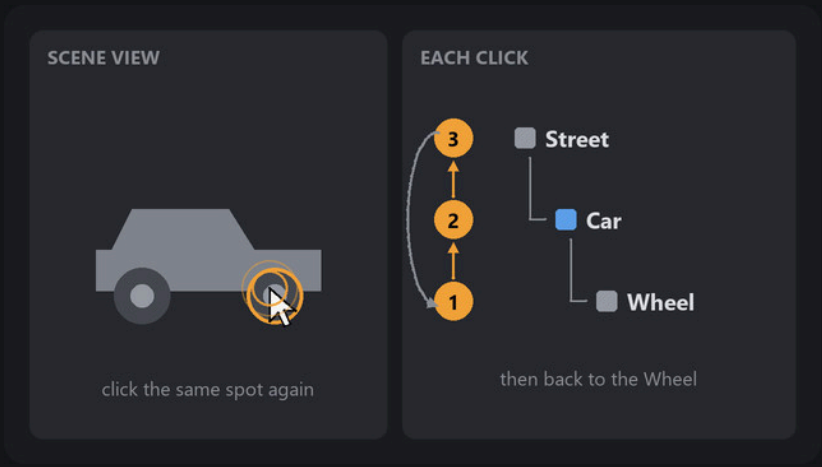


Figure 7.24 Click the same spot again, and the selection climbs to the parent, then the next one up.

7.5.2 Hierarchy keys

Blender's **[** and **]** walk the hierarchy from the Scene view. Objects with nowhere to go, with no parent or no children, stay selected. The keys have their own switch in the **Hierarchy Keys** panel; off, Blendon leaves them alone. Their **Blendon Context Only** row, off by default, keeps them to the Blendon tool context.



Figure 7.25 The hierarchy keys, shown on the stack from the pictures above.

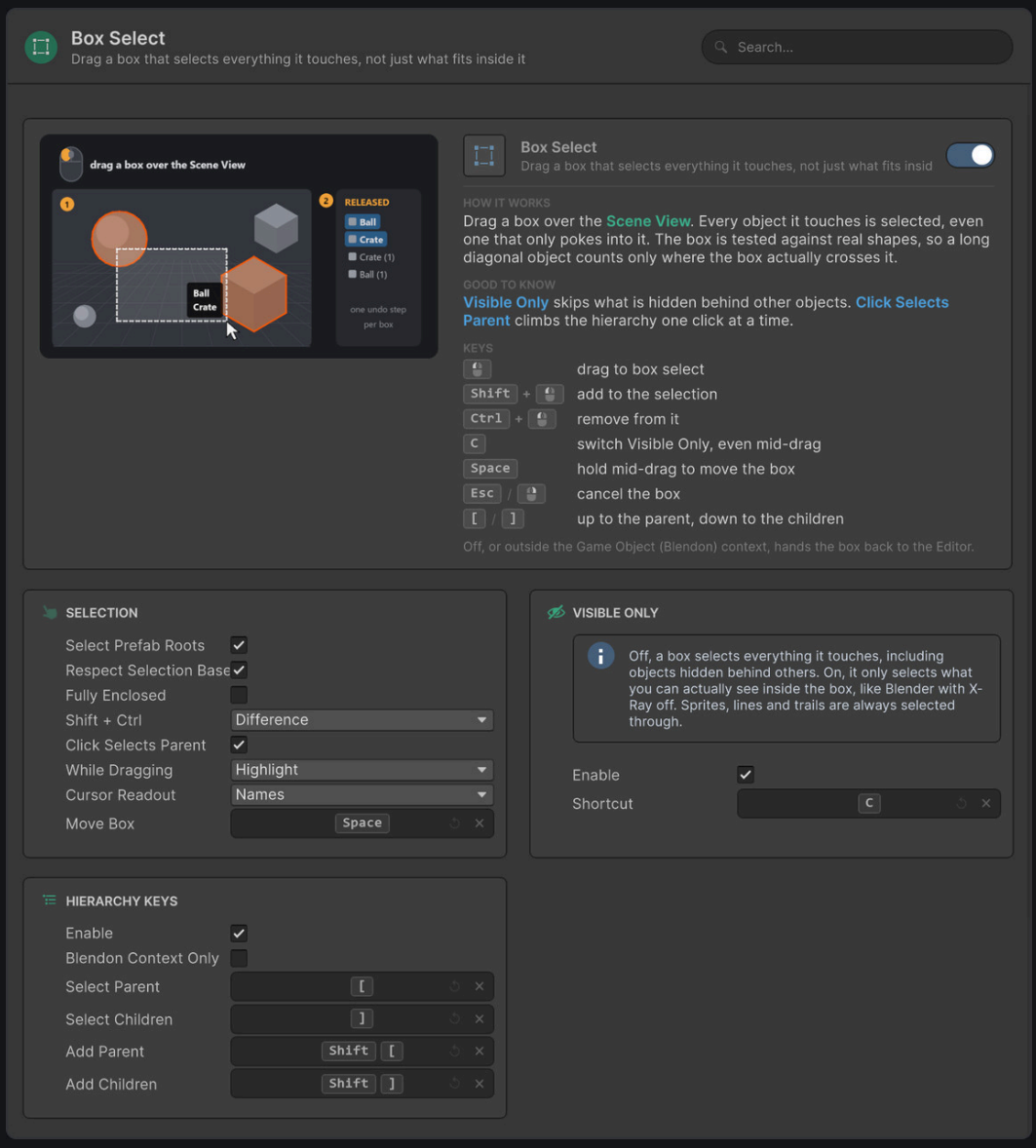


Figure 7.26 The Box Select page: its header card, then the Selection, Visible Only and Hierarchy Keys panels.

8 Pie menus

Blender's radial menus, with both of its interaction styles on one key.

- **Flick.** Hold the key, move the mouse in a direction, then release. The item in that direction runs immediately.
- **Tap.** Press and release the key quickly without moving, however short the tap. The menu stays open, you can move freely, and a left-click runs an item.
- **Cancel.** Press **Esc** or right-click. Releasing the key, or left-clicking an open menu, also cancels while the cursor is inside the center deadzone or points at an empty or greyed-out slot.

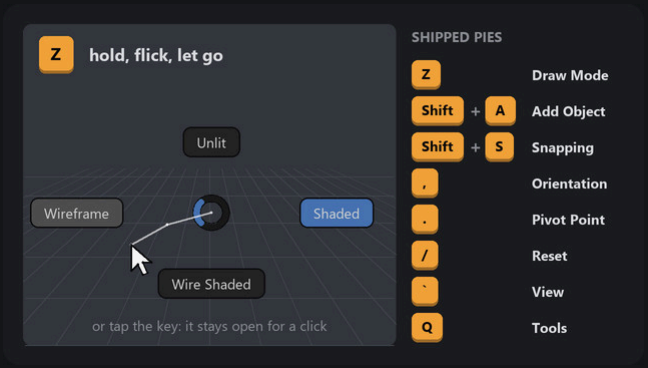


Figure 8.1 Hold a pie's key, flick toward an item, let go. A quick tap keeps the menu open for a click.

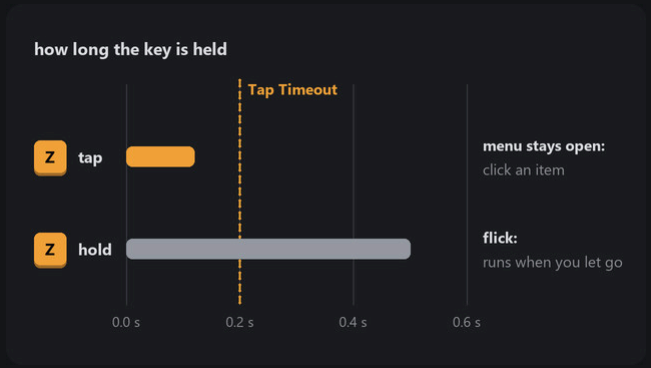
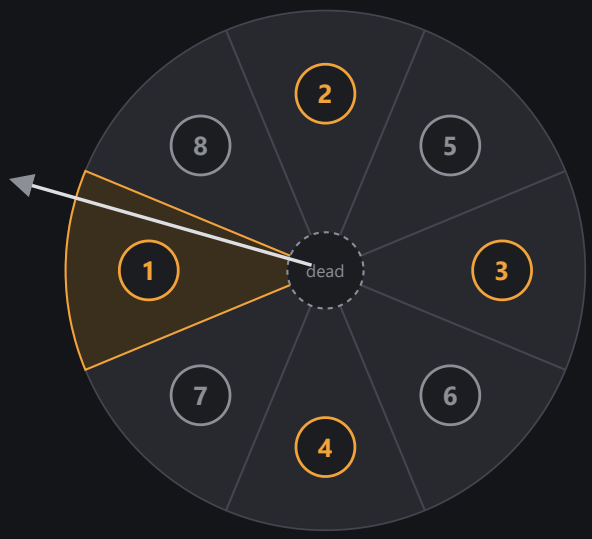


Figure 8.2 How long the key is held decides: a tap leaves the menu open, a hold flicks.



Slots 1-4
left, up, right, down

Slots 5-8
upper-right, lower-right, lower-left, upper-left

Selection by angle
only the direction counts, not the distance

Wedges
90 deg with four items, 45 deg with eight

Deadzone
release inside it and nothing runs

Figure 8.3 How a pie is laid out. The first four items take the sides, the next four the diagonals, and only the direction of the cursor counts.

Selection is by angle alone, measured from the menu's center, so you never have to land on a button: a flick 500 pixels past an item still picks it. A four-item menu gives each item a 90-degree wedge, and adding diagonals narrows every wedge to 45 degrees, exactly as Blender does.

While a menu is open it owns the Scene view. There is no marquee select, no orbit or zoom, no gizmo drag, and every Blendon shortcut stands down. Items are numbered, and their number keys, on the top row or the keypad, run them directly.

An item that matches the Editor's current state, such as the active tool, the draw mode or a snap toggle that is on, is tinted blue. The item under the cursor is lit. An item that cannot run right now, such as a Reset with nothing to reset or a tool whose package is not installed, is greyed out in place rather than removed, so the rest of the ring never shifts.

8.1 Built-in pies

Blendon ships eight pies. Each one can be switched off on its own, rebound, and edited exactly like a pie of your own; see Custom pie menus.

Pie	Default	Items
Draw Mode	Hold **Z**	Wireframe, Unlit, Shaded, Wireframe Shaded
Add Object	Hold **Shift + A**	Cube, Empty, Sphere, Camera, Cylinder, Light, Particle System, Plane
Snapping	Hold **Shift + S**	Incremental, Angle, Scale, Grid
Orientation	Hold **,**	Global, Grid, Local
Pivot Point	Hold **.**	Center, Median, Active, Pivot
Reset	Hold **/**	Position, Scale, Rotation, All
View	Hold **`**	Left, Top, Right, Bottom, Front, Perspective / Orthographic, Camera, Back
Tools	Hold **Q**	Move, Scale, Rotate, Transform, Rect, Hide Gizmo, Create Spline, View

Items fill the ring in Blender's own order: the first four go **left, up, right and down**, and the next four go to the **upper-right, lower-right, lower-left and upper-left** diagonals. An empty item keeps its slot blank, so the items after it stay where they are. Orientation ships just three items, so the slot below them stays empty, where Blender keeps an orientation Unity has no match for.

8.2 Draw Mode pie - Z

Blender's Z menu. Unity's Scene view has neither Blender's matcap Solid mode nor its HDRI Material Preview, so the pie switches between the four modes Unity does have: **Wireframe** to the left, **Unlit** above, **Shaded** to the right and **Wireframe Shaded** below. The mode the view is already in is tinted, and a mode the active render pipeline does not offer is greyed out.

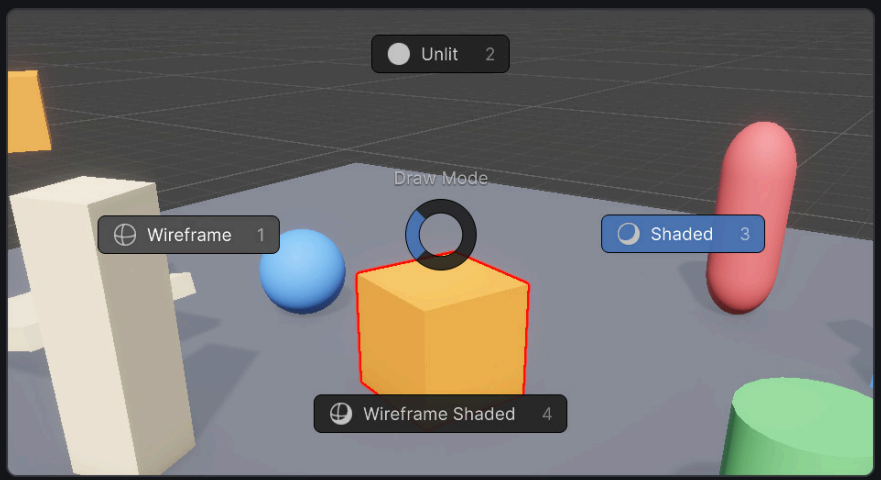


Figure 8.4 The Draw Mode pie. Shaded is tinted because the view is already in it; Wireframe is lit because the cursor points left.

Item	Unity equivalent
Wireframe	Wireframe draw mode; nothing else is touched
Unlit	Shaded draw mode with Scene Lighting off, so the view is lit from the camera; skybox, particles and visual effects on; post-processing, fog and flares off
Shaded	Shaded draw mode with scene lighting, skybox, post-processing, fog, flares, particles and visual effects on, and Always Refresh
Wireframe Shaded	Shaded Wireframe draw mode, with the same lighting and effects as Shaded

Unlit and Shaded are the same draw mode told apart by the Lighting toggle, so flipping that toggle by hand switches which one the menu considers current. Every other effect toggle you change is remembered per view and per mode for the rest of the Editor session, so leaving a mode and coming back restores your own toggles rather than the defaults.

In Unity's **Edit -> Shortcuts** window this pie is listed as `Blendon/Pie Menus/Draw Mode`.

8.3 Toggle Wireframe - Shift + Z

The pie's other half, the way Blender pairs the two keys. **Shift + Z** drops the Scene view straight into Wireframe, and pressing it again returns to the mode you came from, or to Shaded if there is no earlier mode to go back to, such as in a fresh session or after a debug draw mode the pie does not name. What each mode remembers is unchanged, so the round trip keeps your own effect toggles.

Toggle Wireframe has a toggle of its own on the Pie Menus page, so switching every pie off leaves the key working. During a grab the key means Blender's plane lock instead, and the running manipulation receives it.

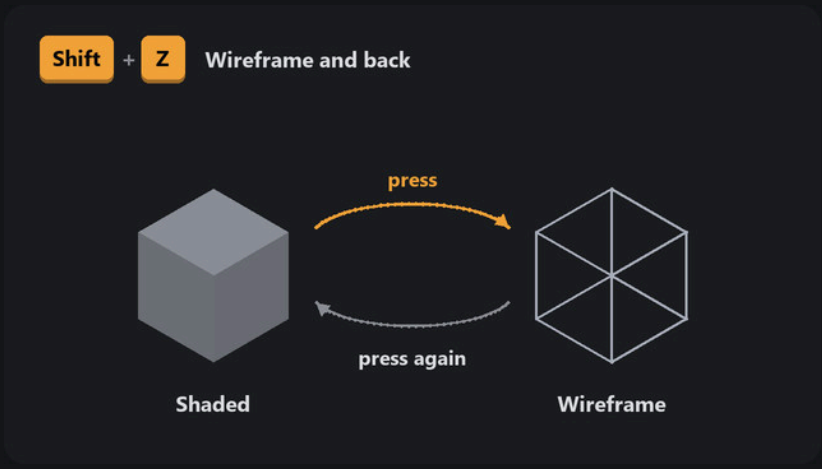


Figure 8.5 Shift + Z drops into Wireframe; pressing it again goes back.

8.4 Add Object pie - Shift + A

Blender's Shift + A menu: **Cube** to the left, **Empty** above, **Sphere** to the right, **Camera** below, then **Cylinder**, **Light** (a Point Light), **Particle System** and **Plane** on the diagonals.

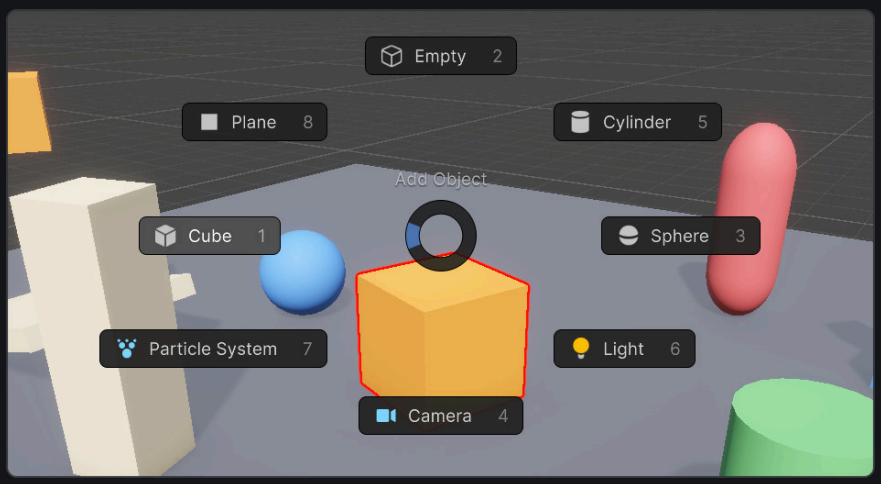


Figure 8.6 The Add Object pie, with eight items.

Each item runs Unity's own **GameObject** menu command, then moves the new object to where the pie was opened: onto the surface under the cursor, or, over empty space, onto the camera-facing plane through the view's pivot. With **Place On Floor** on, which is the default, an object with a renderer is lifted by half its height so that it rests on that point instead of sinking halfway into it. Creation and placement are a single undo step, and the new object is left selected.

The placement works for these items in any pie, not only in this one.



Figure 8.7 Place On Floor: on, a new object rests on the point instead of sinking halfway into it.

8.5 Snapping pie - Shift + S

Toggles Unity's own snap settings, the same state the Scene view toolbar shows and keeps: **Incremental** to the left, **Angle** above, **Scale** to the right and **Grid** below.

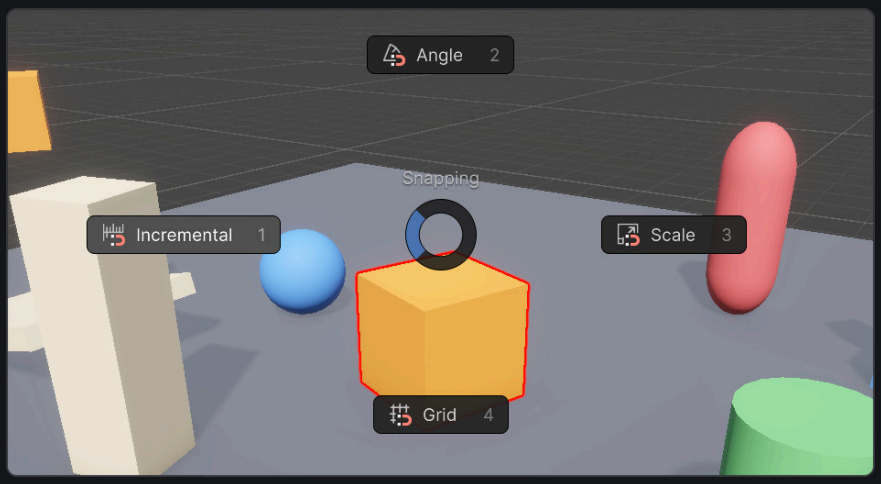


Figure 8.8 The Snapping pie.

- **Incremental.** Snap by the increment values from Unity's snap settings, relative to where the drag started. Turns Grid off.
- **Grid.** Snap to the Scene grid itself.
- **Angle and Scale.** Snap rotation and scale by their increments. Before Unity 6000.4 the Editor has a single toggle for all three increments, so there Incremental, Angle and Scale all flip that same toggle.

8.6 Orientation pie - Comma

Blender's Comma menu, set on Unity's own handle rotation toggle: **Global** to the left, **Local** to the right and **Grid** above, where Blender keeps Gimbal. Grid is greyed out on Unity versions whose toggle has no Grid, such as 6000.0.

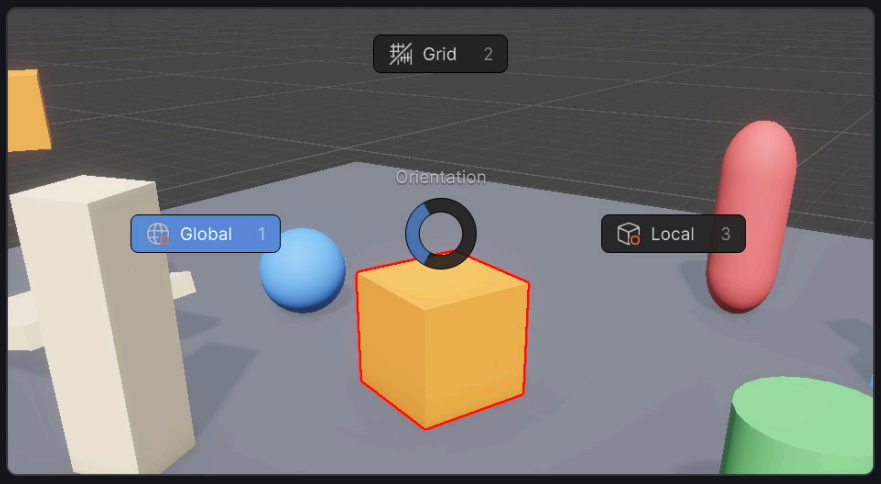


Figure 8.9 The Orientation pie. Global is tinted as the current orientation.

8.7 Pivot Point pie - Period

Blender's Period menu, laid out as Blender lays it out: **Median** (Median Point) above, **Center** (Bounding Box Center) to the left, **Active** (Active Object) to the right and **Pivot** (Individual Origins) below. The ring uses the toolbar button's short names; the action picker lists the full ones. Active sits where Blender keeps 3D Cursor, which Unity has no match for.

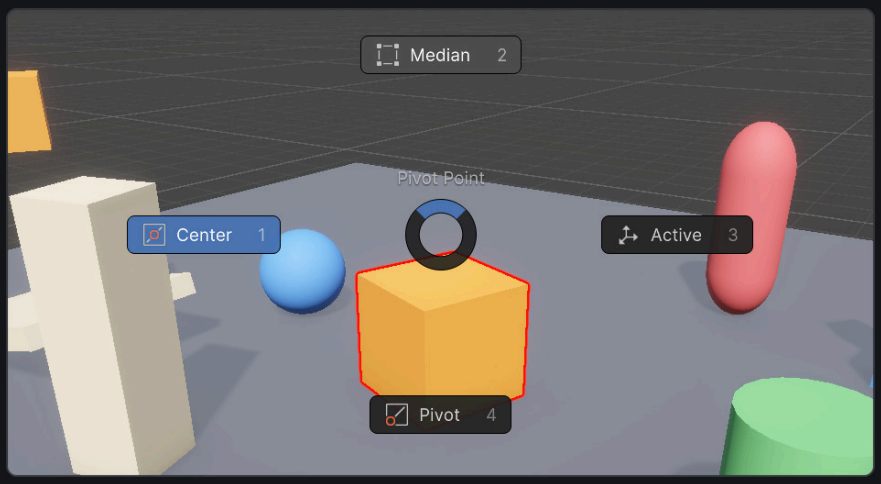


Figure 8.10 The Pivot Point pie. Center is tinted as the current mode.

Picking a mode flips Unity's Pivot/Center toggle to its side, the same as the Pivot Point dropdown. With Pivot Point Menu switched off, Median and Active are greyed out and only Unity's own Center and Pivot are left. Outside the Blendon tool context the same two are left, and they set Unity's own toggle; the mode picked in Blendon's context waits there for your return.

8.8 Reset pie - Slash

Blender's Alt + G, Alt + R and Alt + S, on the **/** key: **Position** to the left, **Scale** above, **Rotation** to the right and **All** below. The pie resets the *local* transform of every selected object, putting position to zero, rotation to none and scale to one, so a child returns to its parent's origin rather than the world's.

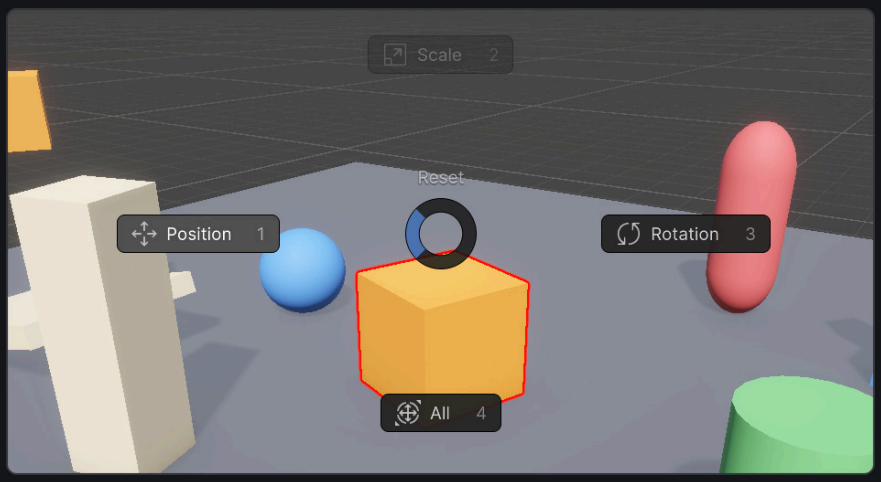


Figure 8.11 The Reset pie. Scale is greyed out because the selected crate is already at scale one.

It is fully undoable, and each item is greyed out while it has nothing to do: **Rotation** when every selected object already sits at zero, **Scale** when every one of them is at one, and the whole pie with nothing selected.

8.9 View pie - Backtick

The numpad's camera moves under the mouse, on the ` key: **Left**, **Top**, **Right** and **Bottom** on the four sides, **Front** upper-right, a projection toggle lower-right, **Camera** lower-left and **Back** upper-left. Each item runs the same move as its numpad key, on the Scene view the pie was opened in.

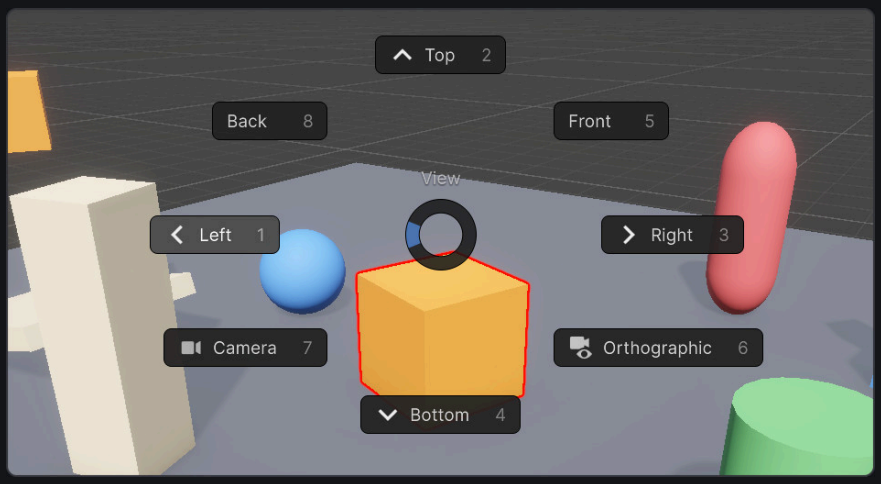


Figure 8.12 The View pie.

The projection toggle is labelled with the projection it switches *to*, **Perspective** or **Orthographic**. **Camera** aligns the view to the active camera like **Numpad 0**, and an axis item is tinted while the view is exactly aligned to it.

8.10 Tools pie - Q

Unity's own tools, without reaching for the toolbar: **Move** to the left, **Scale** above, **Rotate** to the right and **Transform** below, then **Rect**, **Hide Gizmo**, **Create Spline** and **View** on the diagonals. The active tool is tinted. While the pie sits on **Q**, the View tool's own key, **Q** is the View tool again outside the Blendon tool context, as **Z** is Toggle Pivot Position for the Draw Mode pie.

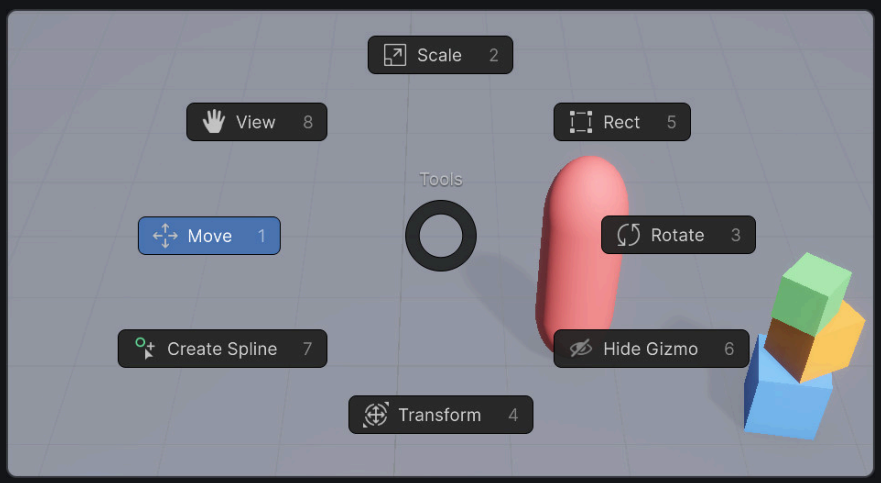


Figure 8.13 The Tools pie.

- **Hide Gizmo** is Blender's Show Gizmo turned around. Turned on in the Blendon tool context, it hides Blendon's gizmos and Unity's Rect handle, so the selection can be seen with nothing over it. Keyboard

G / **R** / **S** still work. Unity's own tool in the slot of a switched-off gizmo stays visible. Outside the Blendon context the item is greyed out, since the gizmos there are Unity's or a package's. It is tinted while the gizmos are hidden and turns off again when Unity restarts.

- **Create Spline** needs the Splines package (`com.unity.splines`) and is greyed out without it. Picking it while it is already active drops back to Move.

8.11 Custom pie menus

The **Pie Menus** settings page lists the built-in pies above your own.

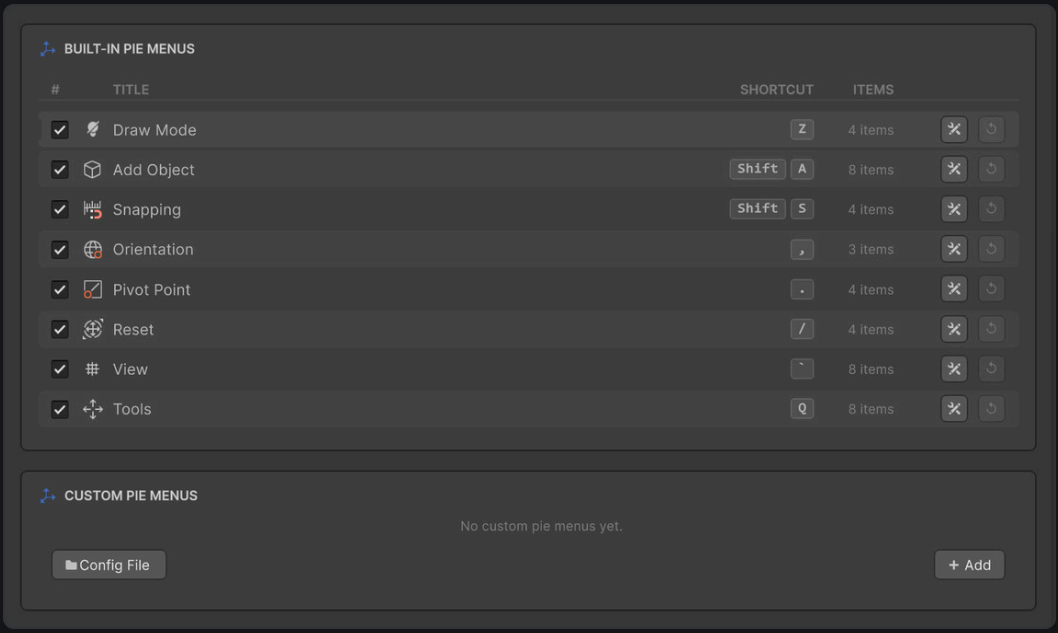


Figure 8.14 The pie lists on the Pie Menus page. Each row has a switch, the title, the key, how many slots run something, and Edit and Reset buttons.

Each row has:

- an on/off switch, which frees the pie's key for the rest of the Editor when off;
- the title, which opens the pie editor when clicked;
- the pie's key, highlighted if it clashes with another command, and how many of its slots run something;
- **Edit**, plus **Reset** for a built-in pie, which puts back the shipped title and items and the keyboard preset's key, or **Remove** for one of your own.

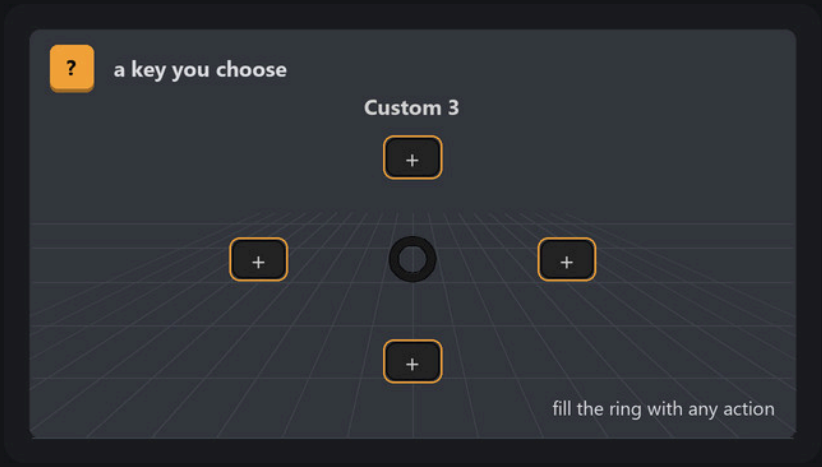


Figure 8.15 A pie of your own: any key, and any Blendon action or Unity menu command in its slots.

Add creates a new pie. There are eight custom slots, each with its own entry in Unity's Shortcut Manager (`Blendon/Pie Menus/Custom 1` to `Custom 8`), unbound until you give it a key. **Config File** reveals the file every pie on the page is saved in; see Data and files.

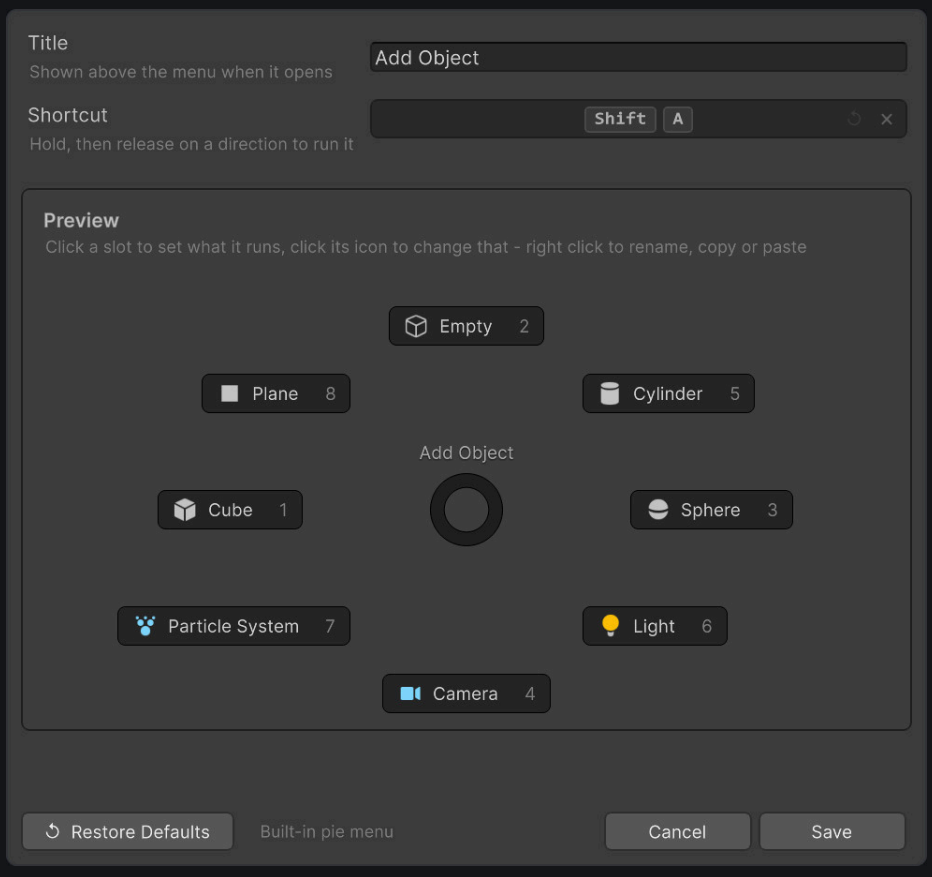


Figure 8.16 The pie editor, open on the built-in Add Object pie.

The pie editor edits one menu, built-in or custom:

- **Title.** Drawn over the center of the menu.
- **Shortcut.** Click the field and press the key to hold. A clash is flagged inline, naming the command it collides with. For a built-in pie this is the same binding that **Edit -> Shortcuts** shows.

- **The ring.** Both the preview and the editor, with all eight slots drawn in their real directions. Left-click a slot to choose what it runs, or click its icon to choose an icon. Right-click a slot for **Change Action**, **Set Icon**, **Rename**, **Copy**, **Paste** and **Clear Action**. A copied slot keeps its icon and name, and can be pasted into a different pie.
- **The action picker.** Lists every Blendon action under a **Blendon** branch: everything the built-in pies are made of. Every command in Unity's main menu follows, including commands added by other packages and assets. Type to search, or use the picker's shortcut-capture button and press a key to find the menu command bound to it. A menu command greys out and tints in the ring exactly when the menu bar would grey it out or tick it.
- **Save and Cancel.** Save (**Create** for a new pie) keeps the edits. Cancel, or closing the window, discards all of them, the shortcut included. A built-in pie also has **Restore Defaults**.

8.12 Shared pie settings

The top of the Pie Menus page tunes every pie at once.

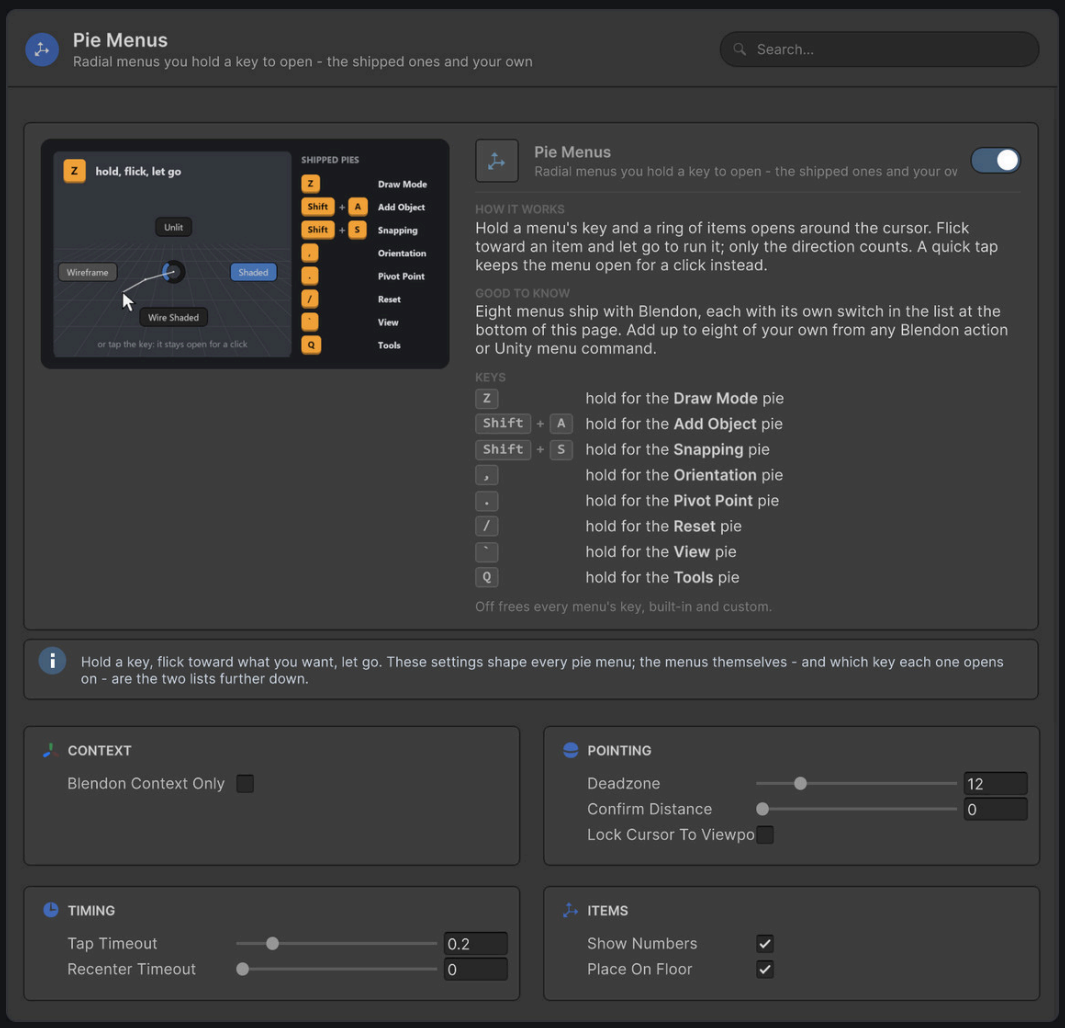


Figure 8.17 The top of the Pie Menus page: the header card with every shipped pie's key, then pointing, timing and items.

Setting	Default	Effect
Enable	On	Master switch. Off switches every pie's key off, built-in and custom, handing the keys back to the Editor; the bindings themselves stay as they are. Toggle Wireframe keeps its own switch.
Blendon Context Only	Off	On, every pie key and Toggle Wireframe answer in the Blendon tool context only, and do what they do in plain Unity everywhere else. Off, the Draw Mode and Tools pies still hand **Z** and **Q** back outside it.
Deadzone	12 px	How far the cursor has to move from the center before it points at an item.
Confirm Distance	0 (off)	Run the highlighted item once the cursor is this far past the deadzone and stops moving, without waiting for the key to be released.
Lock Cursor To Viewport	Off	Keep the cursor inside the Scene view while a pie is open.
Tap Timeout	0.2 s	Release the key faster than this without moving, and the menu stays open for a click.
Recenter Timeout	0 (off)	A menu opened near a view edge is nudged inward to fit. For this long afterward, directions are still measured from where the cursor really was.
Show Numbers	On	Number the items so that their digit keys run them. The keypad digits need NumLock on.
Place On Floor	On	Lift an object created from a pie by half its height, so that it rests on the point instead of sinking into it.
Radius	100 px	How far the items sit from the center. Under **Appearance**.
Unfold Time	0.06 s	How long the items take to spring out from the center; 0 shows them in place at once. Under **Appearance**.
Colors	-	Item, Highlight, Current, Text and Center Ring. Under **Appearance**.

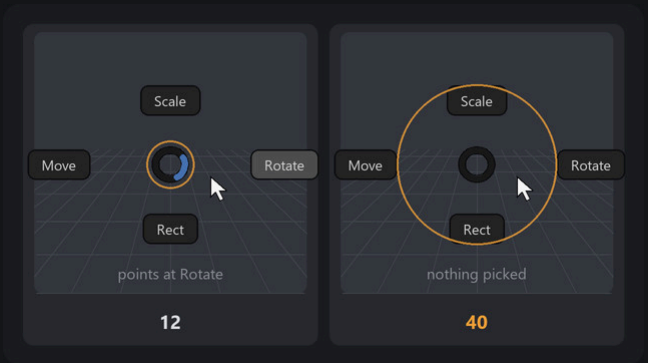


Figure 8.18 Deadzone: how far the cursor moves before it points at an item.

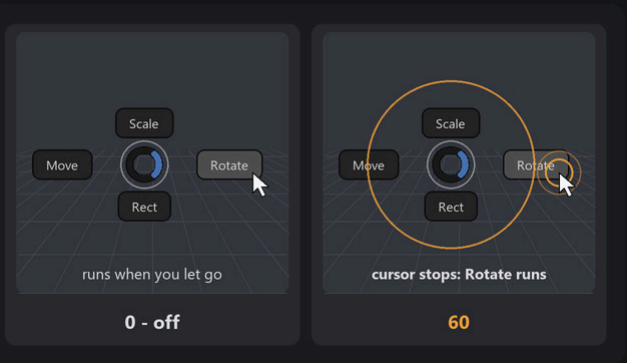


Figure 8.19 Confirm Distance: stop past this distance and the item runs without letting go.

9 Context menu

Right-clicking the Scene view opens Blendon's context menu instead of the Editor's: a compact, searchable panel drawn directly in the viewport, the way Blender draws its menus.

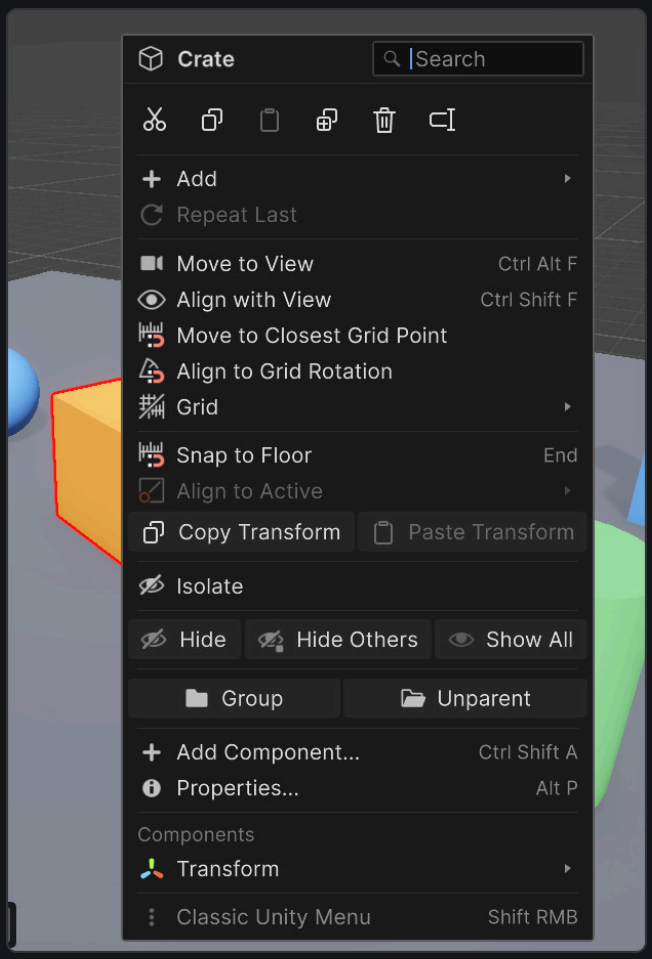


Figure 9.1 The context menu for a single object: the header with its name and search field, the quick-action icons, then the grouped entries.

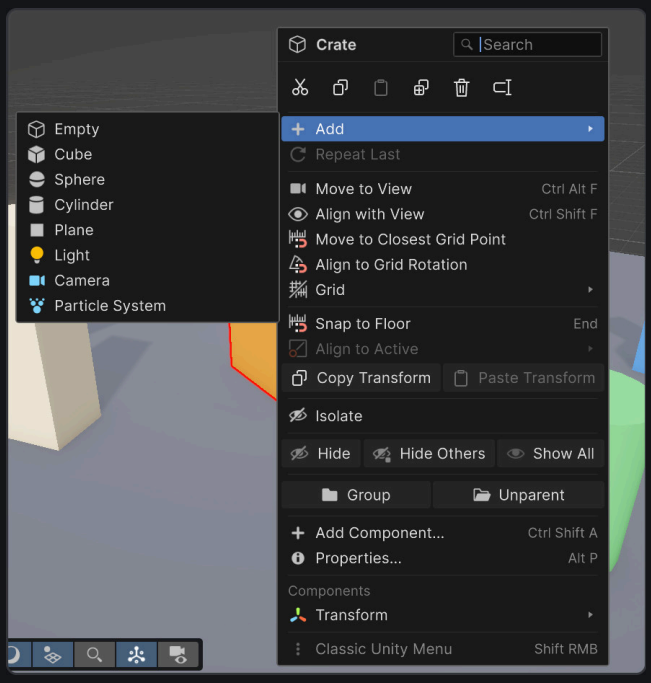


Figure 9.2 Hovering Add opens its submenu. Near the right edge of the view it flips to the left.

- **Every entry.** The menu is built from exactly the entries Unity's own right-click menu would show for the current selection and active tool, including anything that other packages and assets add to it. Blendon only arranges and decorates them: an entry it does not recognise still appears, under **Extensions** or **More**, in the Editor's own order. Greyed-out and ticked states match the Editor's menu as well.
- **Quick actions.** Cut, Copy, Paste, Paste As Child, Duplicate, Rename and Delete sit in a single row of icon buttons along the top, whichever of them Unity offers for the selection, with Blendon's own **Rename** closing the row. Hover over one to see its name and key. Set **Edit Actions** to **List** to show them as ordinary rows instead.
- **Sections.** The remaining entries are grouped: aligning to the view and the grid, isolation, object commands, prefab commands, the selection's components, and whatever the active tool or tool

context adds. Rows show an icon and, where the command has one, its key.

- **Submenus.** A submenu opens beside its row as you hover over it, and flips to the left near the edge of the view.
- **Search.** The header names the selection and holds a search field that already has the caret. Start typing to filter every entry, submenus included, then press **Enter** to run the highlighted match. While you type, the Editor's own shortcuts stand down, so letters land in the field instead of framing the view or switching tools.

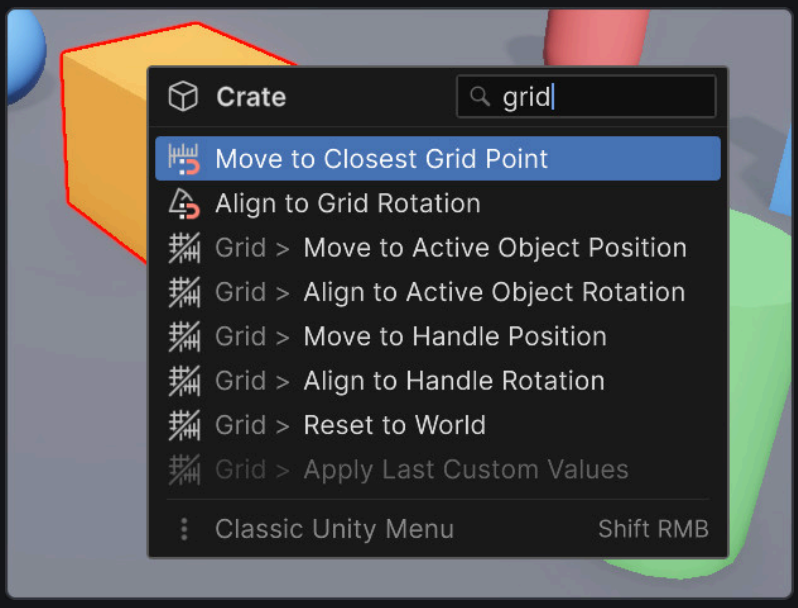


Figure 9.3 Typing "grid" filters every entry, submenus included. Entries from a submenu show their path in grey.

- **Rename.** Click the name in the header, or the **Rename** entry in the edit row, and the rest of the header line, beside the selection's icon, becomes a text field holding the current name, selected and ready to be typed over. **Enter** takes the new name as a single undo step, **Esc** leaves it as it was; either way the menu stays open and the header goes back to showing the name. It is there whenever one object in the scene is selected.
- **Extra entries.** Blendon can add entries the Editor's menu does not carry; see the table below. Each joins the group it belongs to rather than piling up at the end.
- **Keyboard. Up** and **Down** move the highlight, **Right** opens a submenu or steps along the quick actions, **Left** closes the submenu, **Enter** runs the highlighted entry, and **Esc** closes a submenu, then clears the search, then closes the menu.
- **Closing.** Running an entry, clicking outside the menu, changing the selection or switching to another window closes it.

9.1 The Editor’s own menu

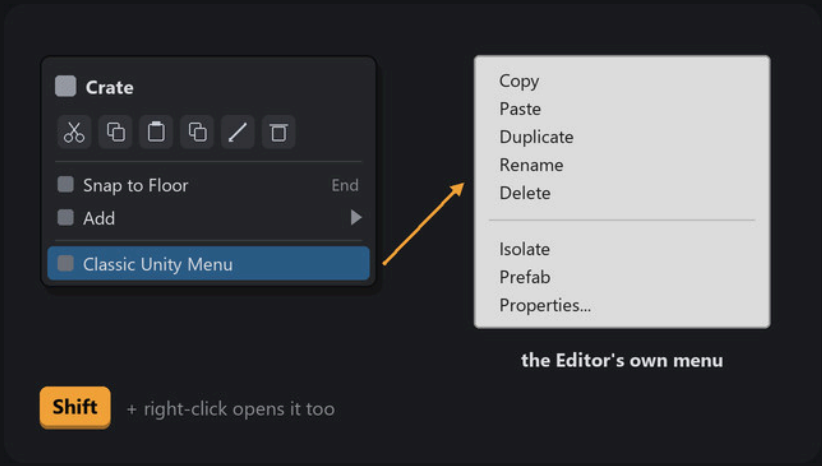


Figure 9.4 The Editor’s own menu is one row away, or Shift + right-click.

The Editor’s own menu is always one step away. **Shift + right-click** opens it instead, and so does the **Classic Unity Menu** row at the bottom of Blendon’s menu, unless **Classic Menu Row** is switched off. Both hold the very same entries.

A right-click that drags is left alone, so holding the right mouse button to fly through the scene works exactly as before. A right-click with **Ctrl**, **Alt** or **Cmd** held also goes to the Editor.

9.2 Context menu settings

The **Context Menu** page, under **Keys & Menus**, holds the menu’s shortcut, three appearance settings and a table of extra entries.

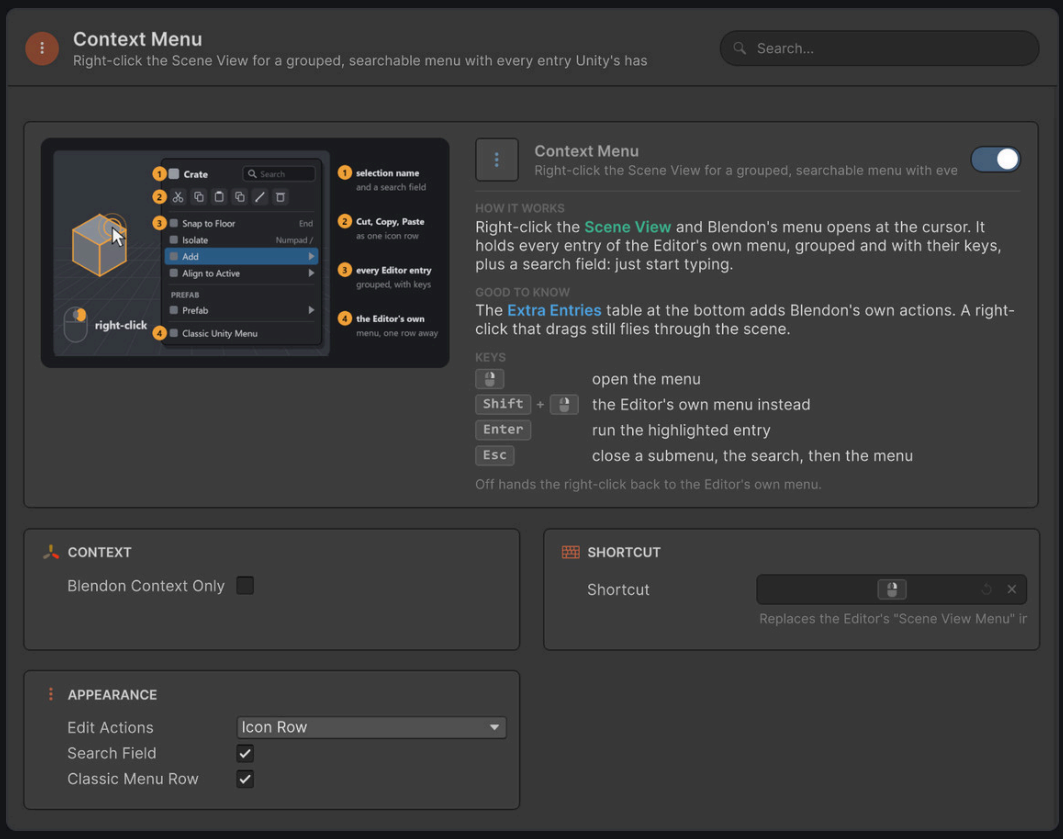


Figure 9.5 The Context Menu page: the header card, the shortcut and the three appearance settings.

Setting	Default	Effect
Enable	On	Off hands the right-click back to the Editor's own menu.
Blendon Context Only	Off	On, the right-click opens the Editor's own menu outside the Blendon tool context, the same one **Shift + right-click** opens. The Editor's `Scene View/Menu` stays unset either way, so a change of context never writes your shortcut profile.
Shortcut	Right-click	The button that opens the menu, `Blendon/Context Menu` in Unity's Shortcut Manager. Two menus on one button would make Unity ask which to open, so while this one holds the right button Blendon unsets the Editor's own `Scene View/Menu` and gives its key back when the page goes off, the master switch goes off, this shortcut moves to another key or the Editor closes; the next start takes it again. That is the only key Blendon ever writes by itself. On Unity's read-only **Default** profile it cannot, so the Editor's menu keeps the button until the Overview page makes an editable copy. On the Unity keyboard preset the right-click stays the Editor's, and Blendon's menu opens on **Alt + right-click** instead; an Alt + right-drag still zooms the view.
Edit Actions	Icon Row	**Icon Row** puts Cut, Copy, Paste, Paste As Child, Duplicate, Rename and Delete in one row of buttons along the top. **List** shows them as ordinary rows with their names and keys.

Setting	Default	Effect
Search Field	On	Shows the search box in the header. Off leaves the whole header to the selection's name; the Editor's shortcuts still stand down while the menu is open.
Classic Menu Row	On	Shows the **Classic Unity Menu** row at the bottom. **Shift + right-click** opens the Editor's own menu either way.

Each extra entry adds something of Blendon's own. An entry the selection cannot take is greyed out rather than hidden.

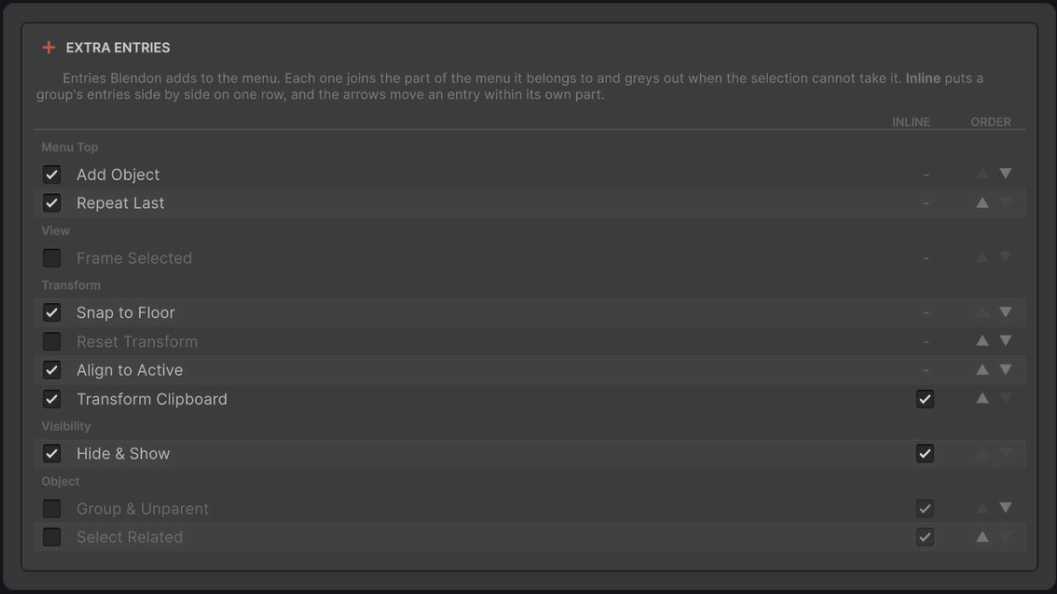


Figure 9.6 The Extra Entries table. The Inline column puts a group's entries side by side on one row, and the arrows reorder an entry within its group.

Extra Entry	Default	What it adds
Add Object	On	An **Add** submenu with the usual objects: empty, primitives, light, camera, particles. Whatever it creates lands at the point you right-clicked, on the surface under the cursor or on the grid, the way the Add pie menu places things.
Repeat Last	On	Repeats the entry you ran last, Blendon's own as well as the Editor's, and names it on the row.
Frame Selected	Off	Frames the selection, through Blendon's own framing cycle where that feature is on and the Editor's otherwise.
Snap to Floor	On	Drops the selection onto whatever is below it, the same action as Blendon's **Snap to Floor** shortcut.
Reset Transform	Off	A **Reset** submenu for position, rotation, scale, or all three at once. A channel already at rest, rotation at zero or scale at one, is greyed out, the same as in the Reset pie.

Extra Entry	Default	What it adds
Align to Active	On	Puts the rest of the selection where the active object stands: its position, its rotation, or both.
Transform Clipboard	On	**Copy Transform** takes the active object's position, rotation and scale. **Paste Transform** hands them to whatever is selected next. Inline by default.
Hide & Show	On	**Hide**, **Hide Others** and **Show All**, over the Editor's own scene visibility, the same thing the eye column in the Hierarchy toggles. Inline by default.
Select Related	Off	**Children**, **Hierarchy** and **Parent**, which walk the hierarchy from the Scene view: the children, the whole subtree, or the parent. Inline by default; with Inline off they move into a **Select** submenu.
Group & Unparent	Off	**Group** puts the selection under a new empty parent. **Unparent** moves it back out to the top of the scene. Inline by default.

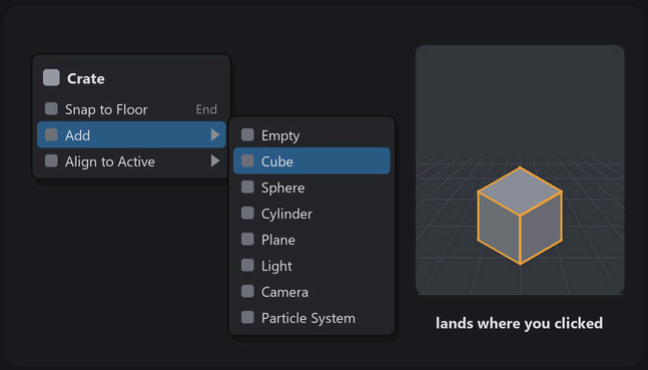


Figure 9.7 Add: the new object lands where you right-clicked.

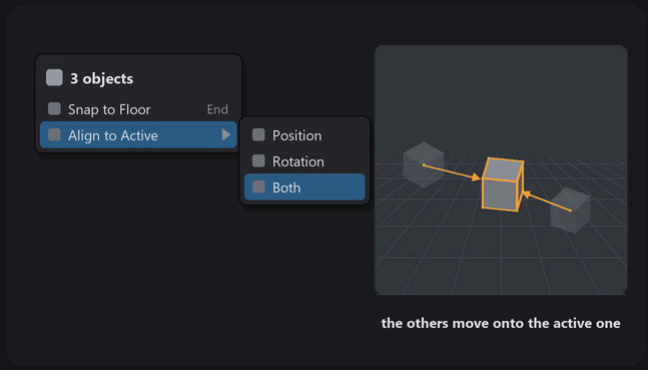


Figure 9.8 Align to Active: the others move onto the active one.

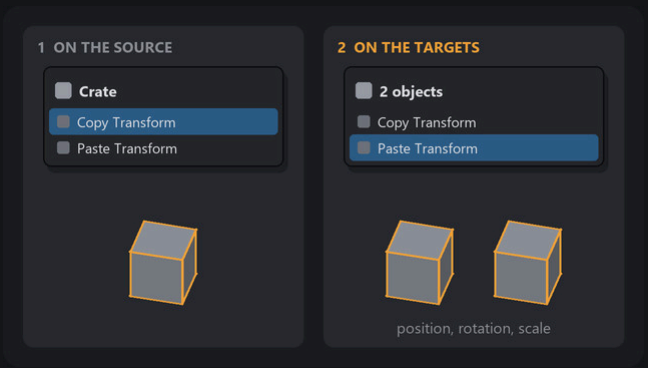


Figure 9.9 Copy Transform on the source, Paste Transform on the targets.

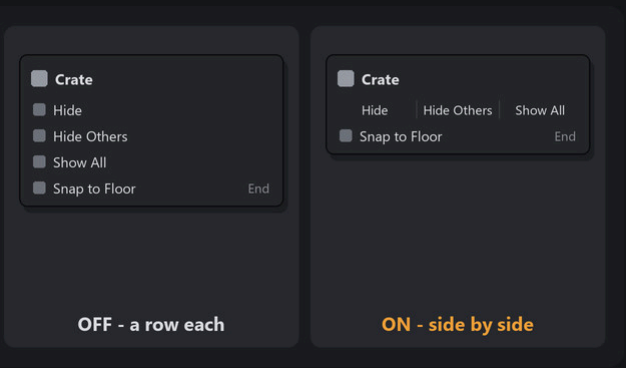


Figure 9.10 Inline: a group's entries side by side on one row.

The **Essentials** and **Tools Only** presets switch the context menu off, and **Full** switches it on.

10 Scene tutorial

The tutorial teaches Blendon by doing. It is one journey through a small scene you build yourself: a floor, a cube, then a second cube to place against the first, with each task leaving the scene ready for the next one. It lives in two places: a small badge in the Scene view, and the checklist on the **Tutorial** page of the settings window.

10.1 The badge

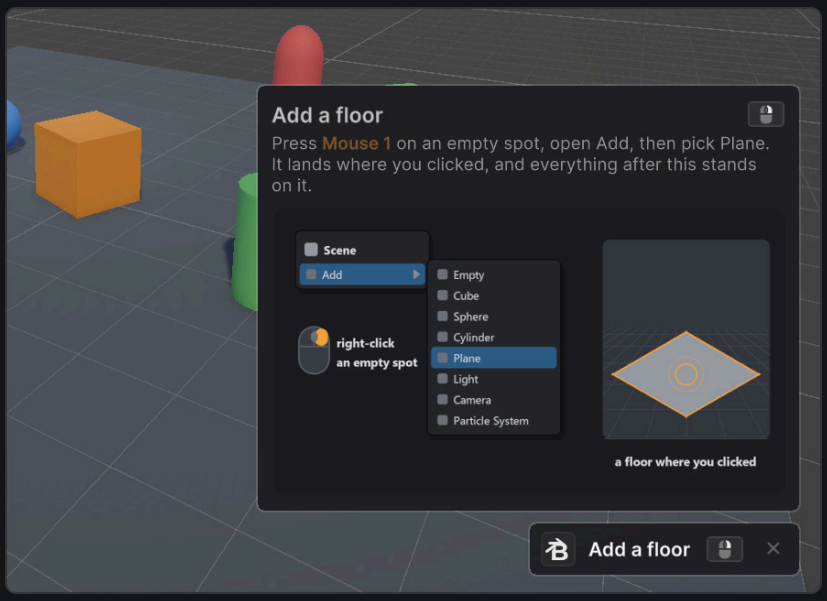


Figure 10.1 The tutorial badge in the corner of the Scene view. It names the task asked for next, with its keys, and hovering over it opens a card with a hint and a picture.

The badge sits in the bottom-right corner of the Scene view you are working in.

- **Starting.** At first the badge reads **Blendon Tutorial**. Click it to start: the settings window opens on the Tutorial page, and from then on the badge names the task asked for next, with its keys.
- **Hint.** Hover over the task's name for a card with a short hint and a picture of the task. The Blendon website shows the same cards with a short clip of each task. When a task needs something first, the badge says so instead, for example "Select the cube first".
- **Ticking off.** A task ticks off only once you actually perform the gesture, not by reading about it. The badge then shows the finished task's name with a green tick for a moment.
- **Outside the Blendon context.** The tasks teach Blendon's gestures, which Game Object and the packages' contexts leave to the Editor. There the badge reads **Switch to Blendon**, with the Toggle Blendon key; clicking it opens that key's row on the Overview page. The Tutorial page shows the same note.
- **Moving and closing.** Drag the badge by its logo to put it somewhere else; the spot is kept per machine. Its close button hides it without asking. That pauses the tutorial and keeps your progress; the

switch at the top of the Tutorial page brings the badge back.

10.2 The Tutorial page

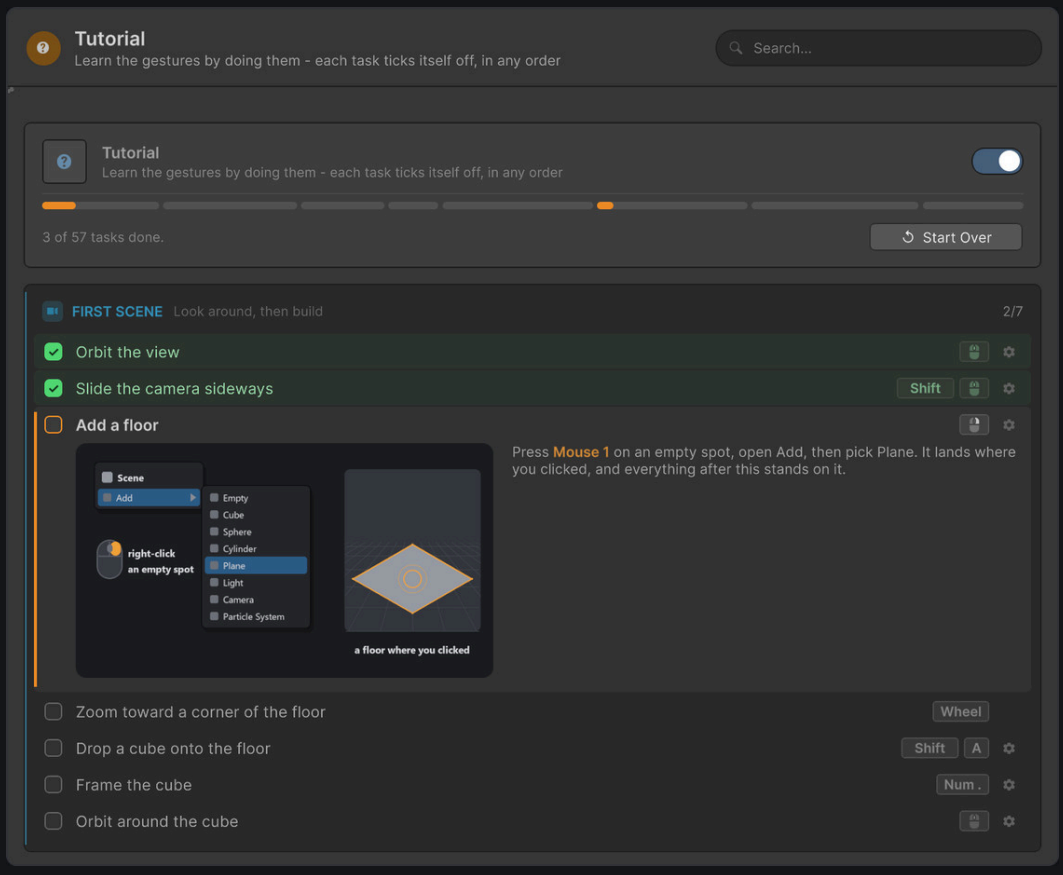


Figure 10.2 The Tutorial page. One card per chapter, the current task open with its picture and hint, and the progress bar with Start Over at the top.

The page is under **Get Started** in the sidebar. At the top, the switch beside the page title shows or hides the badge, a progress bar shows one segment per chapter, and a line counts the tasks done. Below, each chapter has a card that lists all of its tasks.

- **The current task** is open, with its picture, hint and keys. A finished task turns green.
- **Any order.** The order is a suggestion, not a gate. Click any open task to do it next; once it is done, the tutorial goes back to where it was. A gesture done early ticks the first open task that asks for it.
- **The gear** beside a task opens the settings page where its keys are set, with the row marked.
- **Live keys.** Every key shown is read from your current bindings, so a key you rebound shows as you set it.

There are eight chapters and 56 tasks, in order:

1. **First Scene.** Orbit and pan, add a floor, zoom toward one of its corners, drop a cube onto it, frame the cube and orbit around it.
2. **Grab and Move.** Grab, drop and cancel, slide along one axis, lift the cube exactly two metres by typing, drop it back with Snap to Floor, slide it across the floor with the plane lock, then grid steps and

precision.

3. **Rotate and Scale.** The rotate grab, an exact 45 degrees, the cube's own axis, the scale grab, and the Reset pie to put the size back.
4. **A Second Cube.** Add or duplicate a second cube, snap a corner onto a corner, and stand it on top of the first with surface snap.
5. **Tools and Handles.** The Tools pie, arrows and plane squares, Local orientation, rotation rings, look-at, a virtual pivot, and keeping that pivot as an empty.
6. **Selecting.** Box select and its modifiers, the hierarchy keys, Click Selects Parent, selection history, the Pivot Point pie and Align to Active.
7. **Views.** The numpad views, perspective and orthographic, orbit steps, axis snap, quick roll, the orientation gizmo, the View pie, the camera view, view history and isolate view.
8. **Pie Menus and You.** The Draw Mode pie flicked and tapped, Toggle Wireframe, running a pie item by its number, the Snapping pie, and giving a shortcut a key of your own.

Unity's own ways count wherever they do the same job: **W** and **E** or the Tools overlay for the tool tasks, and **Ctrl + D** for the second cube.

10.2.1 Tasks that follow your setup

The tutorial teaches the setup you have, not the one Blendon ships with.

- **Skipped tasks.** A task whose feature is switched off, or whose key is unbound, is skipped. It stays in its chapter, dimmed and tagged **OFF**, and its tooltip says what to switch on to get it back.
- **Another way.** Where Blendon has a second way to the same result, the task takes that way rather than being skipped. For example, on Unity's read-only **Default** profile, where Blendon's right-click menu waits, the floor comes from the Add Object pie instead of the context menu, and with the Tools pie off, the tool tasks take Unity's own keys.
- **Tasks that build on another.** The hierarchy tasks need a parent: the empty that **Keep that corner as a pivot** left behind. When that task was skipped, they are skipped too.

Skipped tasks count toward the progress bar, and the tutorial is finished once every task is done or skipped. The badge then goes away.

Progress is kept per machine rather than per project, so it survives across projects, and it is left untouched by the settings window's **Reset All Pages**. **Start Over** on the Tutorial page asks first, then un-ticks every task and starts again from the first one.

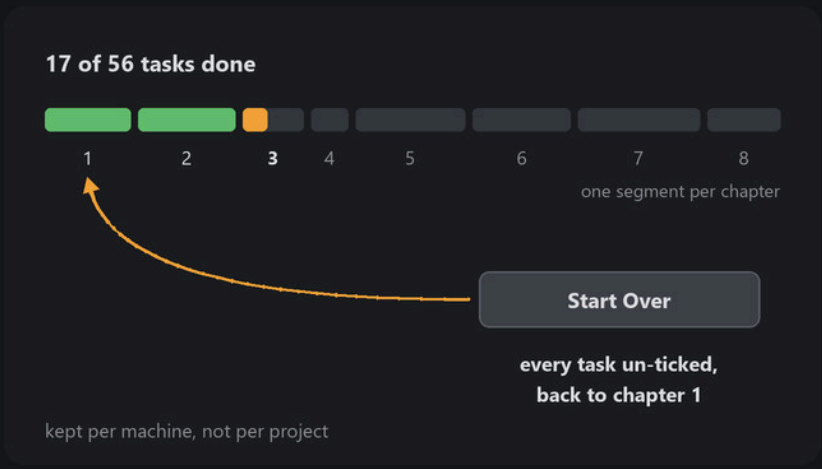


Figure 10.3 Start Over un-ticks every task and goes back to chapter 1.

10.3 Shortcut tips

The first time you press one of Blendon's keys that an Editor command also wants, a small card slides into the bottom-right corner of the Scene view. It has one line per command: its keys, the logo of its owner, its name and where it answers. Most often the Editor's command is covered: Blendon answers the key in the Scene view, and the Editor's command keeps it everywhere else. If you moved the Editor's command to a key of its own, its line shows the key it had before. If several commands really clash on one key, the card lists them all and frames their keys in amber. For a key that follows the Blendon tool context, such as **G**, Blendon's line says that it answers in the tool context. Orbit Selected and Camera Pan never show a card.

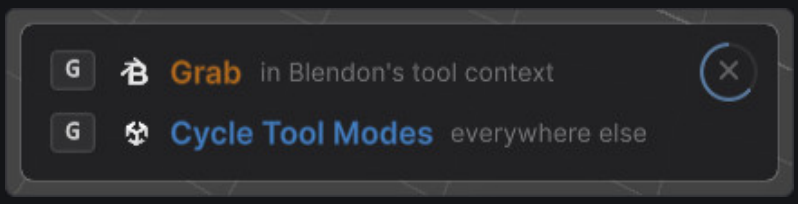


Figure 10.4 A shortcut tip after the first press of G: Blendon's Grab answers the key in its tool context, and the Editor's Cycle Tool Modes keeps it everywhere else.

Each name is a link to where its key is changed. Blendon's orange names open the **Keyboard** page, and the Editor's blue ones open Unity's **Shortcuts** window, with the name copied for its search field. The ring around the close button shows how long the card stays, about ten seconds, and the card does not leave while the pointer is over it.

Each card is shown once per key per Editor session, and armed again whenever a binding changes. While the tutorial badge is showing, the card appears just above it, and it slides away when the badge's hover card opens. **Shortcut Tips** on the Overview page switches the cards off, and **Show Again** arms every one of them again.

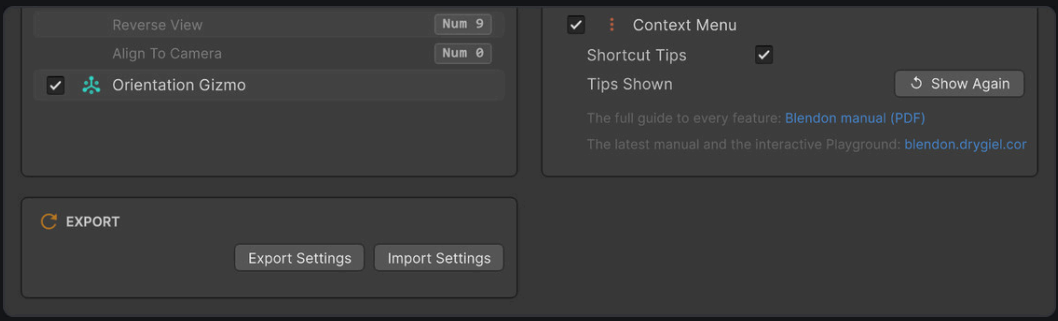


Figure 10.5 The Shortcut Tips switch and the Export buttons on the Overview page.

11 Modifier keys

Blendon's hold-to-engage modifiers have no entry in Unity's Shortcut Manager, so they are configured as modifier dropdowns in the settings window. Each can be set to **None** to switch the gesture off. The Box Select row is the exception: those are Unity's own shortcuts.

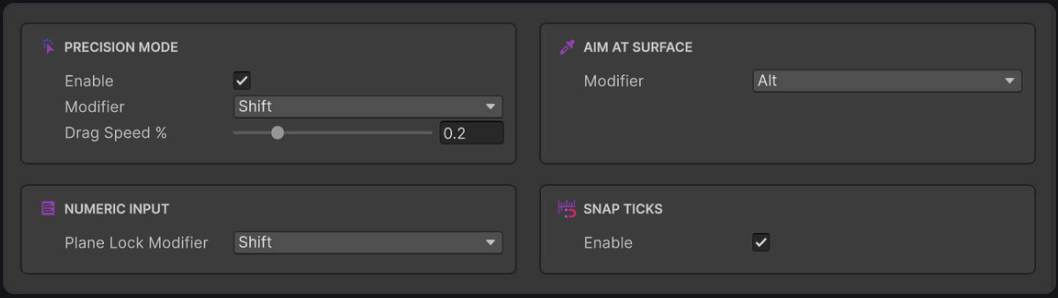


Figure 11.1 The modifier dropdowns on the All Tools page.

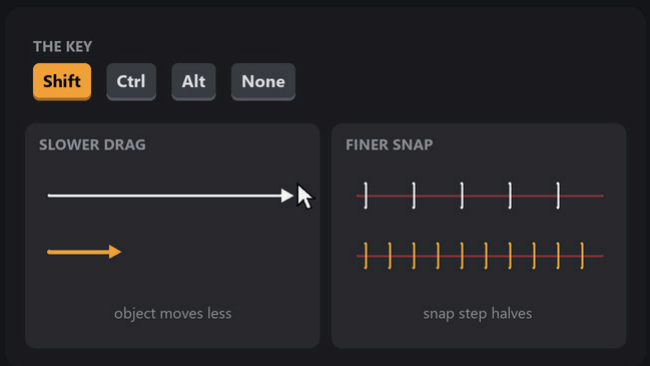


Figure 11.2 The precision modifier slows the drag and halves the snap step.

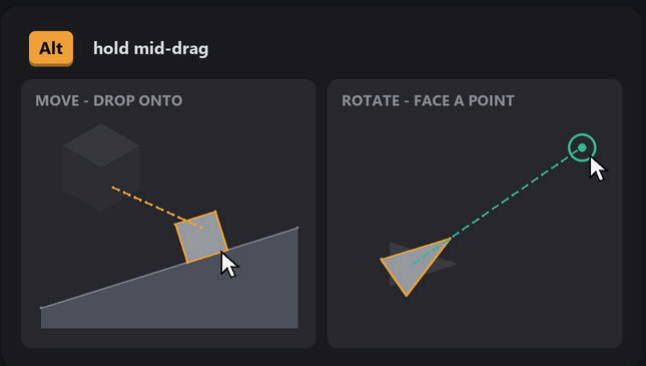


Figure 11.3 The aim modifier: Move drops onto a surface, Rotate faces a point.

Gesture	Default	Where to change it
Precision drag (slow drag, half snap increment)	**Shift**	All Tools -> Precision Mode
Aim at surface (Move surface snap, Rotate look-at)	**Alt**	All Tools -> Aim at Surface, or the **Modifier** under Move's Surface Snap and Rotate's Look At

Gesture	Default	Where to change it
Plane lock (with X / Y / Z during a drag or grab)	**Shift**	All Tools -> Numeric Input
Orbit axis snap	**Alt**	Orbit Selected -> Snap view to world axis
Box Select add / remove	**Shift** / **Ctrl**	Unity's Shortcuts window: *Scene View/Add Box Select* and *Invert Box Select*

Labels are platform-correct: macOS users see Cmd and Option rather than Ctrl and Alt.

Unity's **Ctrl** for grid snapping works throughout. Vertex snapping is Blendon's own, on **V** (hold) and **Shift + V** (pick a virtual pivot). The Editor ships its own pair on those same keys. Blendon's two answer them in the Blendon tool context only and replace the Editor's pair there, because Blendon's is the same mode with the Editor's limits removed. In any other context the Editor's pair answers. Both keyboard presets keep these keys.

12 Default shortcuts

Every Blendon shortcut declares its Blender key, so the Blendon preset is what a fresh install has on any profile. The table lists the keys under each preset, with "same" where the two agree. The grabs, vertex snapping, Box Select with its **C** and **Space**, and the Draw Mode and Tools pies on **Z** and **Q** answer in the Blendon tool context only. The other keys work in every context, unless their page has **Blendon Context Only** on.

Action	Blendon preset	Unity preset
Orbit Selected	Hold Middle Mouse	Hold Ctrl + Middle Mouse
Quick Roll	Alt + Middle Mouse, flick a direction	Ctrl + Shift + Middle Mouse
Pan	Shift + Middle Mouse	Alt + Middle Mouse
Zoom	Scroll Wheel	same
Zoom In / Out	Numpad + / Numpad -	same
Move Gizmo	W (Move tool)	same
Rotate Gizmo	E (Rotate tool)	same
Scale Gizmo	R (Scale tool), outside the Scene view	R
Transform Gizmo	Y (Transform tool)	same
Toggle Blendon (tool context)	B, in any window	same

Action	Blendon preset	Unity preset
Grab / Rotate Grab / Scale Grab	G / R / S	Shift + G / Shift + R / Unbound
Vertex Snap	Hold V	same
Pick Virtual Pivot	Shift + V	same
Frame Selected	Numpad .	same
Snap to Floor	End	same
Isolate View	Numpad /	same
Box Select	Left Mouse drag	same
Box Select: Visible Only	C	same
Box Select: Move Box	Hold Space while dragging	same
Select Parent / Children	[/]	same
Add Parent / Children to selection	Shift + [/ Shift +]	same
Draw Mode Pie	Hold Z, flick a direction	Hold Alt + Z
Toggle Wireframe	Shift + Z	same
Add Object Pie	Hold Shift + A	same
Snapping Pie	Hold Shift + S	same
Orientation Pie	Hold ,	same
Pivot Point Pie	Hold .	same
Reset Pie	Hold /	same
View Pie	Hold `	Hold Ctrl + `
Tools Pie	Hold Q	Hold Alt + Q
Custom Pies 1 to 8	Unbound	Unbound
Context Menu	Right-click, without dragging	Alt + Right-click
Editor's own context menu	Shift + Right-click	Right-click
Selection History Back / Forward	Mouse 3 / Mouse 4	same
View History Back / Forward	Shift + Mouse 3 / Shift + Mouse 4	same
View Front / Back	Numpad 1 / Ctrl + Numpad 1	Back: Ctrl + Numpad 5 with URP or HDRP

Action	Blendon preset	Unity preset
View Right / Left	Numpad 3 / Ctrl + Numpad 3	Left: Ctrl + Numpad 6 with URP or HDRP
View Top / Bottom	Numpad 7 / Ctrl + Numpad 7	same
Toggle Perspective/Ortho	Numpad 5	same
Orbit Up / Down / Left / Right	Numpad 8 / 2 / 4 / 6	same
Reverse View	Numpad 9	same
Align To Camera	Numpad 0	same

URP and HDRP put their decal handles on Ctrl + Numpad 1 and 3, so in a project with either one the Unity preset moves Back and Left to Ctrl + Numpad 5 and 6; without them they stay put.

All of these except the scroll wheel, the box drag and Shift + right-click are rebindable, either from Blendon's settings window or from Unity's own **Edit -> Shortcuts** window. Blendon's own are the `Blendon/...` entries; W, E, R and Y are Unity's own tool shortcuts. A row whose key differs from the keyboard preset's is marked as modified, and its restore button puts the preset's key back. Disabling a feature switches its keys off without touching their bindings: the Editor command on the same key answers again at once, and the feature takes the key back when it comes back on. Unity's Shortcuts window shows a Blendon key covering an Editor command as no conflict, so the Overview page lists every such pair.

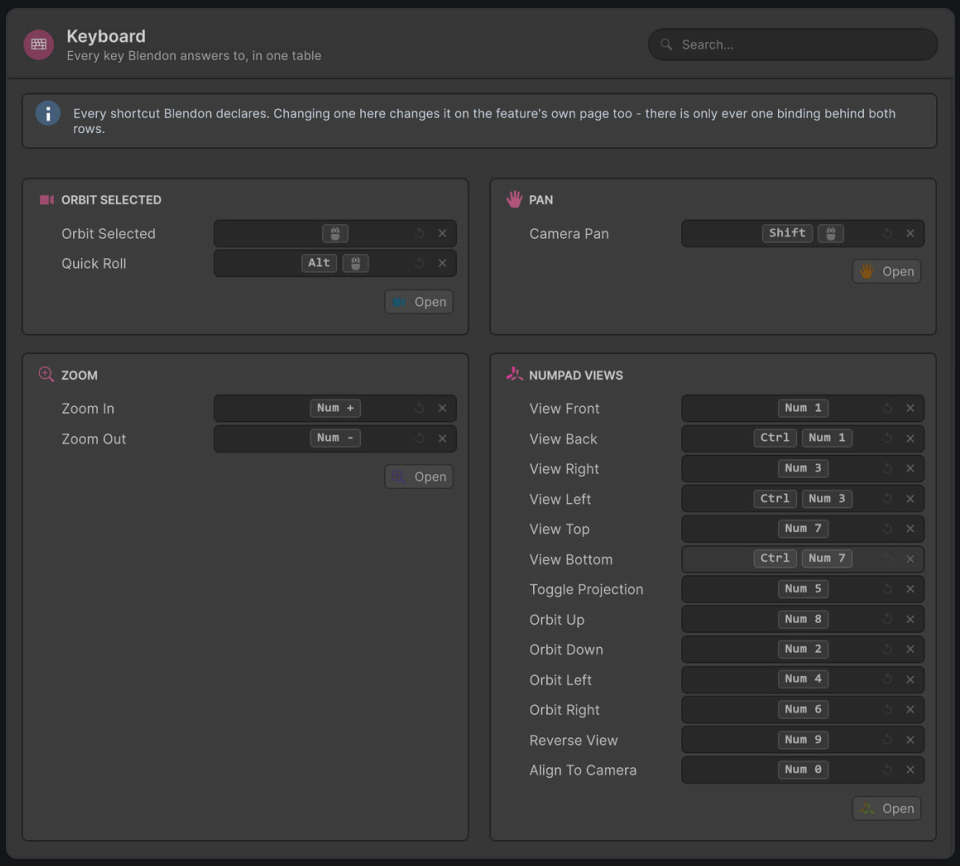


Figure 12.1 The Keyboard page lists every shortcut Blendon declares, grouped by feature.

13 Settings window

Open the settings window from **Tools -> Blendon -> Settings**.

The sidebar groups every feature into **Get Started**, **Navigating**, **Transform Tools**, **Scene Tools** and **Keys & Menus**. Each feature page can enable or disable its feature, rebind its shortcut and tune its options. The search field at the top finds a specific setting, and a per-page **Reset to Defaults** button restores defaults. Its shortcuts go back to the keyboard preset's keys. On the transform tool pages the defaults are the picked gizmo look's values; see Shared gizmo settings. Changes apply immediately and persist across sessions in `EditorPrefs`. Only a value that differs from its default is stored: a setting you never changed, or put back to its default, follows the default of the Blendon version you run, so an update that changes a default reaches it. A number within 0.0005 of its default counts as the default: it is not stored and its row is not marked. A stored value outside a setting's range, for example one written by a newer Blendon in another project, is read as the nearest value in range and left as it is.

A setting changed from its default is marked with an accent bar at the start of its row. Right-click the row and pick **Revert to Default** to put back just that one setting. **Esc** clears the search field. Drag the sidebar's edge to resize it, or double-click the edge to collapse the sidebar to a rail of icons; the window remembers both.

Sidebar group	Pages
Get Started	Blendon (the Overview page), Tutorial
Navigating	Orbit Selected, Pan, Zoom, Numpad Views, Orientation Gizmo
Transform Tools	All Tools, Move, Rotate, Scale, Transform
Scene Tools	Frame Selected, Isolate View, Snap to Floor, Navigation History, Box Select
Keys & Menus	Keyboard, Pie Menus, Context Menu

13.1 Overview page

The window opens on this page the first time, and after that on the page you last viewed. It is shown in the setup chapter.

- **Master switch.** Turns every Blendon feature off at once and switches all of its keys off, so every Editor command on the same keys answers again and the Editor behaves as if the plugin were not installed. The only key written is the Editor's own Scene view menu, which gets the right mouse button back, and the Scene view goes back to Unity's Game Object tool context. Nothing is lost: every setting and every binding is exactly where you left it when you switch back on.
- **Presets.** Three feature presets: **Full**, **Essentials** and **Tools Only**. They only switch features, and switch Blendon on if it is off; the one your switches match is lit. See Choosing a starting preset.
- **Keyboard Presets.** See Keyboard presets.
- **Feature list.** One row per feature, grouped by category, with the pie menus and the context menu together in a **Menus** group. Each row has an on/off toggle and its current key, highlighted if it clashes with a Unity command. A feature that answers to more than one key lists them all underneath its row, as the pie menus do. Clicking a feature's name jumps to its settings page.
- **Shortcut Tips.** The switch for the shortcut tip cards, and **Show Again** once a card has been shown. The Scene tutorial has a page of its own; see Scene tutorial.
- **Tool Context. Toggle Blendon**, the key that switches to the Blendon tool context and back, **B** by default; only the master switch turns it off. **Esc Keeps Context**, on by default, keeps **Esc** in the Scene view from dropping you out to Game Object.
- **Links** to this manual and to the Blendon website, and a line with the Blendon version and **Copy support info**; see Reporting a problem. They work with Blendon switched off too.
- **Export Settings / Import Settings.** Save every Blendon setting, shortcut binding and pie menu to a single JSON file, or load one back, either as a backup or to carry your setup to another machine. The file names the picked gizmo look and carries every value it covers, so an import brings back the same look. See Settings backup.
- **Defaults.** The values the rest of Blendon falls back to; see Defaults.
- **Reset All Pages.** Puts every setting back to what the plugin ships with, the **Blender** gizmo look included. Shortcut keys go back to the keyboard preset's.

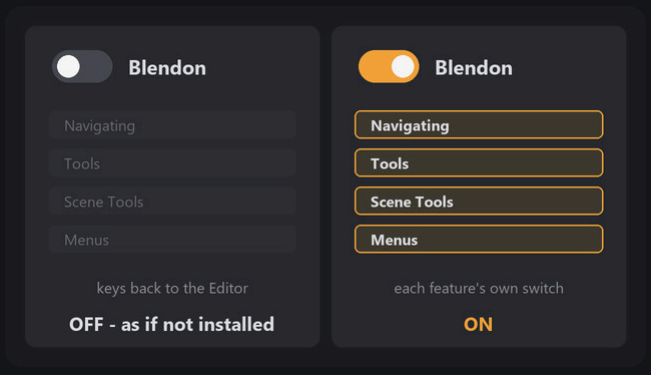


Figure 13.1 The master switch: off hands every key back, as if Blendon were not installed.



Figure 13.2 Export Settings: settings, shortcuts and pie menus in one file.

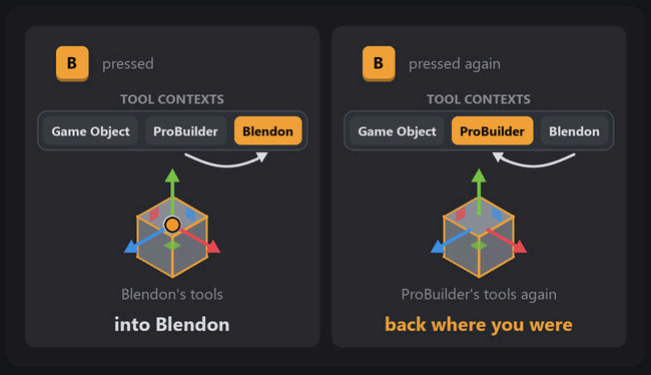


Figure 13.3 Toggle Blendon: B goes to Blendon's context, and back to the one you came from.

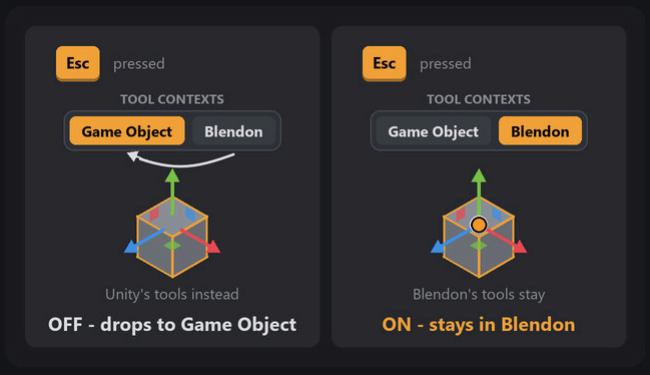


Figure 13.4 Esc Keeps Context: on, Esc leaves Blendon's tools in place; off, it drops to Game Object.

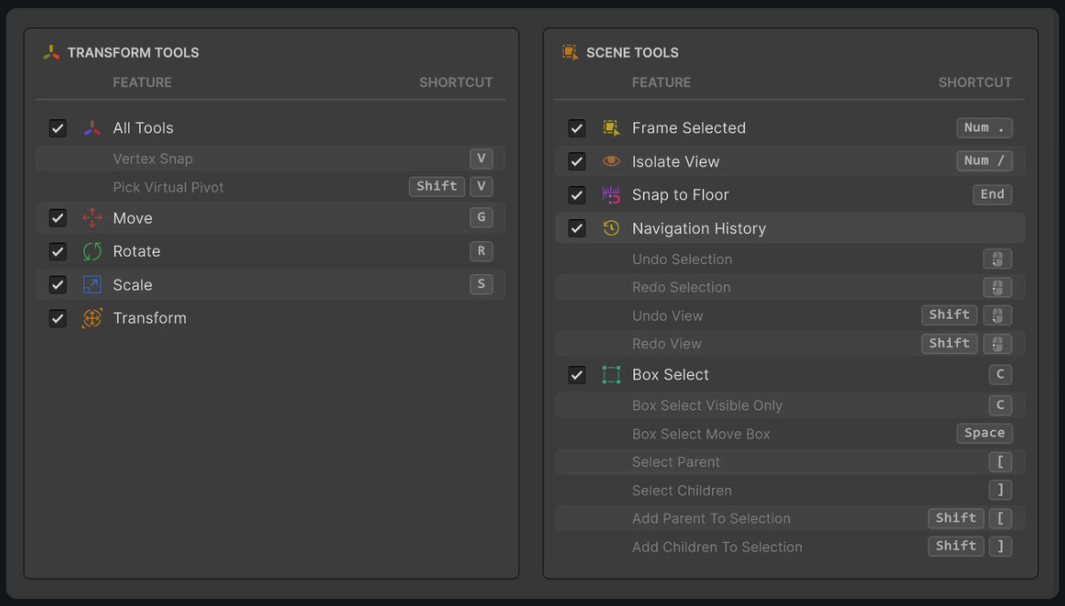


Figure 13.5 The feature list on the Overview page. Rows under a feature list its extra keys.

13.1.1 Settings backup

Export Settings writes one JSON file with every Blendon setting, shortcut binding and pie menu, and records the Blendon and Unity versions that wrote it. Numbers in it use a dot as the decimal point, whatever your system's language.

Import Settings asks first, then replaces the current settings, shortcuts and pie menus with the file's. Its closing message says what it could not use:

- An entry it cannot read, such as a number typed by hand with a comma (`0,35`) or a key Unity does not accept, is skipped, and that setting or shortcut keeps its current value.
- An entry this version does not know, most likely from a newer Blendon, is skipped too. The message names up to three, and the Console lists them all. A gizmo look this version does not ship counts as one, and the current look stays.
- Pie menus in a newer Blendon's format are left out and yours stay as they are, while the rest of the file is imported. So are the file's pie menus while your own pie menu file has not been read yet (see Data and files); import again in a moment.
- A file whose own format is newer than this version can read is not imported at all.
- An import that fails partway puts back the settings, shortcuts and pie menus from before it, and the message says so.

On Unity's read-only **Default** profile the import offers an editable copy for the shortcuts. Declined, everything but the shortcuts is imported.

13.2 Defaults and overrides

A few settings matter to many features at once: animation, dragging, cursor wrap and the axis colors. They are set once, in the **Defaults** panel at the bottom of the Overview page, and every page that carries the same setting shows the default greyed out, with an **Override** box beside it. Tick **Override** to give that one page its own value; changing a default changes it everywhere that has not been told otherwise.

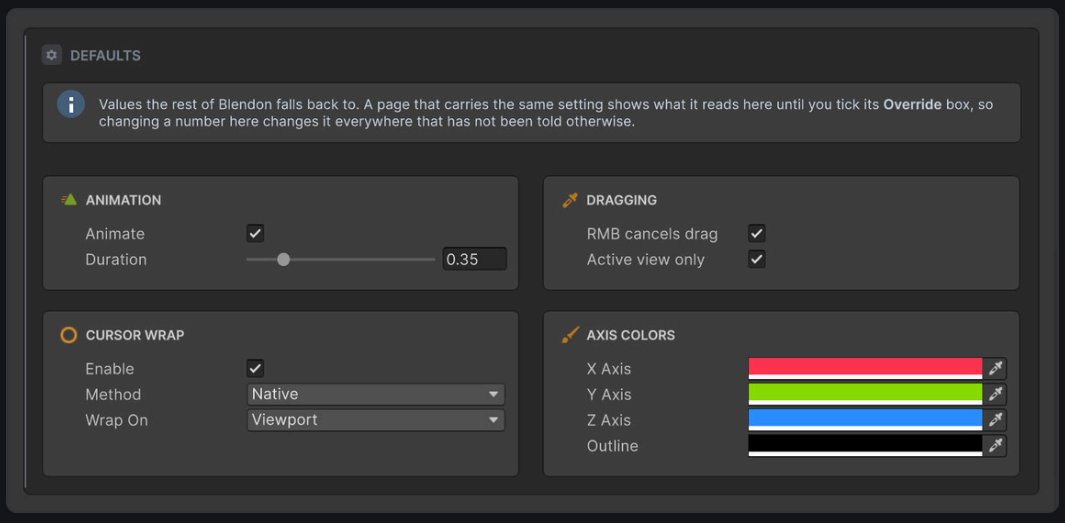


Figure 13.6 The Defaults panel. Changing a value here changes it on every page that does not override it.

Default	Value	Effect
Animate	On	Camera moves (numpad views, the orientation gizmo, frame, history) fly instead of cutting.
Duration	0.35 s	How long each of those flights takes.

Default	Value	Effect
RMB cancels drag	On	A right-click during any Blendon drag abandons it and restores the selection.
Active view only	On	A drag stays in the Scene view it started in instead of being mirrored into every open Scene view.
Cursor Wrap	On	Lets drags run past the edge of the view.
Method	OS Native	**OS Native**: Blendon teleports the cursor to the opposite edge itself. **Unity**: the Editor's own edge jumping. **Clamp To View**: no teleport, and the drag stops at the border.
Wrap On	Scene View Viewport	Where OS Native wraps: the Scene view's 3D area, the whole **Unity Window**, or the **Screen Boundary**. A Scene view in a floating window wraps at its own area under Unity Window. With two or more displays, Screen Boundary is the monitor under the cursor on Windows, and wraps like Scene View Viewport on macOS and Linux.
Axis Colors	The gizmo look's	X, Y, Z and the outline, shared by every gizmo and the orientation gizmo. They start from the colors of the gizmo look picked on the transform tool pages (**Blender** or **Unity+**), and a reset puts that look's colors back.

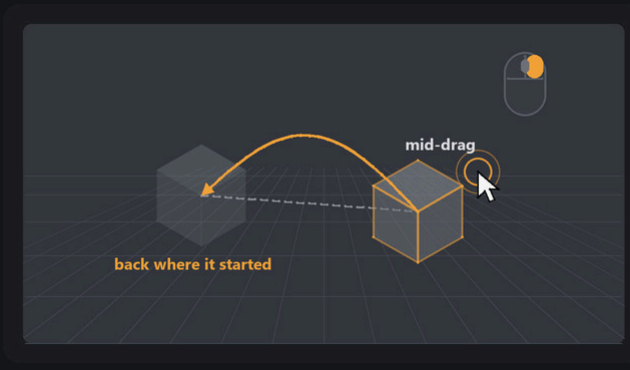


Figure 13.7 RMB cancels drag: a right-click puts the object back where it started.

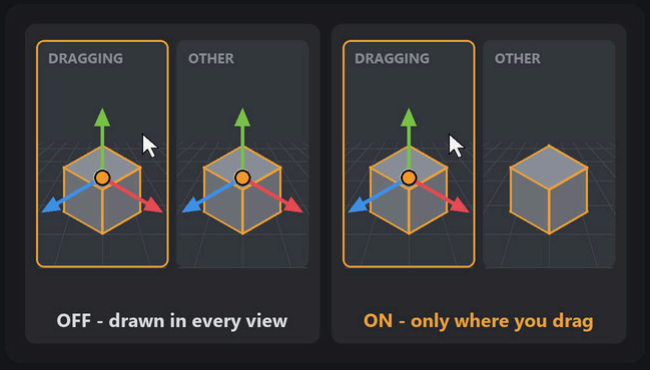


Figure 13.8 Active view only: the gizmo is drawn only in the view you drag in.

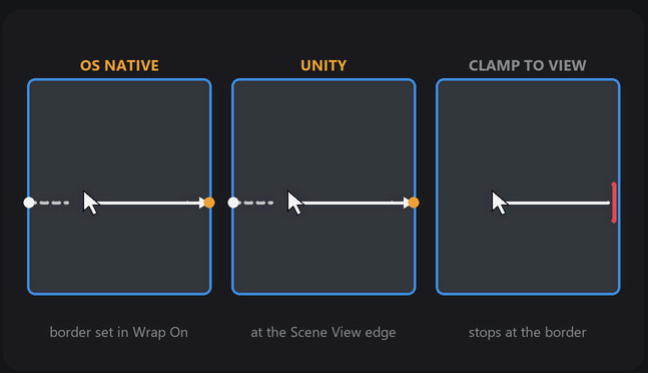


Figure 13.9 The three wrap methods: OS Native, Unity and Clamp To View.

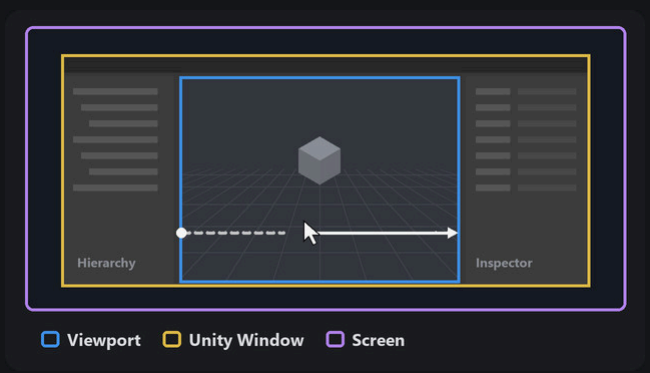


Figure 13.10 Where OS Native wraps: the viewport, the Unity window or the screen.

13.3 Keyboard page

Every shortcut Blendon declares, in one table grouped by the feature that owns it; see Default shortcuts. Each row edits the very same Shortcut Manager entry that the feature's own page edits: there is one binding behind both rows, never a copy. A count of how many bindings clash with something else sits at the top. A switched-off feature keeps its rows, under a note: its keys do not answer and the Editor's commands on them work as usual, but they keep their bindings and can still be changed here.

13.4 Keyboard presets

Blendon and Unity both want some of the same keys: G, R, S, Z, Q, V, the backtick, the numpad views, the middle mouse button and the right-click. The Overview page's **Keyboard Presets** picker settles that, on whatever shortcut profile the Editor is on:

Option	Effect
Blendon (recommended)	Blendon's Blender keys, as they ship. Each Editor command on one of them moves one modifier aside (Cycle Tool Modes to **Shift + G**, the Scale tool to **Shift + R**, ...), so it works in the Scene view too; a switched-off feature leaves its Editor command where it is. Where Blendon's command does the Editor's job (vertex snapping, the middle-mouse pans) both keep the key: in the Scene view Blendon's answers, elsewhere the Editor's. The moves are written into the active profile, and so is the right mouse button (below).
Unity	The Editor keeps every key and Blendon's contested shortcuts move instead: orbit on **Ctrl + Middle Mouse**, quick roll on **Ctrl + Shift + Middle Mouse**, pan on **Alt + Middle Mouse**, grab on **Shift + G**, the context menu on **Alt + right-click**, and so on. Scale Grab is left without a key. Vertex snapping stays on **V** and **Shift + V**: it answers in the Blendon tool context only, and the Editor's pair keeps those keys everywhere else. These are written into the active profile; going back to Blendon clears them again. Every Editor command the Blendon preset moves gets its own key back, and so does one you moved onto a key Blendon's now takes.

Which preset is lit is read from the keys themselves, so a copied profile shows the right one and nothing can drift. Picking a preset, the lit one too, sets every contested pair again. A key you tuned by hand stays tuned through any switch the picker makes, apart from the contested pairs the presets set. Unity's **Default** profile is read-only: picking a preset there, or editing any key, asks first, then creates an editable copy named **Default copy**, the way Unity's own Shortcuts window names it, and switches to it. The first-run question is the one exception: either preset answer already agrees to the copy, so it is made without asking again. Blendon never makes or switches a profile otherwise.

How a feature's keys switch off: each Blendon shortcut carries a tag of its own, which Blendon registers while the feature is on. Rotate Grab (**R**), the Draw Mode pie (**Z**) and the Tools pie (**Q**) share keys with Unity's Global Scale tool, Toggle Pivot Position and View tool, so they use a Scene view context of their own instead: when they are off, or another window has focus, the Editor's command answers. The context

menu and the eight custom pie slots use such a context too: a tag on the right button would also silence the fly-through drag, and an unbound slot has no key to tag. Either way, a switched-off feature writes nothing to your profile.

The right mouse button. Two menus on one button would make Unity ask which one to open on every click, so while Blendon's context menu holds the right button, Blendon unsets the Editor's own `Scene View/Menu` and remembers its key, also through a rename of the profile. The key goes back when the menu or Blendon is switched off, when Blendon's menu moves to another key and when the Editor closes, and the next start takes it again. Unity shares shortcut profiles between all your projects and reads them as an Editor starts, so a project without Blendon that opens on the same profile while Blendon runs has no right-click menu in the Scene view; see Troubleshooting. On Unity's read-only **Default** profile that write is not possible, so the Editor's menu keeps the button and Blendon's waits. The Overview page then shows an **Editable Profile** note, and the sidebar marks the Overview page with a red dot. The note's **Make Editable Copy** copies the profile and switches to the copy, gives Blendon's menu the right button, and moves every Editor command Blendon covers one modifier aside, so each works in the Scene view too. Declining the copy changes nothing. A copy made anywhere else, for example by editing a key on another page, skips those moves, so the Overview offers them in a note of its own: **Move All** moves every covered Editor command, **Keep** leaves them covered. Either answer ends the note, and so does picking a preset.

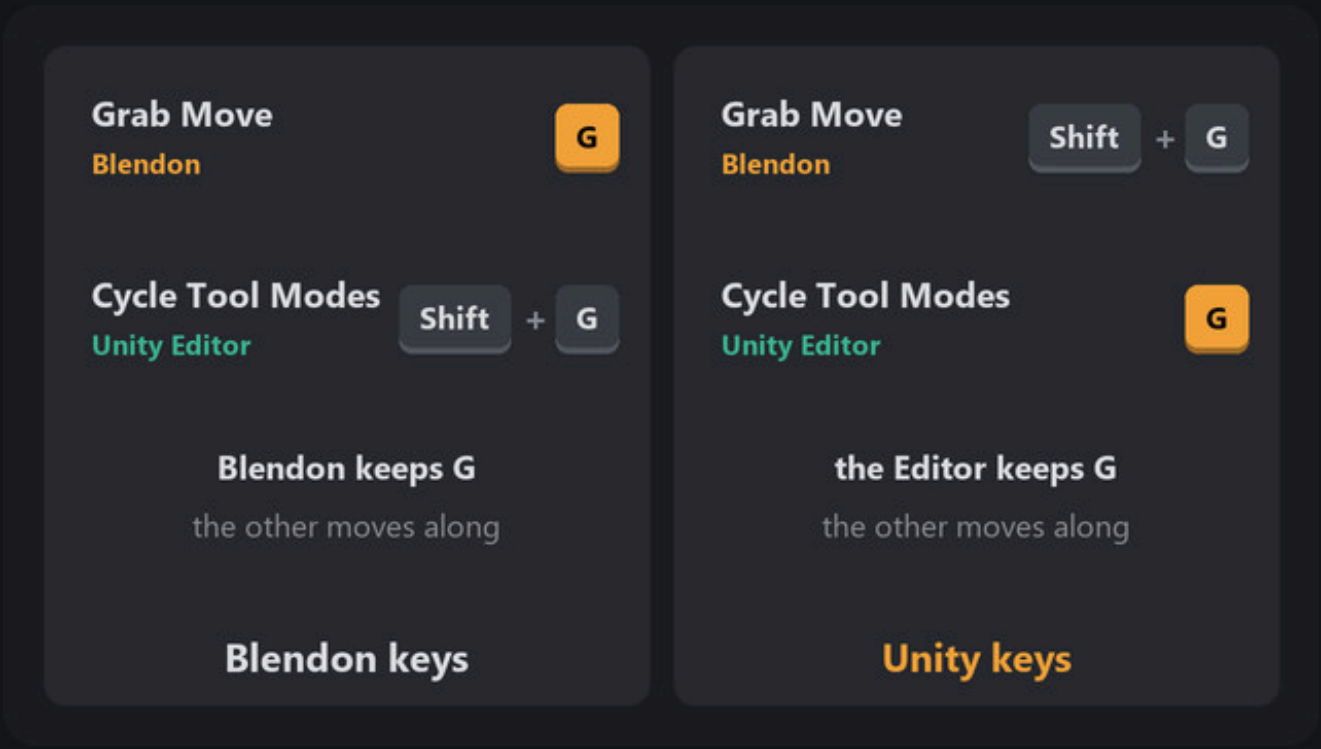


Figure 13.11 One contested key: in the Scene view Blendon's command answers it, the Editor's waits there.

Below the picker, the contested-keys table lists every pair whose Blendon feature is on: Blendon's shortcut next to the Editor command that wants the same key, as shown in Step 2. Unity's own Shortcuts window does not flag a covered Editor command as a conflict, so this table is where to find them. The button on each row swaps the pair, so the two never end up on one key:

- ↺ gives the Editor's command its own key back and moves Blendon's one modifier aside (Grab Move to **Shift + G**, Grab Rotate to **Shift + R**, …). Grab Scale is left without a key, since **Alt + S** is ProBuilder's.

- ↪ gives Blendon's shortcut its key back and moves the Editor's command aside (Cycle Tool Modes to **Shift + G**, the Scale tool to **Shift + R**, …), so it works in the Scene view too. A covered command, still on Blendon's key, offers ↪ as well.

A swap is written once and nothing keeps it there; picking a preset sets every row again. Pairs where Blendon's command does the Editor's job (vertex snapping, the middle-mouse pans) have nothing to swap, so they are listed only once their Editor command was moved by hand, and their ↺ gives it its key back. The Editor's Scene view menu is never listed, it follows Blendon's context menu by itself.

If one of Blendon's shortcuts clashes with something outside this system, for example after a binding has been hand-edited, the settings window still flags it inline wherever that shortcut appears, naming the conflicting command and offering to open Unity's Shortcuts window. A key that works in every window, such as **Toggle Blendon**, counts as a clash too when a Scene view command sits on it: in the Scene view the Editor's command comes first. Unity offers no public way to filter that window, so Blendon copies the command's name to the clipboard and says so; paste it into the window's search box.

14 Data and files

Blendon is Editor-only and adds nothing to player builds. It writes in three places, all of them yours to delete.

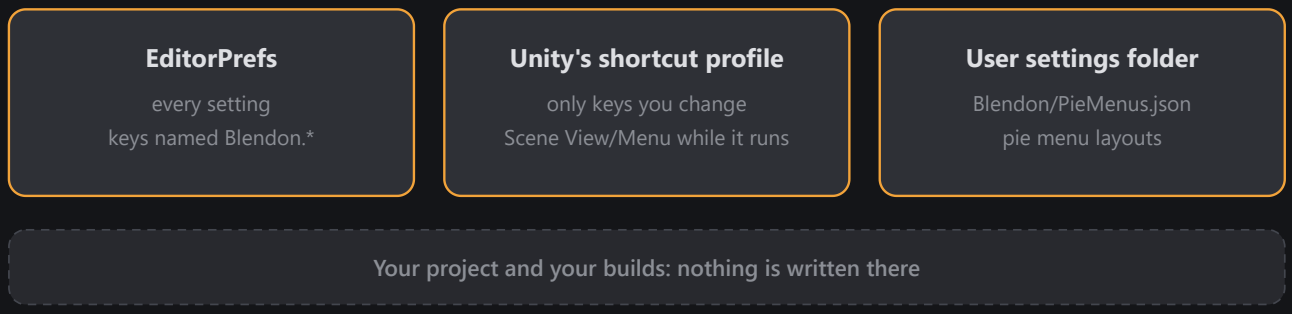


Figure 14.1 Everything Blendon writes, and where.

- **EditorPrefs.** Every setting in the settings window, and the Scene tutorial's progress, under `Blendon.*` keys.
- **Unity's shortcut profiles.** Only what you change yourself, the Unity preset's keys if you pick it, and the Editor's `Scene View/Menu`, unset while Blendon runs with its context menu on the right button. A `Default copy` only if you edited a key on Unity's read-only Default and agreed to the copy. Unity keeps these profiles for all your projects on the machine.
- **Your user settings folder.** Pie menu layouts, in `Blendon/PieMenus.json` under `%APPDATA%` on Windows, `~/Library/Application Support` on macOS, and `$XDG_CONFIG_HOME` (or `~/.config`) on Linux. These are kept outside the project on purpose: a menu edited once is edited for every project on the machine, which is the same scope as the EditorPrefs the rest of the settings use.

The file holds the menus, not their keys: a pie's key lives in Unity's shortcut profile, like every other Blendon key. A menu of your own is stored under the id of its slot, `custom.1` to `custom.8`; an entry under any other spelling stays in the file with no key and no row.

Every save that changes `PieMenus.json` keeps the previous version beside it as `PieMenus.json.bak`. If the file cannot be reached for a moment, for example while a second Editor or a sync tool such as OneDrive holds it, Blendon keeps your pie menu changes in memory and saves them once the file is free. If it is already out of reach when Blendon first reads it, the pies open as shipped and editing waits: the Pie Menus page and the pie editor say so, and nothing is written until the file has been read. A file that cannot be read is never written over: Blendon sets it aside as `PieMenus.corrupt-<date and time>.json` in the same folder, keeping the five newest such copies, says so in the Console, and carries on from the backup, or from the shipped pies when the backup cannot be read either.

A newer Blendon in another project may add menus to the shared file. This version keeps them in the file but does not list or open them. If a newer Blendon saved the file in a format this version does not know, this version still uses the menus it knows, but it does not save your pie menu changes: they last until the Editor closes, and the Console says so once.

Blendon makes no network requests, collects no analytics and installs no packages.

15 Troubleshooting

The keyboard presets dialog did not appear. It comes once per install in each project, and not at all while the profile is already on the Unity keys. An update keeps the earlier answer. Pick a preset on the Overview page at any time; see Keyboard presets.

None of Blendon's shortcuts does anything. Blendon's keys answer while the Scene view has focus; click into it first. Check the master switch on the Overview page.

G, R, S or V does what it does in plain Unity, and a drag draws Unity's selection box. The Scene view is not in the Blendon tool context. Outside it these keys and gestures are the Editor's on purpose. Press **B**, or pick **Game Object (Blendon)** in the Tool Contexts overlay; see The Blendon tool context.

Esc drops the Scene view to Game Object. Esc Keeps Context is off on the Overview page, or Blendon draws no gizmo: the master switch or all four gizmos are off. Then **Esc** works the Unity way. Press **B** to come back; with the master switch off, switch Blendon on first.

A shortcut does nothing. Open **Tools -> Blendon -> Settings** and check the feature's row on the Overview page. If the row is switched off, its keys belong to the rest of the Editor. If the key is highlighted, it clashes with another command; see Keyboard presets.

G, S, Z, Q or the backtick no longer runs the Editor's command in the Scene view. On the Blendon keyboard preset Blendon's command covers it there while its feature is on; in other windows the Editor's command still answers. For **G**, **S**, **Z** and **Q** that holds in the Blendon tool context only; outside it the Editor's command answers in the Scene view too. Unity's Shortcuts window does not show this as a conflict, but the contested-keys table on the Overview page lists every covered command. Press ↪ there to give the

Editor's command a key of its own, pick the **Blendon** preset to move every one of them, or pick the **Unity** preset. Vertex snapping, the middle-mouse pans and the Scene view menu are not in the table: Blendon's command does that job.

The numpad viewport shortcuts do nothing. NumLock must be on. The Numpad Views settings page warns inline when it is off.

Pressing R scales instead of starting the rotate grab. R answers Rotate Grab only while the Scene view has focus, in the Blendon tool context; anywhere else it is Unity's Scale tool. Click into the Scene view first, and press **B** if the context is Unity's. **Grab From Any Window** on the Rotate page lets **R** start the grab from other windows too. On the Unity keyboard preset Rotate Grab is on **Shift + R**.

Unity's own gizmo is drawn instead of Blendon's. Either that tool is disabled, or the Scene view is not in the Blendon tool context. A disabled transform tool deliberately falls back to Unity's own gizmo; switch it back on from its settings page or from the Overview list. If **Game Object** was picked by hand, it stays until the Editor restarts. A package tool that switches to Game Object by itself while the Blendon context is active, as ProBuilder's Poly Shape can, counts the same. Press **B**, or pick **Game Object (Blendon)** again in the Tool Contexts overlay; see The Blendon tool context.

Blendon's gizmo does not appear while editing a spline or a ProBuilder mesh. This is on purpose. The Splines and ProBuilder tool contexts use their own tools, so they keep working exactly as they do without Blendon, with the Editor's keys. **B** goes to Blendon's context from there, and Blendon takes its own context back once Unity leaves theirs.

The Pivot Point dropdown shows only Center and Pivot. Pivot Point Menu is off on the All Tools page. Switch it on for all four modes.

The Tool Settings overlay shows Unity's Pivot/Center dropdown, not Blendon's Pivot Point. Blendon's dropdown shows only while one of its own Move, Rotate, Scale or Transform tools is active. The Rect tool, a tool whose Blendon gizmo is switched off, and every other tool context keep Unity's own.

Right-clicking the Scene view opens Unity's menu instead of Blendon's. Check the **Context Menu** row on the Overview page, and that the master switch is on. On Unity's read-only **Default** profile Blendon's menu waits for an editable copy; the Overview page offers one. On the **Unity** keyboard preset the right-click stays with the Editor by design, and Blendon's menu is on **Alt + right-click**; pick the **Blendon** preset to have it on the plain right-click. A right-click with **Shift**, **Ctrl**, **Alt** or **Cmd** held goes to the Editor's own menu, unless the binding itself asks for that modifier, as **Alt** does on the Unity preset. If the `Blendon/Context Menu` binding was changed in Unity's Shortcuts window, use the shortcut row's restore button on the Context Menu page to put the preset's key back.

Right-clicking the Scene view does nothing in a project without Blendon. Unity shares shortcut profiles between all your projects. While an Editor with Blendon runs and its context menu holds the right mouse button, the Editor's own **Scene View/Menu** has no key on that profile, so a project without Blendon that opens on the same profile meanwhile has no right-click menu in the Scene view. Closing the Editor with Blendon gives the key back, and the other Editor reads it at its next start. After a crash nothing gave it back, so it stays unset until an Editor with Blendon closes normally. To get it back at once, open **Edit -> Shortcuts** (**Unity -> Shortcuts** on macOS) in that project, find **Scene View/Menu** and reset it to its default. A project with Blendon that opens on the same profile takes the button again. To keep the

Editor's menu on the right button for good, pick the **Unity** keyboard preset, or switch the context menu off, in your projects with Blendon. **Prepare Uninstall…** gives the button back on every profile; see Uninstalling.

An entry is missing from the context menu. Blendon's menu holds exactly what the Editor's own menu holds for the current selection and active tool. Select the object the entry belongs to, or switch to the tool that adds it, and right-click again.

Dragging over the Scene view draws Unity's box, not Blendon's. Box Select is switched off, or the Scene view is not in the Blendon tool context, which is also the case with all four gizmos off. Check its row on the Overview page, or press **B**. Terrain, particle systems, UI and gizmo icons are selected only when the box contains them wholly, as in the Editor.

Box Select misses an object that is behind another one. Visible Only is on, which is the default. Press **C** while dragging to include hidden objects, or switch it off on the Box Select page.

Vertex snapping finds no vertices. Check that the objects you are snapping to have a MeshRenderer or SkinnedMeshRenderer and are not hidden by Scene Visibility or excluded by Isolate View.

I want everything back the way it was. Use the **Master switch** on the Overview page. It switches every Blendon key off, gives the Editor its right-click menu back and returns the Scene view to the Game Object tool context, without losing any of your settings. To remove Blendon, see Uninstalling.

Something is still wrong. See Reporting a problem: **Copy Support Info** puts your versions and setup on the clipboard, ready for the report.

16 Uninstalling

Blendon writes nothing into your project, so removing it comes down to deleting its folder. Unity's shortcut profiles and Blendon's settings outlive the folder, though, and every project on the machine shares them. **Prepare Uninstall…** puts the profiles in order while Blendon can still do it:

1. Open **Tools -> Blendon -> Prepare Uninstall…** and click **Restore** in the dialog that asks first.
2. A second question offers to delete your Blendon settings as well. **Keep Settings**, which Esc and the close button also give, leaves them for your other projects and a reinstall.
3. Delete the Blendon folder from the Project window.

Prepare Uninstall switches Blendon off until you restart the Editor; the master switch you set stays as it is. Then, on every shortcut profile that can be edited, it:

- gives the Editor's own **Scene View/Menu** its right mouse button back, the one key Blendon writes by itself;
- gives every Editor command that the Blendon preset or a ↪ button on the Overview page moved its own key back;
- clears every Blendon key stored there: keys you changed, and the Unity preset's keys.

Unity's read-only **Default** profile is never written, and holds none of these. A closing message says what was done and where your pie menus are. Since the profiles are shared, another project that uses Blendon on the same profile goes back to the Blender keys too.

What stays behind:

- **Settings**, unless you picked **Delete Settings**. They live in `EditorPrefs` (the registry on Windows), which every project on the machine shares, so another project with Blendon keeps them and a reinstall finds them. So do the Scene tutorial's progress, a few remembered view states such as where the tutorial badge sits, and the Blendon version that last ran. **Delete Settings** removes all of these for every project on the machine.
- **Pie menus**, either way. The `Blendon` folder in your user settings folder: `%APPDATA%` on Windows, `~/Library/Application Support` on macOS, or `$XDG_CONFIG_HOME` (or `~/.config`) on Linux. It holds `PieMenus.json`, its `.bak` backup and the `PieMenus.corrupt-...` copies set aside there. The folder is kept outside the project on purpose, so that a reinstall finds your own menus where you left them; delete it only if you want them gone for good.
- **A few small `EditorPrefs` entries**, even after **Delete Settings**: one per project that remembers the keyboard preset question was asked, and the Blendon version that last ran in your other projects. They need no cleanup: only Blendon reads them.

Nothing else is left: no files inside the project, and nothing that was ever part of a player build.

If you keep the folder after all, Blendon is back after the restart, with its settings, or its defaults if you deleted them, and on the Blender keys. To switch it back on at once, flip the master switch on the Overview page or pick a feature preset.

Deleting the folder without Prepare Uninstall. Deleting the Blendon folder, or a folder that holds it, from Unity's Project window gives the Editor its keys back by itself: the Scene View menu and the commands a ↪ button moved get their own keys back, on the active profile and on every profile Blendon took the menu on. Blendon's own keys stay in the profiles, where they answer to nothing and do no harm. Deleted the folder outside Unity, for example in a file manager? Then nothing gave the keys back: open **Edit -> Shortcuts** (**Unity -> Shortcuts** on macOS), find **Scene View/Menu** and reset it to its default, and do the same for any Editor command you moved. A project without Blendon on a profile that Blendon still holds needs the same step; see Troubleshooting.

Updating. To update Blendon, import the new version over the old one. Do not delete the folder first: that counts as an uninstall, so the Editor commands you moved get their own keys back, and on Windows the keyboard preset question comes again.

17 Support

Questions, bug reports and feature requests: blendon@drygiel.com

With a bug report, include the support info described below, or at least your Unity version and your Editor platform. Most of what Blendon touches (shortcut profiles, the Scene view overlay set, the native cursor and the NumLock read) differs between Windows, macOS and Linux, and between Unity 6.0 and the current release.

The latest manual, the interactive Playground and news about updates are on the website: blendon.drygiel.com.

17.1 Reporting a problem

Copy Support Info puts a short report of your setup on the clipboard, ready to paste into an email. It is on the Overview page, as **Copy support info** in the line that names your Blendon version, and in the menu bar as **Tools -> Blendon -> Copy Support Info**. Both work while Blendon is switched off, and so do the Overview page's links to this manual and to the website.

The report is plain text, one line per item:

- **Blendon version**, and the highest version that ever ran on this machine.
- **Unity**, **OS**, **Render pipeline** and **Editor skin**.
- **Shortcut profile**, whether it is read-only, and the **Keyboard preset** its keys add up to.
- Whether the **Blendon tool context** is active, the **Master switch**, and which features are on and which are off.
- **Optional packages**: whether the Terrain module, the Video module and the Input System are there.
- **Pie menu file**: whether it can be saved, and its folder.
- **Recent exceptions**: the last five errors Blendon caught, newest first, each with its time, its type, its first line and where it was thrown.

Folders are cut down to tokens such as `<project>`, `%APPDATA%` and `~`, and your user name and machine name are blanked, so the report names no project path, user or machine. Beyond each feature's on/off state it holds no setting values, and it holds no keys. Read it before you send it.

Blendon's messages in the Console start with **Blendon**. When one part of Blendon fails, the Console shows a line in this form, followed by the error itself with its stack trace: `Blendon: <part> threw <error>. Repeats of this are not logged.` Each part logs each kind of error once, until scripts reload or the Editor restarts. Other errors Blendon catches read `Blendon: <what> failed with <error>.`, again followed by the error. Copy these entries into your report too.

Send the report to blendon@drygiel.com, with what you did, what you expected and what happened instead.